Electronic Displacement Effects

Permanent Effects (Polarisation Effects)

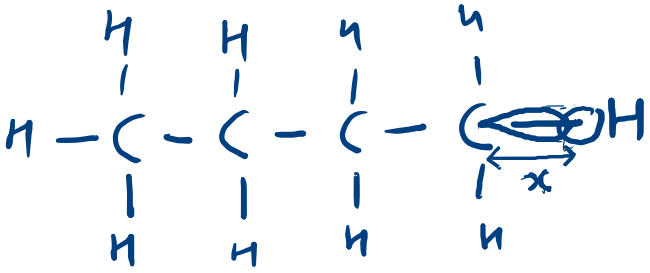
1. Inductive Effect
2. Resonance
3. Mesomeric Effect
4. Hyperconjugation
5. SiR Effect
6. Ortho Effect for acidic strength
7. Ortho Effect for Basic Strength
8. Steric Effect
9. field Effect.

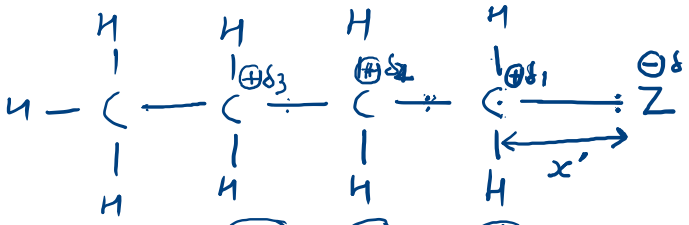
Temporary Effect (Polarizibility) Effects

10. Electromeric Effect
11. Inductomeric Effect.

Inductive effect

Partial and permanent displacement of σ electrons towards more electronegative group or atom is known as inductive effect





EN

H < Z

2.1

$x < x'$ $-I$

$\oplus\delta_3 < \oplus\delta_2 < \oplus\delta_1$

$\ldots\ldots + \oplus\delta_3 + \oplus\delta_2 + \oplus\delta_1 = \ominus\delta$

Important points related to Inductive effects

1. It is partial displacement of σ electrons, therefore partial charge develop on adjacent atoms.
2. It is permanent displacement of σ-electrons because external reagent is not required.
3. Approx 10% effect transferred to the next carbon
4. It is observed upto 3 carbons, after 3 carbons it's effect is negligible. (# It is distance dependent)
5. it is weak or poor effect
6. # It is distance dependent effect. I.e as the distance in carbon chain increases from inductive group, I-effect decreases exponentially.
7. Net charge remain constant in a molecule having inductive effect, (Because it is intra-molecular displacement of σ-electrons)

9. I Effect are two types

(A) - I Effect

(B) + I Effect

(A) - I Effect / - I Group

The group which attrat σ-éle is called - I group. and effect is k/a - I Effect

order of strongness of - I group

$-\overset{\oplus}{N}F_3 > -\overset{\oplus}{N}R_3 > -\overset{\oplus}{N}H_3 > -NO_2 > -CN > -SO_3H >$

$-\overset{O}{\overset{\|}{C}}-H > -\overset{O}{\overset{\|}{C}}- > -\overset{O}{\overset{\|}{C}}-OH > -F > -Cl > -Br > -I >$

$-OR > -OH > -NH_2 > -C\equiv CH > -Ph > -CH=CH >$

Trick (N) (C) (C) (C) halide

$-NO_2$ $-CN$ $-CHO$ $-\overset{O}{\overset{\|}{C}}OH$ F Cl Br I

(8) causes for Inductive effect

(a) Difference in electronegativity.

eg. (1) $CH_3 \rightarrow F$ ($\oplus\delta$ on CH_3, $\ominus\delta$ on F)

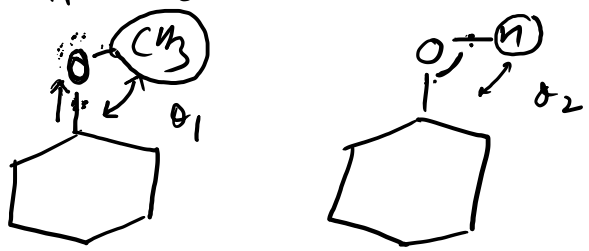
(b) Difference in Hybridisation state

eg. (2) $CH_3 - CH_2 - CH = CH_2$ ($\oplus\delta$, $\ominus\delta$)

sp^3 sp^2

	Compound	Hybridisation state	% S	EN
eg. (3)	CH_3-CH_3	sp^3	25%	2.48
eg. (4)	$CH_2=CH_2$	sp^2	33.33%	2.75
eg (5)	$HC\equiv CH$	sp	50%	3.25
eg (6)		sp^3	25%	3
eg (7)		sp^3	25%	3.5

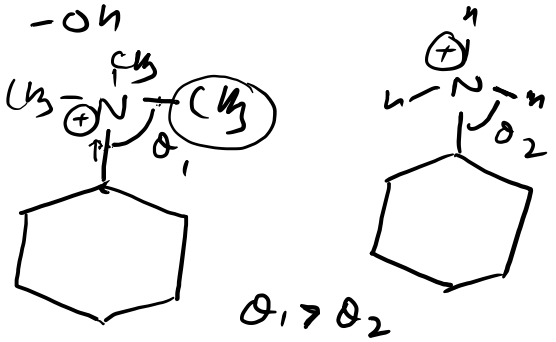
-I effect $-OR > -OH$



$\theta_1 > \theta_2$

$-OCH_3 > -OH$

-I



-I order

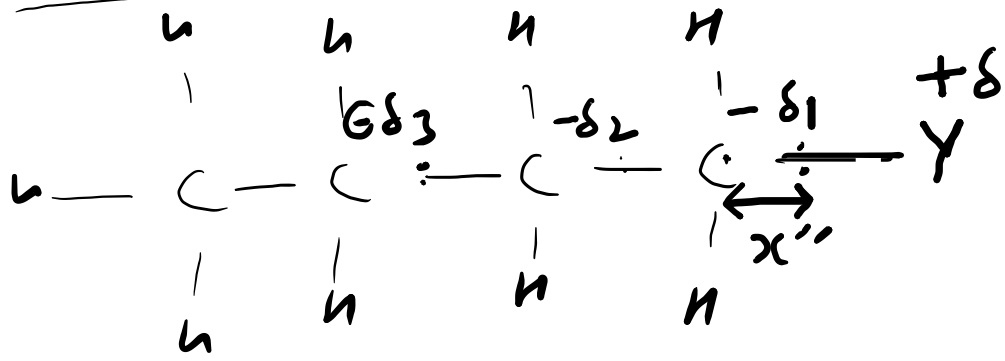
$-\overset{\oplus}{N}R_3 > -\overset{\oplus}{N}H_3$

$\theta_1 > \theta_2$

(B) +I Group or +I Effect

The group which repeal the σ-$\bar{e}$les is called +I group and effect is

K/a +I effect



EN

H > Y

$x > x''$

$+\delta_3 < -\delta_2 < -\delta_1$

$\ldots + \ominus\delta_3 + \ominus\delta_2 + \ominus\delta_1 = \oplus\delta$

order of strongness of +I Groups

$-\overset{\ominus}{\underset{\cdot\cdot}{C}}H_2 > -\overset{\ominus}{\underset{\cdot\cdot}{N}}H > -\overset{\ominus}{\underset{\cdot\cdot}{O}} > -\overset{O}{\overset{\|}{C}}-\overset{\ominus}{O} > -\overset{CH_3}{\underset{CH_3}{\overset{|}{C}}}-CH_3 > -CH\begin{matrix}CH_3\\CH_3\end{matrix} >$

$-CH_2-CH_3 > -CT_3 > -CD_3 > -CH_3 > -T > -D >$

H
||
O
NO +I
NO -I

Que. The I Effect of $-\overset{O}{\overset{\|}{C}}-OH$ group is

(A) In acidic medium (−I) Group

(B) In Basic medium (+I) Group

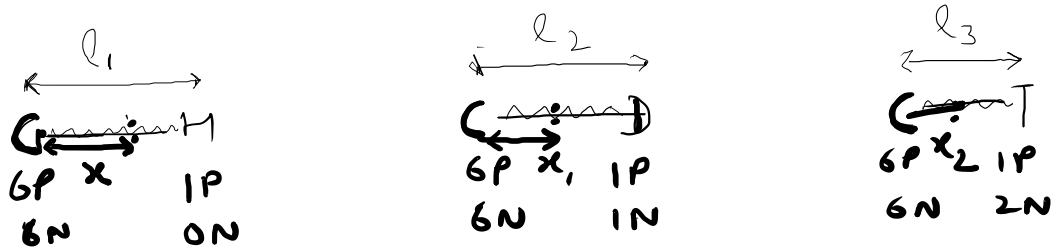
$-\overset{O}{\overset{\|}{C}}-OH \xrightarrow{H^{\oplus}} -\overset{O}{\overset{\|}{C}}-OH$ (In acidic Medium)

$-I$

$-\overset{O}{\overset{\|}{C}}-OH \overset{\overset{\ominus}{OH}}{\rightleftharpoons} -\overset{O}{\overset{\|}{C}}-\overset{\ominus}{O} + H_2O$

Acid — +I

Que. order of +I $-CT_3 > -CD_3 > -CH_3$?



$l_1 > l_2 > l_3$

$x > x_1 > x_2$

C–H C–D C–T

$x > x_1 > x_2$

order of +I

H < -D < -T

Bond order (B.O)

⇓

No. of bonds present b/w atoms

	B.O
A — B	1
A = B	2
A ≡ B	3
A ⋯ B	1 < B.O < 2
A ⋯ B	2 < B.O < 3

* **Bond energy** B.E

B.E ∝ B.O

* **Bond length** (B.L)

$$B.L \propto \frac{1}{B.O}$$

---

order of B.L

C-H > C-D > C-T

order of B.O or B.E

C-H < C-D < C-T

Order of +H

$-CH_3 > -CD_3 > -CT_3$

Order of +I

$-CT_3 > -CD_3 > -CH_3$

Application of I. Effect

(A) To Explain the stability of Intermedi-ates

(B) To Explain acidic and Basic Strength

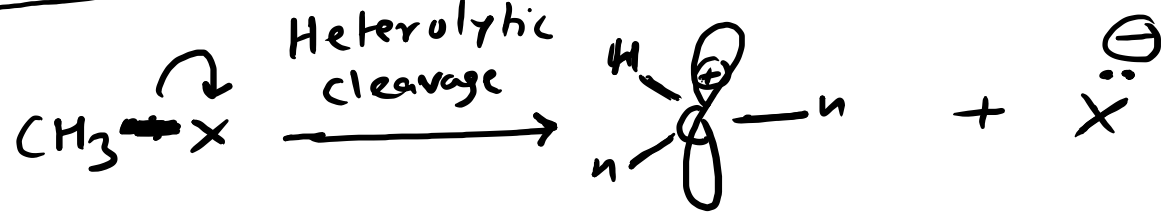
(A) To Explain the stability of reaction intermediate

(A) Reactant → (B) Intermediate → (C) product

Intermediate

* carbocation
* Free Radical
* carbanion
* carbene
* Nitrene
* Benzyne.

① To Explain the Stability of carbocation



Carbocation

sp^2

6 Ele in V.shell.

* Incomple octate
* 14 Di Diamagnetic
* Spin multiplicity

$= 2S + 1 = 2 \times 0 + 1 = 1$

* EDG (Electron donating Groups

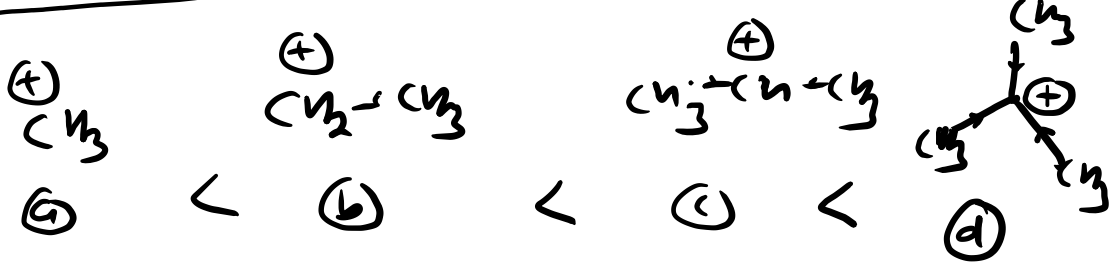
+R, +H, +I increases the Stability of carbocation

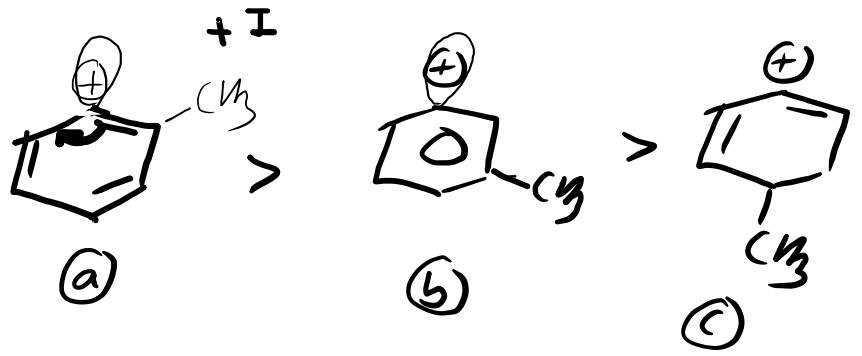
* EWG (Electron withdrawing Group)

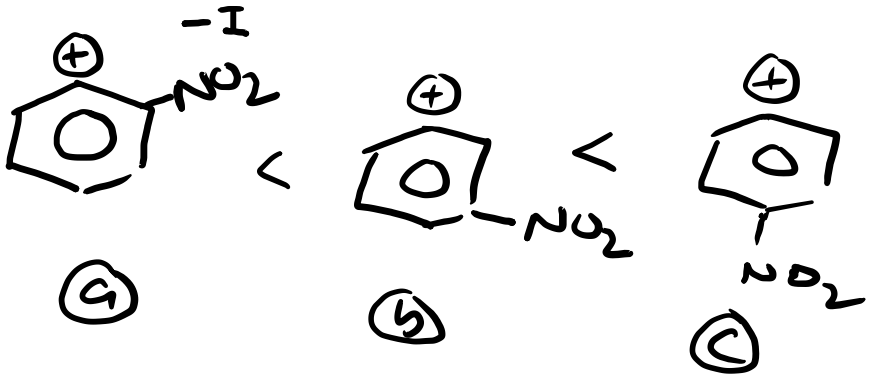
-R, -H, -I decreases the Stability of carbocation.

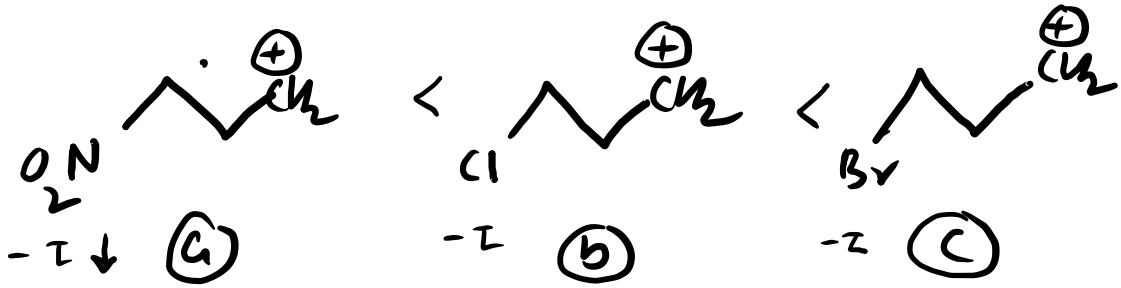
Compare the Stability of carbocation

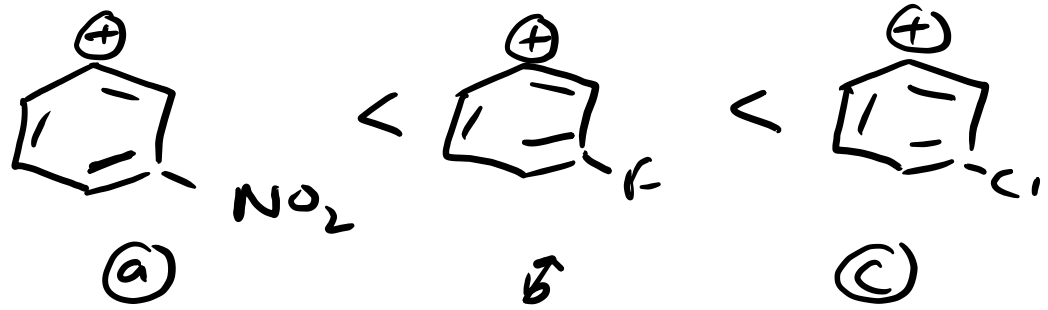
eg. ①

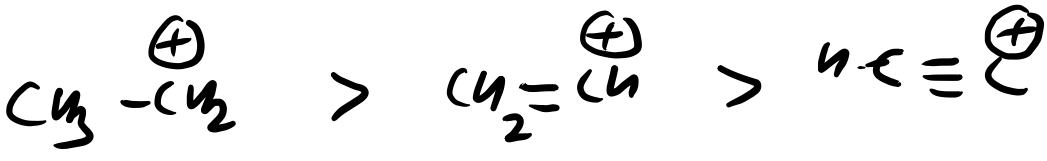


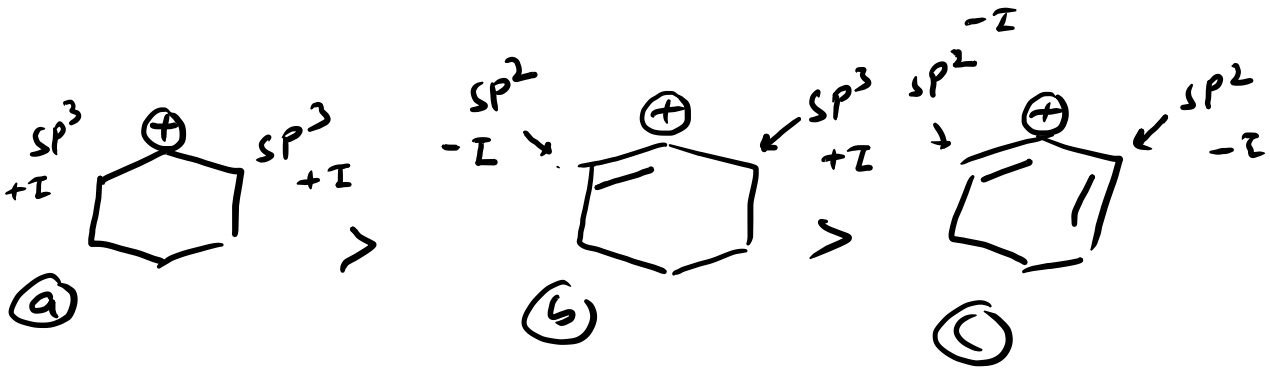
(2) 

(3) 

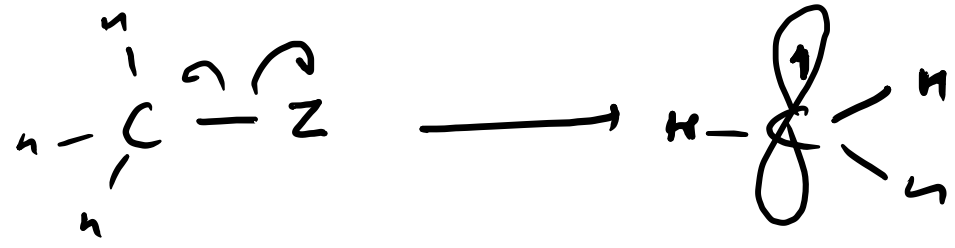
(4) 

(5) 

(6) 

(7) 

EDG ↑ the stability of FREE Radical



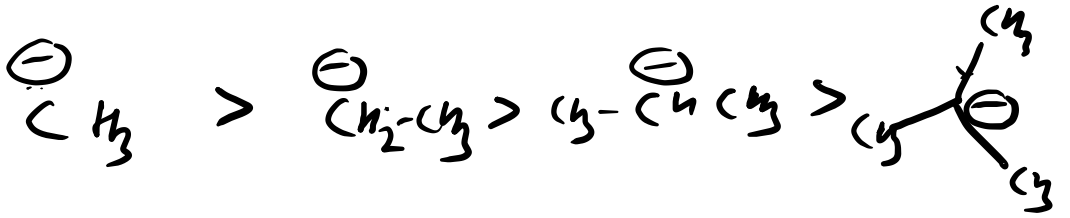
EWG ↓

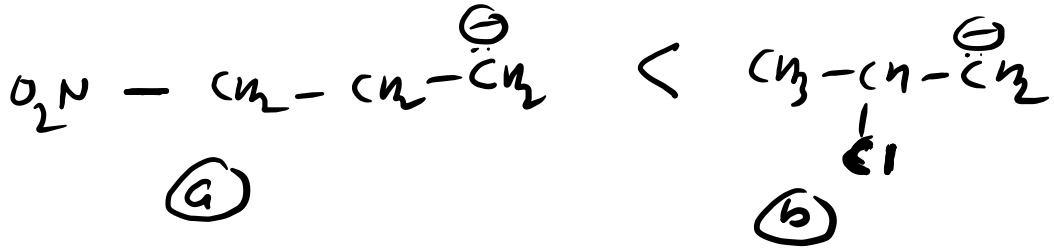
FREE Radical

+ paramagnetic

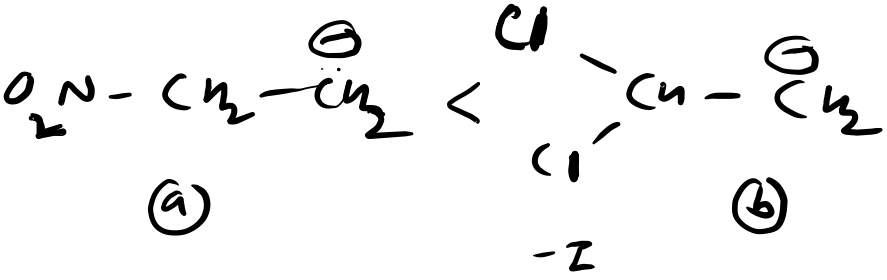
EWG ↑ increases the stability of carbanion

EDG ↓



eg. 

(a) (b)

eg. 

(a) (b)

-I

Race

Que-1

A- → Q Dia

R

S

B- → Q Dia

T SP^3

3+1 = 4

C- → P Para

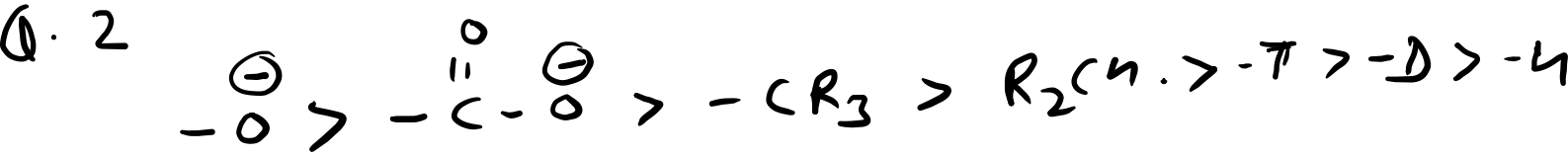
R

S

D –

Q Dia

T SP^3

Q. 2 

Ans. A

Que.

Que. No 4.

(A) $3^{\circ}C < 2^{\circ}C < 1^{\circ}C$

(B) $1^{\circ}C < 2^{\circ}C < 3^{\circ}C$

(C) $1^{\circ}C < 2^{\circ}C < 3^{\circ}C$

(D) X

5 (D) X

Ans B.

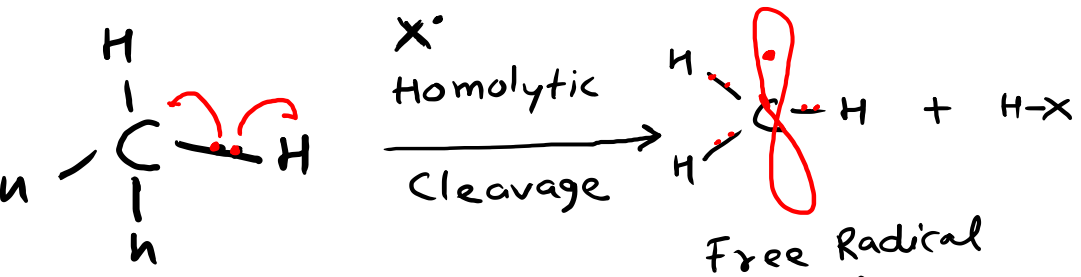
Q.7 00C

Q.8. A, B. D

Q.9 005 (a, b, d, f, i)

Q.10 $x = 3$ $y = 6$ Ans 036.

To Explain the Stability of Free Radical



Free Radical

SP^2

Trigonal planer.

7 ē le. in V. shell

Incomple octate

H.S = $\frac{\sigma \text{ bond} + \text{LLP}}{3 \quad + 0}$

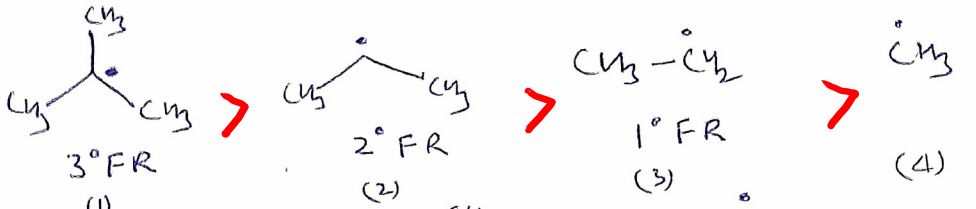
EDG or ERG (+R, +H, +I) ↑

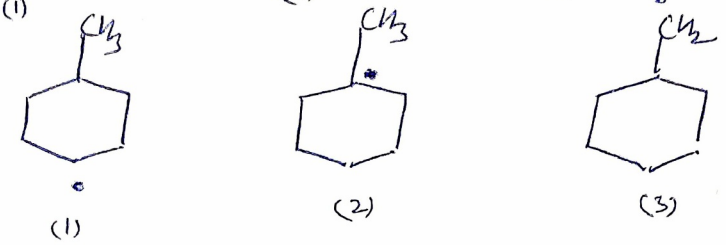
increases stability of FR

EWG (-R, -H, -I) ↓

decreases stability of FR

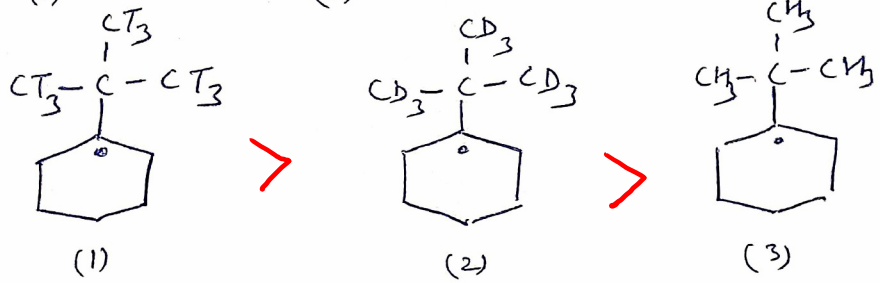
Compare stability of Free Radicals

① 

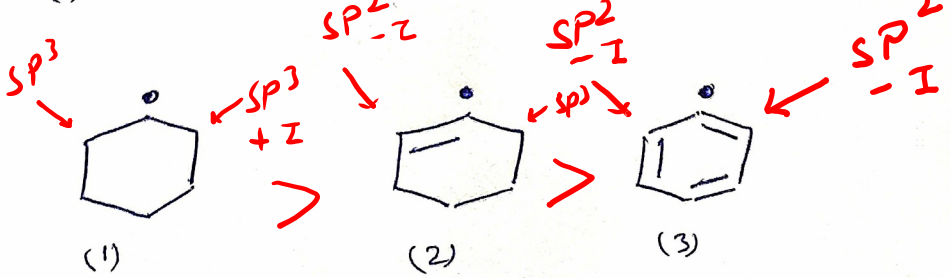
② 

2 > 1 > 3

③ $\dot{C}H_3$ (1) < $\dot{C}D_3$ (2) < $\dot{C}T_3$ (3)

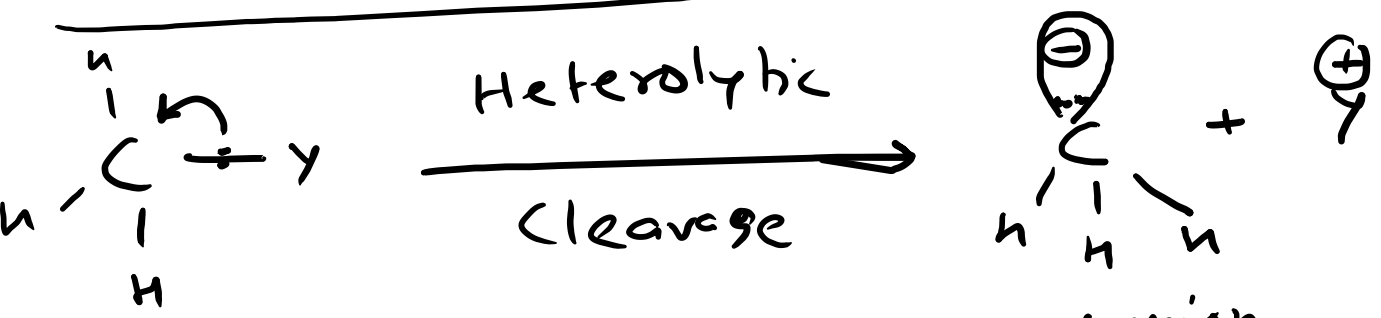
④ 

⑤ $CH_3-\dot{C}H-CH_2-NO_2$ (1) < $CH_2(NO_2)-CH_2-\dot{C}H_2$ (2)

⑥ 

⑦ $\overset{sp^3}{CH_3-\dot{C}H_2}$ (1) > $\overset{sp^2}{CH=\dot{C}H_2}$ (2) > $\overset{sp}{HC\equiv\dot{C}}$ (3)

To Explain the stability of Carbanion

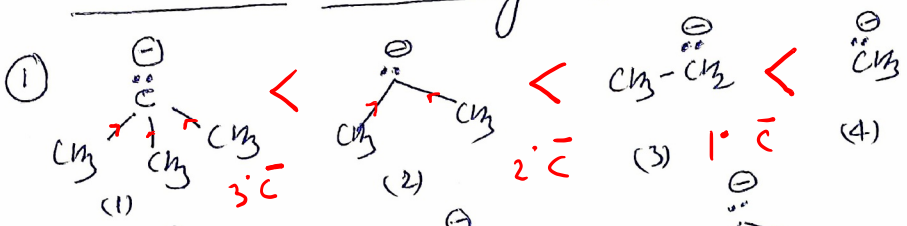


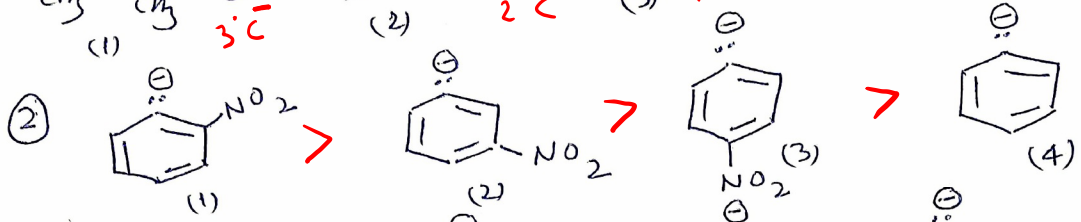
Carbanion
sp^3
Trigonal Pyramidal
Dia

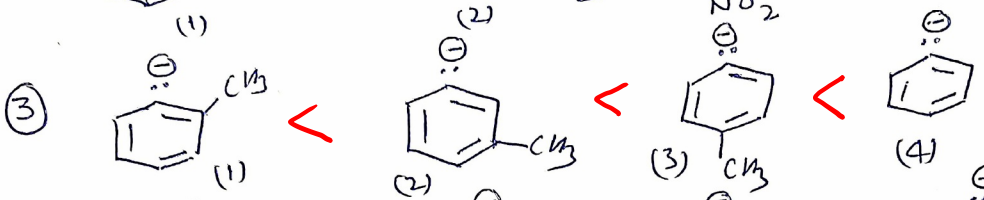
EWG ($-R$, $-H$, $-I$) increases stability of carbanion

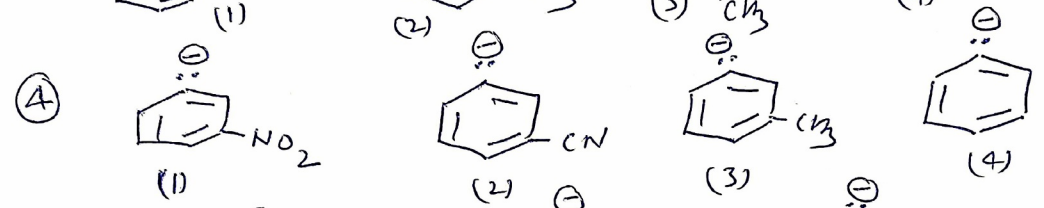
EDG or ERG ($+R$, $+H$, $+I$) decreases stability of carbanion.

Compare stability of carbanion or Anion

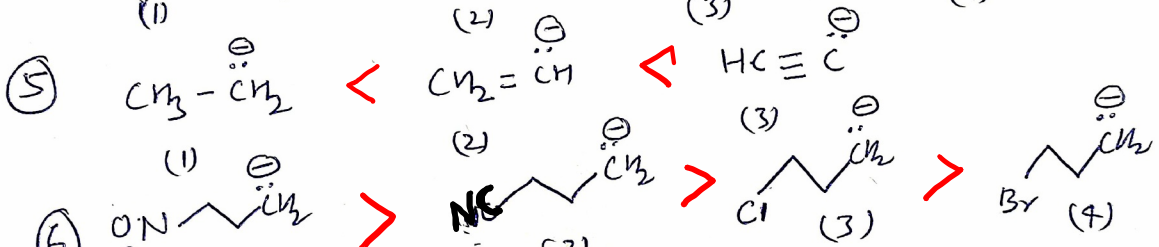


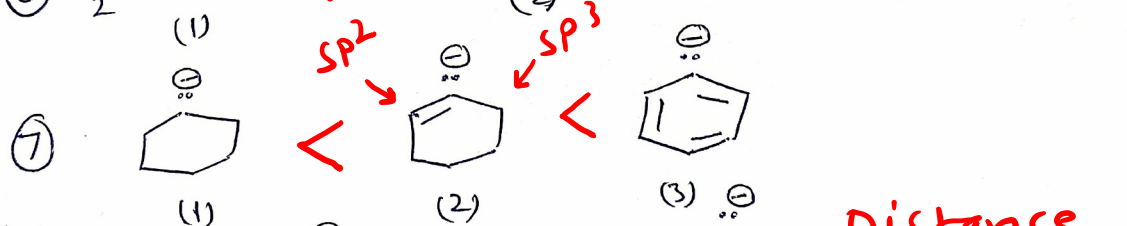




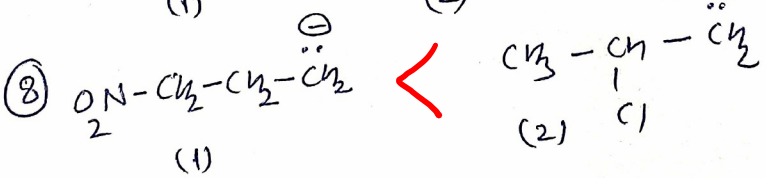


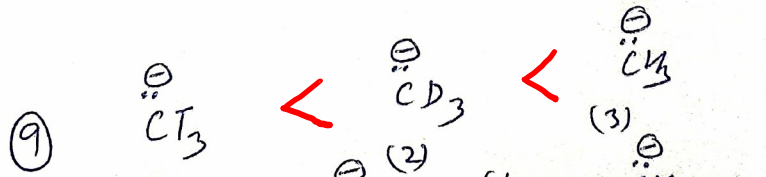
1 > 2 > 4 > 3



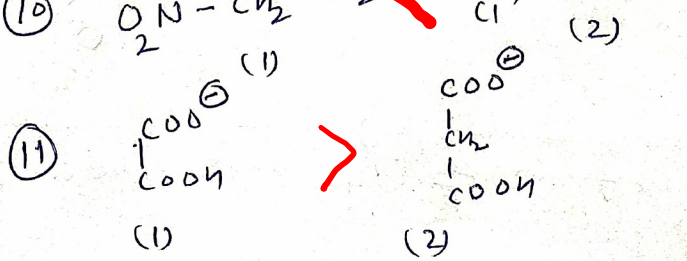


Distance factor dominates





Two weak groups dominates over one strong group.



To Explain Acidic and Basic Strength

Acid Base Concept

(A) Arhenius Theory

In aq medm $H^{\oplus}$ release → Acid

$OH^{\ominus}$ release → Base

(B) Bronsted lowry proton theory

$H^{\oplus}$ proton donor → Acid

$H^{\oplus}$ proton acceptor → Base

(C) Lewis concept

Lone pair donor → Base

Lone pair acceptor → Acid

Lone pair

To Explain Acidic strength / Acidic Nature / or Acidity

$$H-A \rightleftharpoons H^{\oplus} + \ddot{A}^{\ominus}$$

Acid — conjugate Base

(CBI) conjugate Base identification

Note: More stable the conjugate Base (i.e $\ddot{A}^{\ominus}$), more will be the forward reaction, which results more acidic nature of H-A

$$K_a = \frac{[H^{\oplus}][\ddot{A}^{\ominus}]}{[H-A]}$$

K_a ↗ dissociation constant for acid

** **Acidic Strength $\propto K_a$**

$$pK_a = -\log K_a$$

$$\text{Acidic strength} \propto \frac{1}{pK_a}$$

Note EWG (-R, -H, -I) ↑ stability of conjugate Base

so they increases Acidic Strength

EDG (+R, +H, +I) ↓ Decreases Acidic strength.

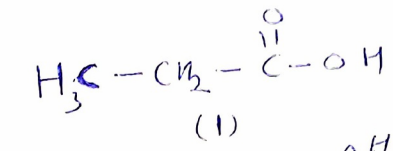
One ① compare acidity

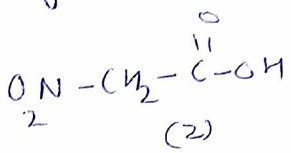
(a) $H-\ddot{O}-H \rightleftharpoons H^{\oplus} + \ddot{O}H^{\ominus}$ (b) $H-F \rightleftharpoons H^{\oplus} + \ddot{F}^{\ominus}$

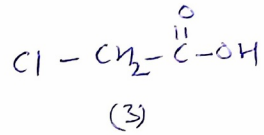
$\ddot{O}H^{\ominus} < \ddot{F}^{\ominus}$ EN

H-O-H < HF

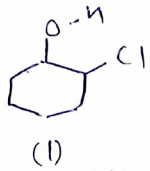
Compare Acidic strength

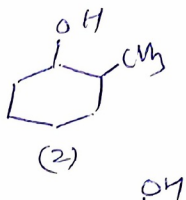
②  (1)

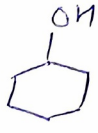
 (2)

 (3)

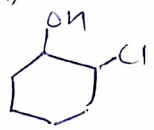
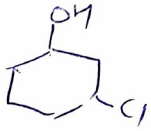
2 > 3 > 1

③  (1)

 (2)

 (3)

1 > 3 > 2

④  (1) >  (2) > (3) > (4)

⑤ CH_4 (1) < NH_3 (2) < $H-O-H$ (3) < $H-F$ (4)

(1) 1L H^+ CH_3^-

(2) 1L H^+ NH_2^-

(3) 1L H^+ OH^-

(4) 1L $H^+ + F^-$ EN

⑥ $H-F$ (1) < $H-Cl$ (2) < $H-Br$ (3) < $H-I$ (4)

(1) 1L H^+ F^-

(2) 1L H^+ Cl^-

(3) 1L H^+ Br^-

(4) 1L $H^+ + I^-$

SIZE

⑦ $H-S-H$ (1) < $H-Cl$ (2) EN

⑧ $H-O-H$ (1) < $H-S-H$ (2) SIZE

(9) (1) $O_2N-CH_2-CH_2-OH$

(2) $F-CH_2-CH_2-OH$

(3) $Cl-CH_2-CH_2-OH$

(4) $Br-CH_2-CH_2-OH$

1 > 2 > 3 > 4

(10) (1) $CH_3-CH_2-\underset{Cl}{CH}-C(=O)-OH$ (-I) > (2) $CH_3-\underset{Cl}{CH}-C(=O)-OH$ (-I) > (3) $Cl-CH_2-CH_2-C(=O)-OH$ (-I)

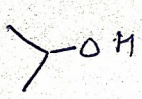
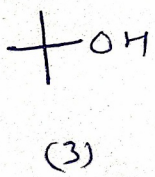
(11) (1) $Cl-CH_2-C(=O)-OH$ < (2) $CHCl_2-C(=O)-OH$ < (3) $CCl_3-C(=O)-OH$

(12) (1) CH_3-CH_3 < (2) $CH_2=CH_2$ < (3) $HC\equiv CH$

$H^{\oplus} + CH_3-CH_2^{\ominus}$ < $H^{\oplus} + CH_2=CH^{\ominus}$ < $H^{\oplus} + HC\equiv C^{\ominus}$

(13) (1) $H-O-H$ < (2) $H-S-H$ < (3) $H-Cl$

(1) → (2): SIZE; (2) → (3): EN

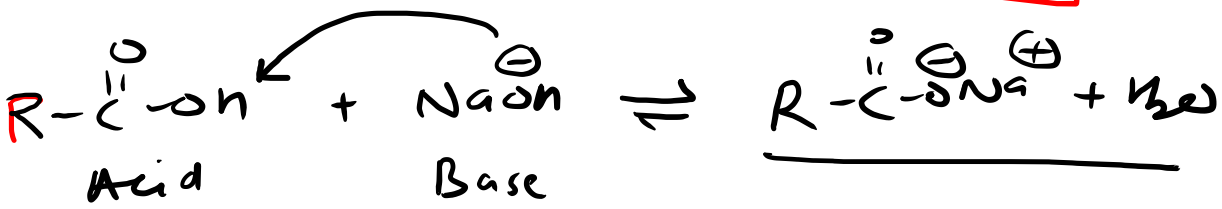
(14) (1) CH_3-CH_2-OH > (2)  > (3) 

Basic strength / Basic Nature / Basicity

The tendency to donate the pair of electrons to $H^{\oplus}$ is known as Basicity.

① Basic Strength $\propto$ ēle density on an atom

② Basic strength $\propto \frac{1}{\text{Stability}}$



$$K_b = \frac{[R-\overset{\overset{O}{\|}}{C}-O^{\ominus}Na^{\oplus}]\,[H_2O]}{[R-\overset{\overset{O}{\|}}{C}-OH]\,[NaOH]}$$

Basic strength $\propto K_b \propto \frac{1}{pK_b}$

$pK_b = -\log K_b$

EDG (+R, +H, +I) ↑ increases Basic strength.

EWG (-R, -H, -I) ↓ Decreases Basic strength.

Basic strength / Basic Nature / Basicity

Basic strength $\propto$ ēle density on an atom

If atom is same.

EDG ↑ Basic strength

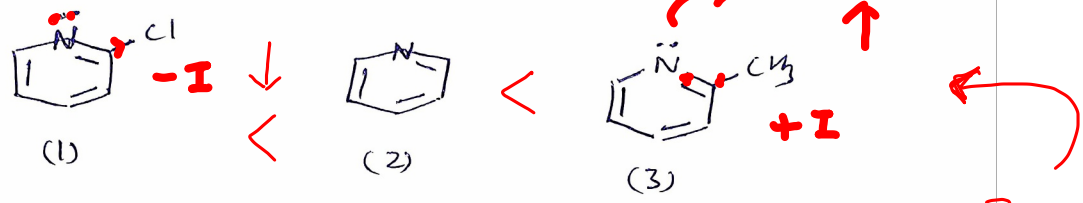
If atoms are different

Basic Strength $\propto \frac{1}{\text{Stability}}$

It is also applicable for same atoms

EWG ↓ Basic strength.

Compare Basic strength

① 

1 < 2 < 3

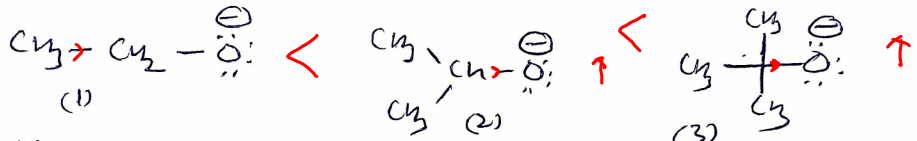
② -I

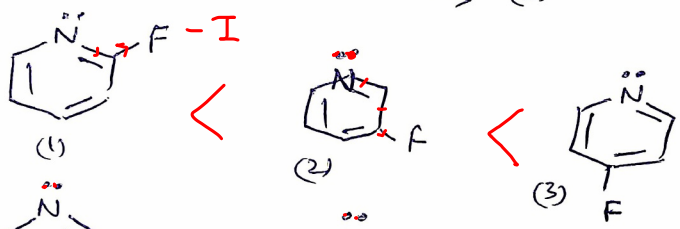
(1) $NO_2 - CH_2 - CH_2 - NH_2$

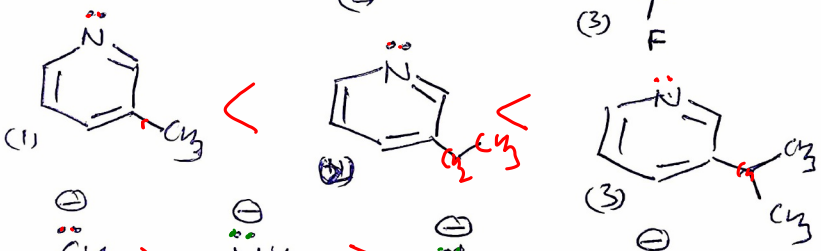
(2) $F - CH_2 - CH_2 - NH_2$

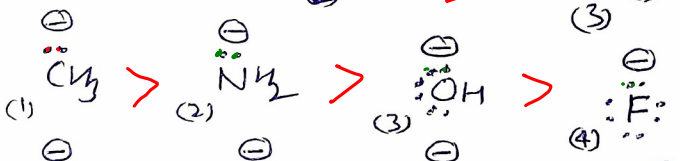
(3) $Cl - CH_2 - CH_2 - NH_2$

3 > 2 > 1

③ 

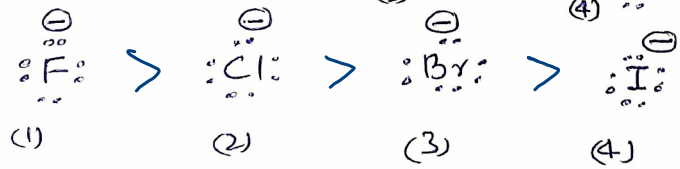
④ 

⑤ 

EN ⑥ 

Stability $\bar{C}H_3 < \bar{N}H_2 < \bar{O}H < \bar{F}$

Basicity $\propto \frac{1}{\text{Stability}}$

SIZE ⑦ 

Stability

$F^- < Cl^- < Br^- < I^-$

EN ⑧ $CH_3 - \bar{C}H_2 > CH_2 = \bar{C}H > H - C \equiv \bar{C}$

(1) (2) (3)

Stability

⑨ $CH_3 - \bar{C}H_2 > CH_3 - CH_2 - \bar{O}$

(1) (2)

Stability 1 < 2

Basicity 1 > 2

⑩ $CH_3 - CH_2 - \bar{N}H_2$ (1)

+I

$\ddot{N}H_3$ (2)

$CH_3 - CH_2 - \bar{N}H_2 > \ddot{N}H_3$

Resonance

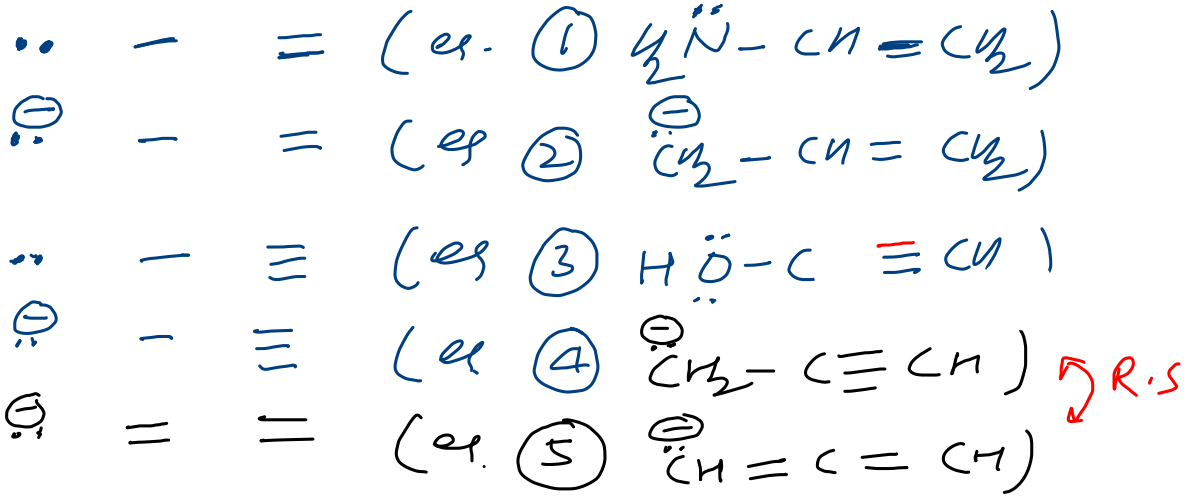
Permanent delocalisation of Lonepair Eles or π Electrons in conjugation is known as Resonance.

* Essential condition for resonance

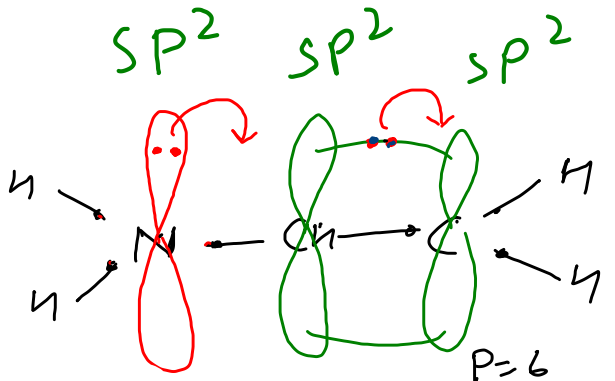
* conjugated system
* Planer

Conjugated System :

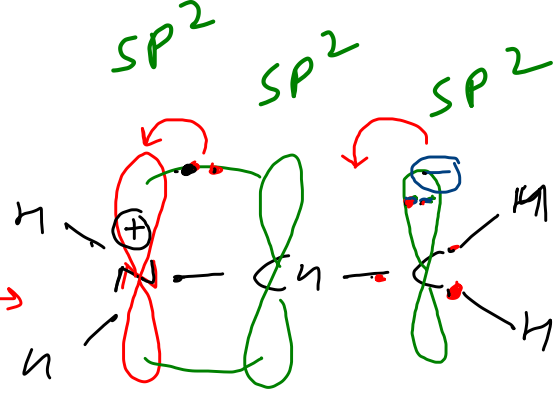
(1) Lone pair conjugated with π bond



eg (1)



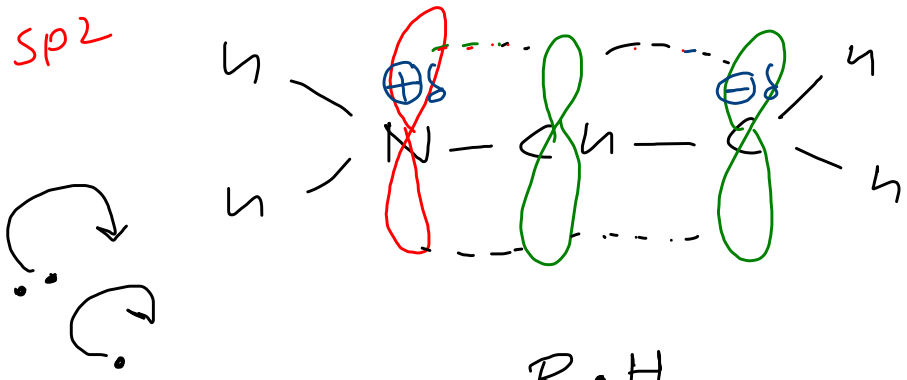
R.S.1



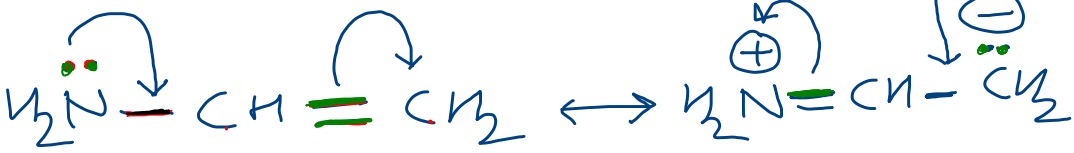
R.S2

δ + LLP

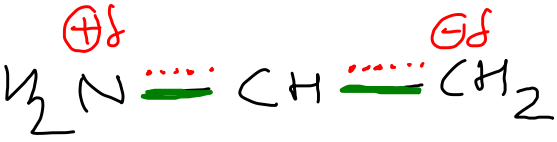
3 + 0 = 3 sp²



R·H

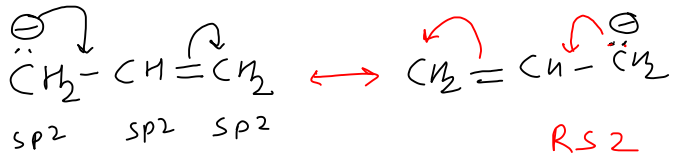


RS1 RS2



R·H

eq. (2)

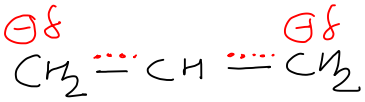


sp2 sp2 sp2

RS 1 RS 2

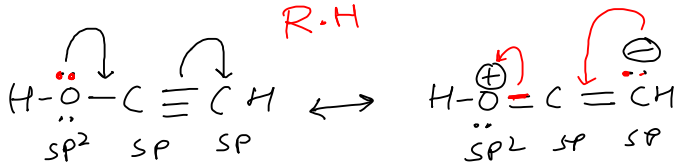
σ = 3 + 0

= 3 sp2



R.H

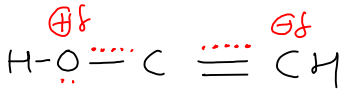
eq. (3)



sp2 sp sp sp2 sp sp

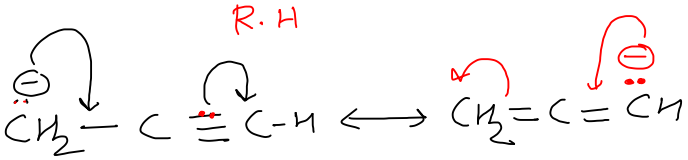
RS 1 RS 2

σ = 2 + 1 = 3



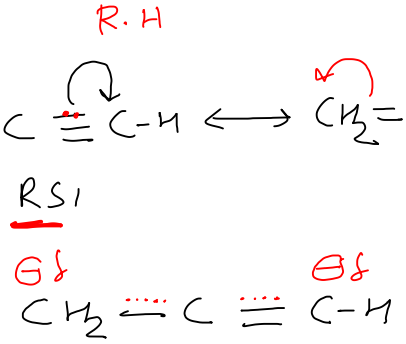
R.H

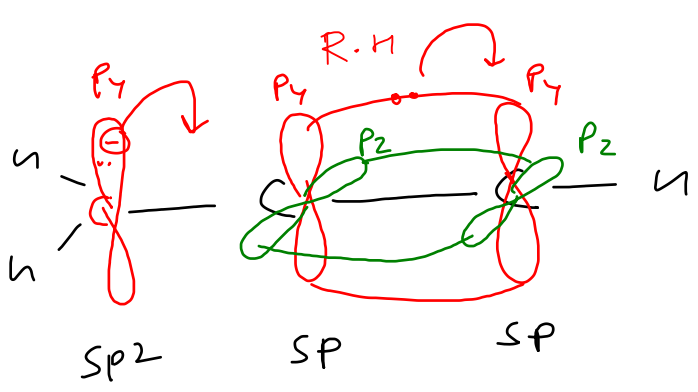
(4)



(5)

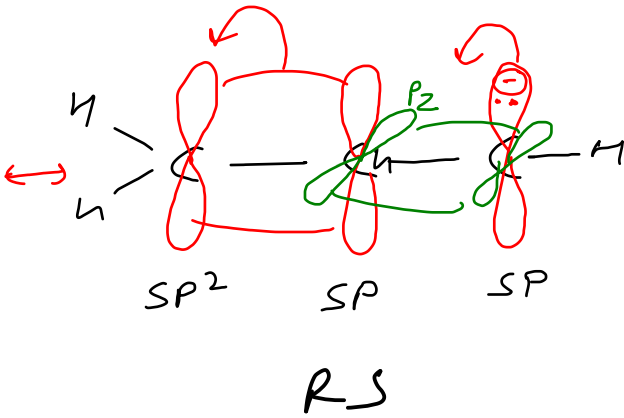
RS 1 RS 2





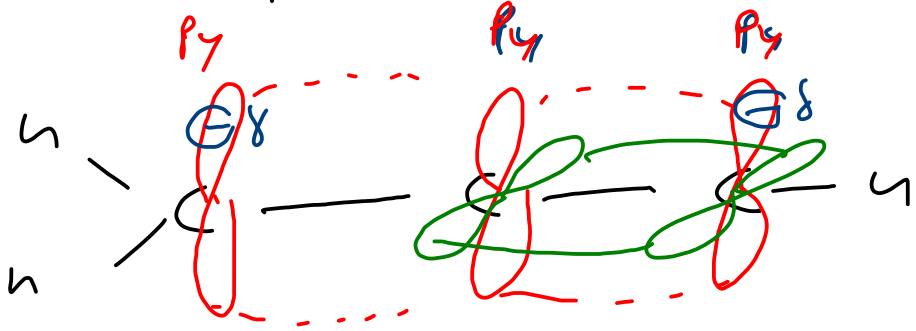
sp2 sp sp

RS 1



sp2 sp sp

RS



R.H

Important points related to Resonance

(1) It is permanent displacement or delocalisation of Lone pair or π electrons. Because external reagent is not required.

(2) Conjugated system must be present in a molecule having Resonance.

(3) The Entire molecule or the part of molecule which participate in resonance must be planer.

(4) The structures, which are generated due to resonance are known as Resonating structures (R.S) or canonical structures.

(5) The intermediate structure of all R.S is known as Resonance Hybrid (R.H)

(6) R.S are the imaginary structures, on the otherhand R.H is the real structure.

(7) All the properties of a molecule are explained by R.H (not by R.S)

(8) Resonance is not distance dependent effect

(9) Net charge remain constant in a molecule having Resonance, because there is NO exchange of electrons from external source.

(10) The contribution of most stable R.S will be more to its R.H.

OR

The structure or characteristics of R.H will be close to the most stable R.S.

(11) The Hybridisation state of a particular atom remain constant in all R.S, and also in it's R.H

(12) Only l.p electrons or π electrons undergoes delocalisation in Resonance, ATOMS never change their position.

(13) The amount of energy released due to Resonance is called Resonance Energy (R.E)

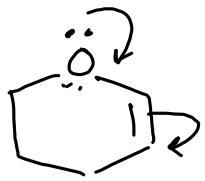
OR

The PE difference b/w R.H. and its most stable R.S is known as R.E.

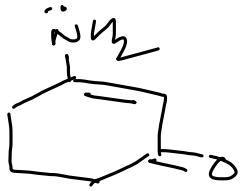
R.E = PE of R.H − PE of most stable R.S

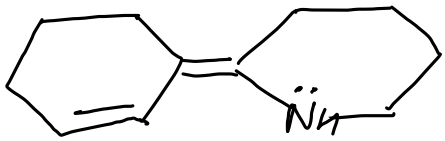
R.E = -ve means it release that energy.

prob Draw RS and RH

6 

7

8 

9 

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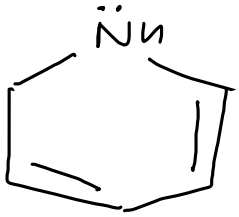
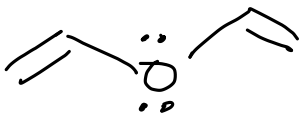
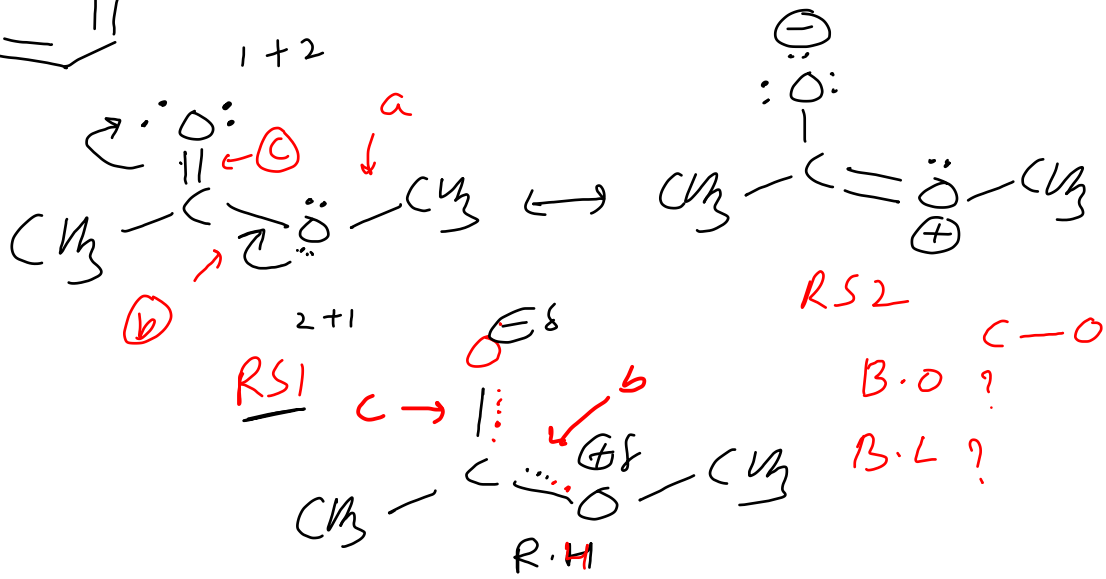
11 

12 

(13)

(14)

(15)



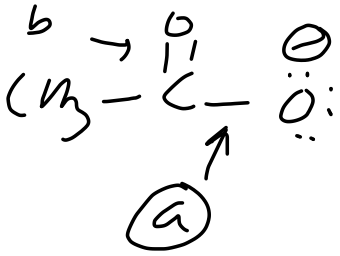
order of B.O b/w C and O

RS1 > RS2 (stability)

order of B.O c > b > a

order of B.L a > b > c

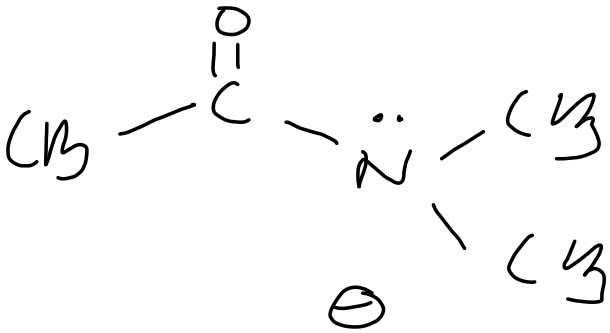
16



order of B.O ?

" " B.L ?

17



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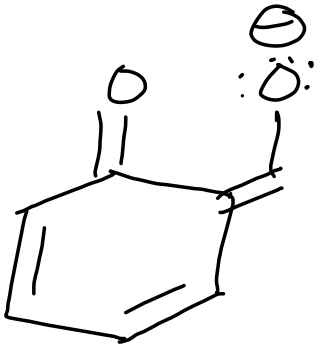

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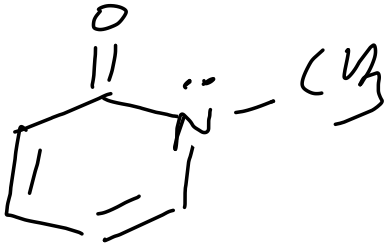

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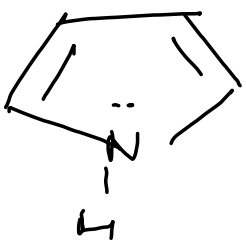


EXERCISE # O-I

1. Write correct order of acidic strength of following compounds :

(i) (a) $NO_2-CH_2-\overset{O}{\overset{\|}{C}}-O-H$ (b) $F-CH_2-\overset{O}{\overset{\|}{C}}-O-H$

(c) $Ph-CH_2-\overset{O}{\overset{\|}{C}}-O-H$ (d) $CH_3-CH_2-\overset{O}{\overset{\|}{C}}-O-H$

(ii) (a) $CH_3-CH_2-\underset{F}{\underset{|}{CH}}-\overset{O}{\overset{\|}{C}}-O-H$ (b) $CH_3-\underset{F}{\underset{|}{CH}}-CH_2-\overset{O}{\overset{\|}{C}}-O-H$

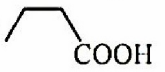
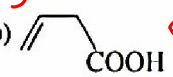
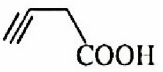
(c) $\underset{F}{\underset{|}{CH_2}}-CH_2-CH_2-\overset{O}{\overset{\|}{C}}-O-H$

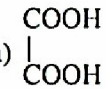
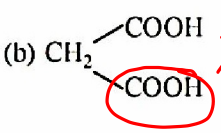
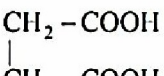
(iii) (a) $Cl-CH_2-\overset{O}{\overset{\|}{C}}-O-H$ (b) $Cl-\underset{Cl}{\underset{|}{CH}}-\overset{O}{\overset{\|}{C}}-O-H$

(c) $Cl-\underset{Cl}{\underset{|}{C}}-\overset{O}{\overset{\|}{C}}-O-H$

(iv)(a) CH_3-CH_2-O-H > (b) $CH_3-\underset{CH_3}{\underset{|}{CH}}-O-H$

(c) $CH_3-\underset{CH_3}{\underset{|}{C}}-O-H$

(v) (a)  < (b)  < (c) 

(vi) (a)  > (b)  > (c) 

(vii) (a) H–F < (b) H–Cl < (c) H–Br < (d) H–I

(viii) (a) CH_4 < (b) NH_3 < (c) H_2O < (d) H–F

(ix)(a) $F-CH_2-CH_2-O-H$ (b) $NO_2-CH_2-CH_2-O-H$

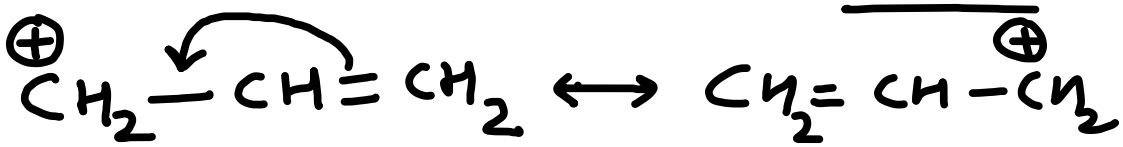
(c) $Br-CH_2-CH_2-O-H$ (d) $NH_3-CH_2-CH_2-O-H$

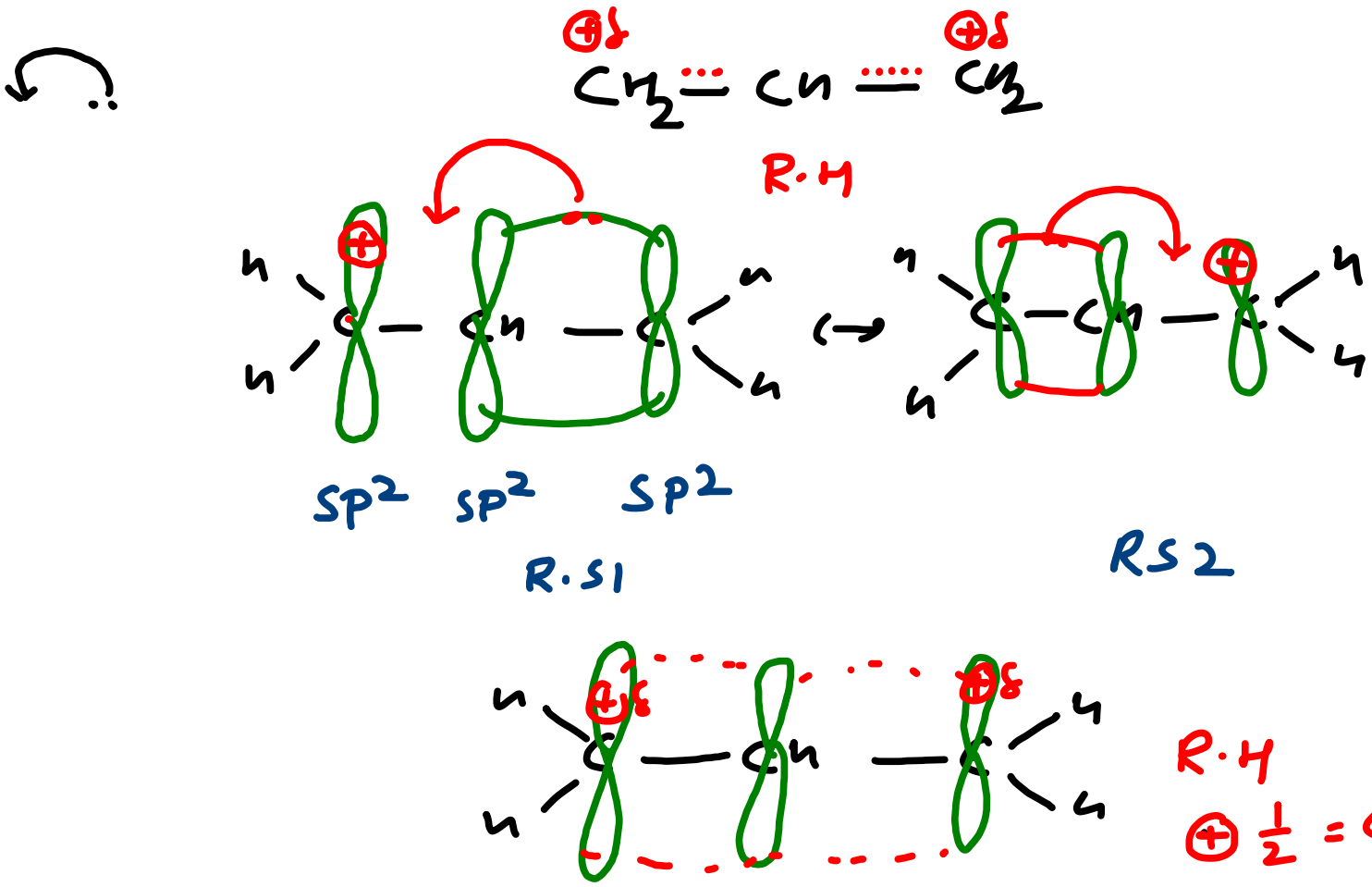
(x) (a) CH_3COOH (b) CH_3CH_2OH (c) C_6H_5OH (d) $C_6H_5SO_3H$

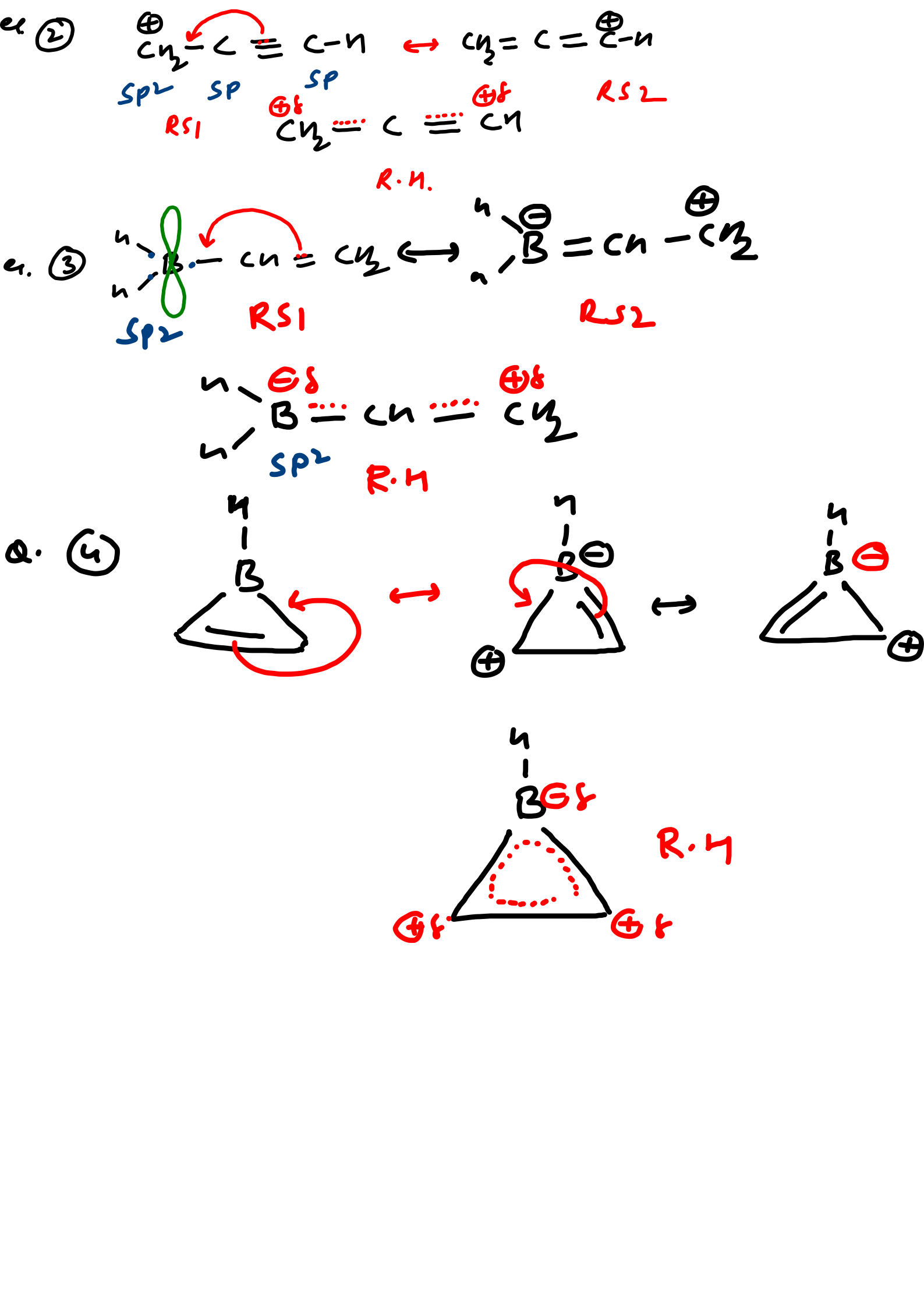
node06\B0B0A\Other Notes\JEE(Advanced)\Notes\Physics\Module#01\Eng\Acidic Strength & Basic Strength

② vacant p- orbital conjugated with π bond

eg. ①





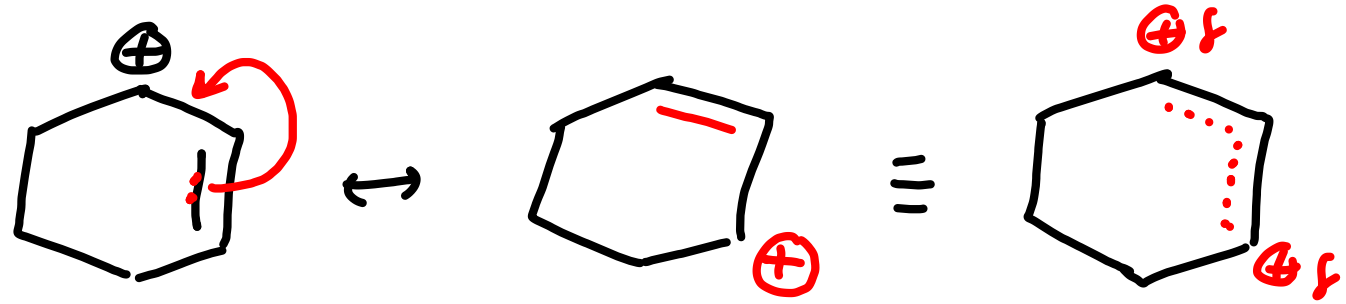

Q. ②
sp²
sp
sp
RS1
RS2
R.H.
Q. ③
Sp²
RS1
RS2
sp²
R.H
Q. ④
R.H

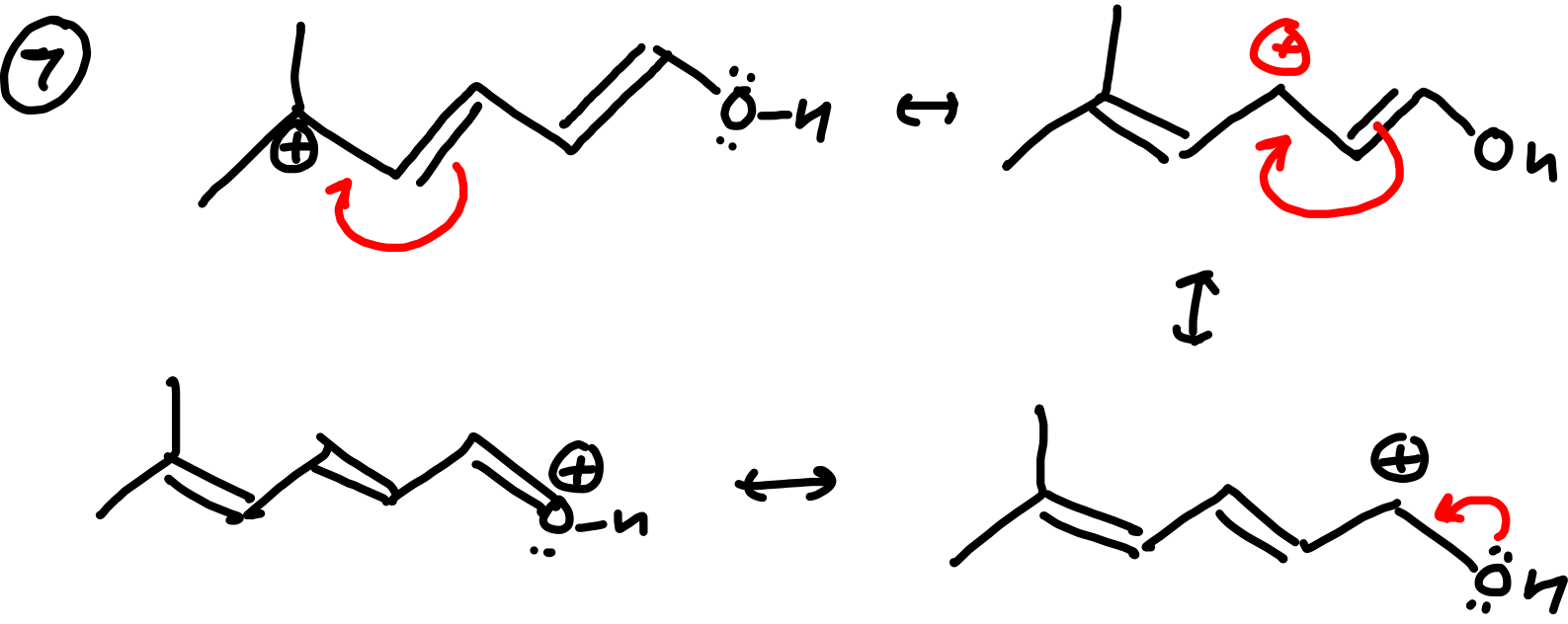
(5)

✗ NO Resonance

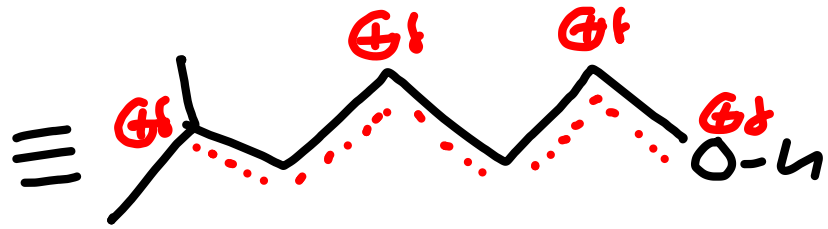
↑ NO vacant p

DRAW RS & RH

(6) 

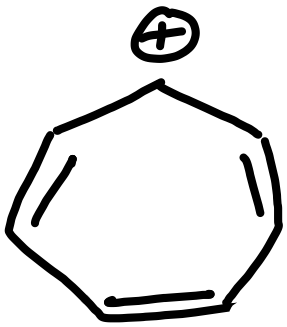
(7) 

I



R.H.

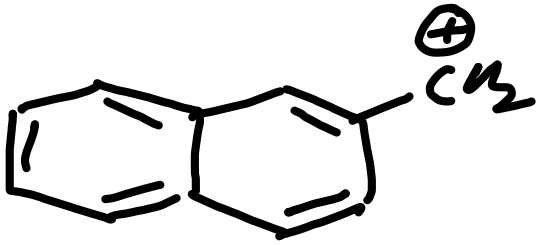
eq. (8) Draw RS & R.H



Tropy

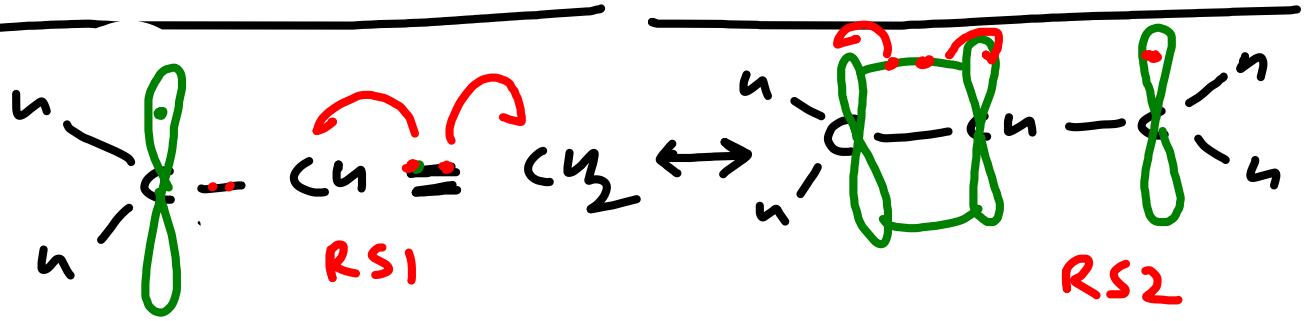
eq (9)

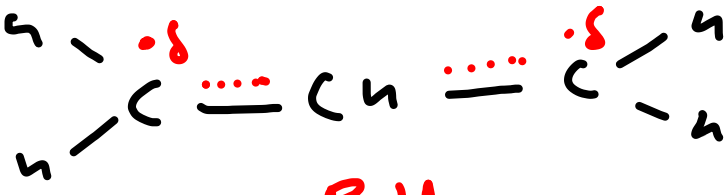
Q. (10)



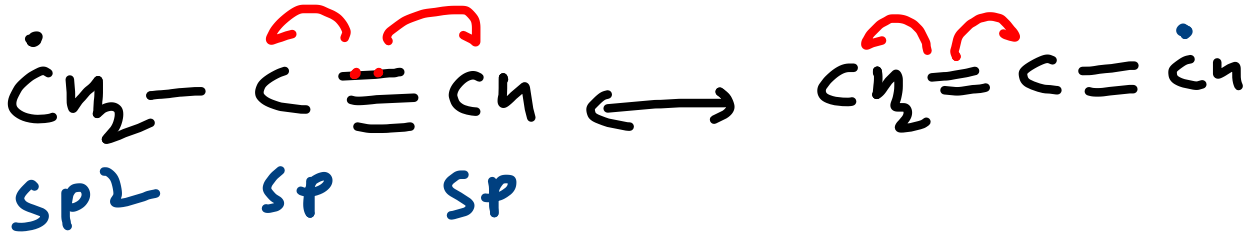
(3) Free Radical conjugated with π bond.

(11)





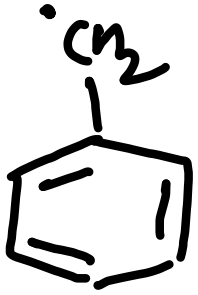
eg (12)



RS & RH ?

HW

eg (13)



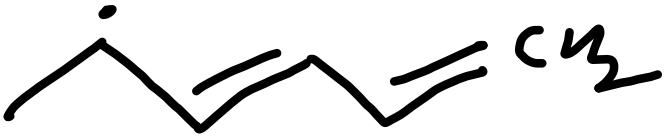
Benzyl FR

HW

(14) Draw RS and R.H

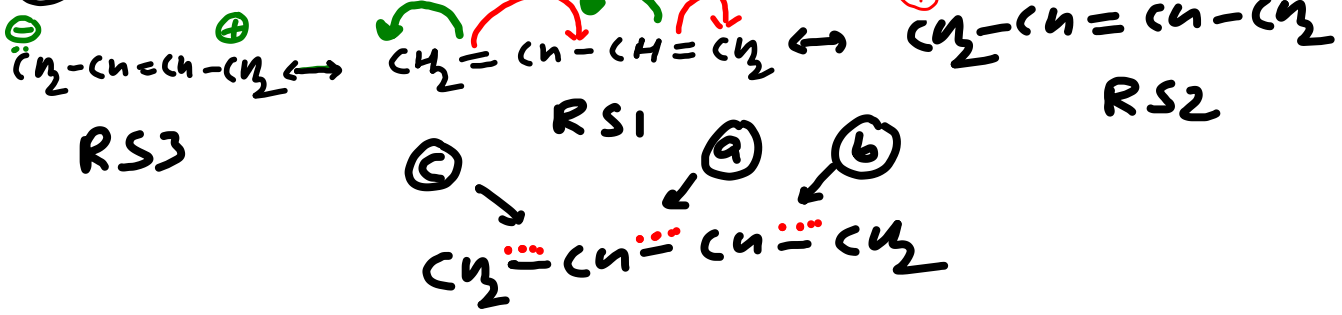
HW

(15)



④ π bond conjugated with π bond

(16)



RS3 RS1 RS2

R.H

stability

RS1 >> RS2 = RS3

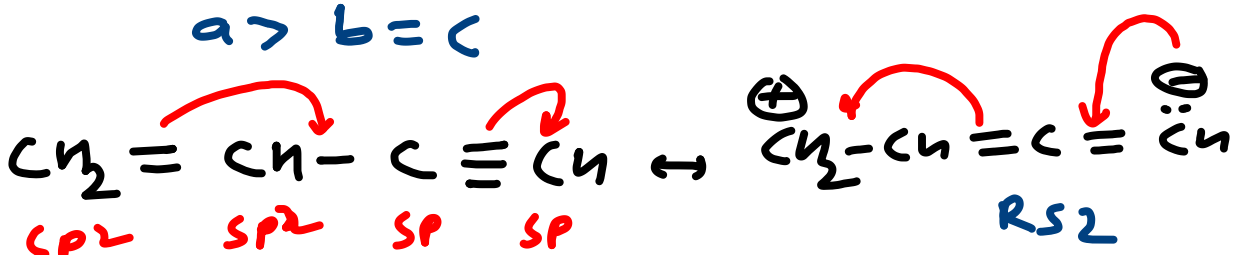
Order of B.O

b = c > a

Order of B.L

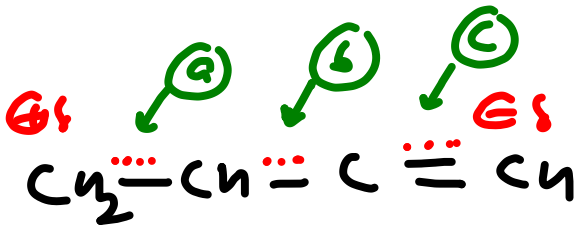
a > b = c

(17)



SP2 SP2 SP SP

RS1 RS2



R.H

stability

RS1 > RS2

Order of B.O

c > a > b.

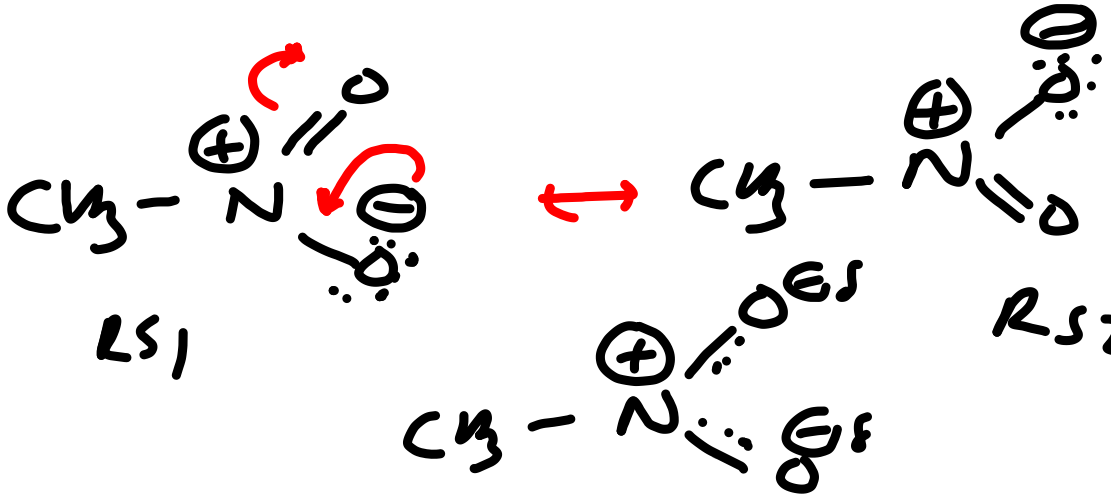
order of B.L

b > a > c

HW Qu (18) Draw RS and RH

$CH_2=CH-CH=O$

(19)

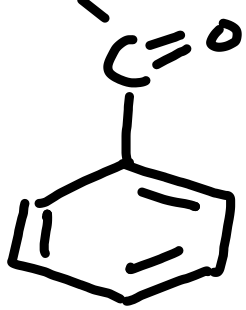


RS1 RS2

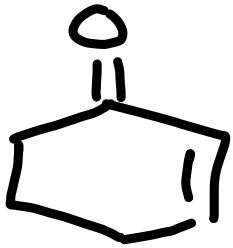
R.H

RS & RH?

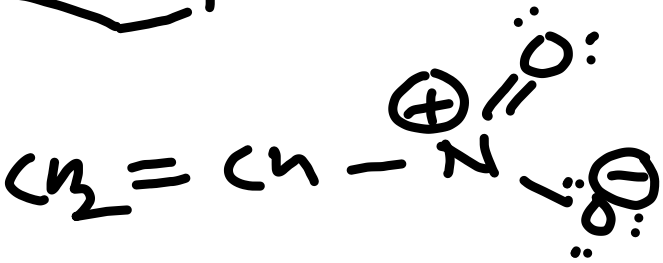
HW (20)



HW (21) Draw RS & RH



(22)



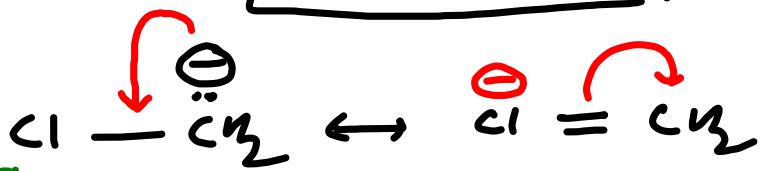
(5) d-orbital Resonance

(Vacant d-orbital conjugated with Lone pair

eg. (23)

Vacant d orbital

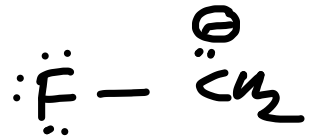
RS1 < RS2 stability.

eg. (24) 

Vacant d.

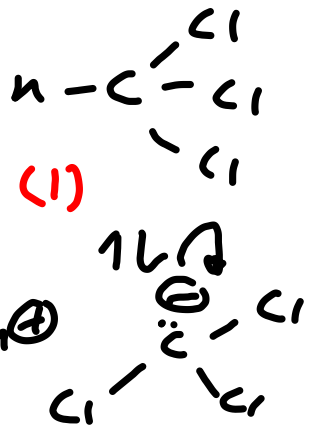
≡

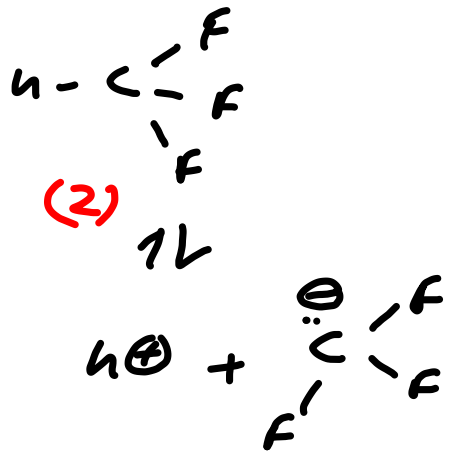
eg. (25) will it show d-orbital Resonance



No vacant No d-orbital Resonance.

(26) compare acidic Strength





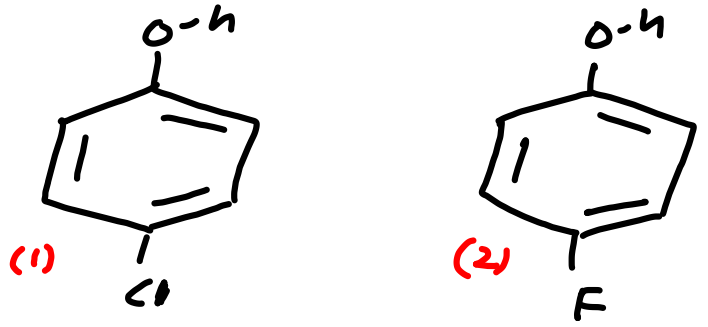
d-orbital ✓

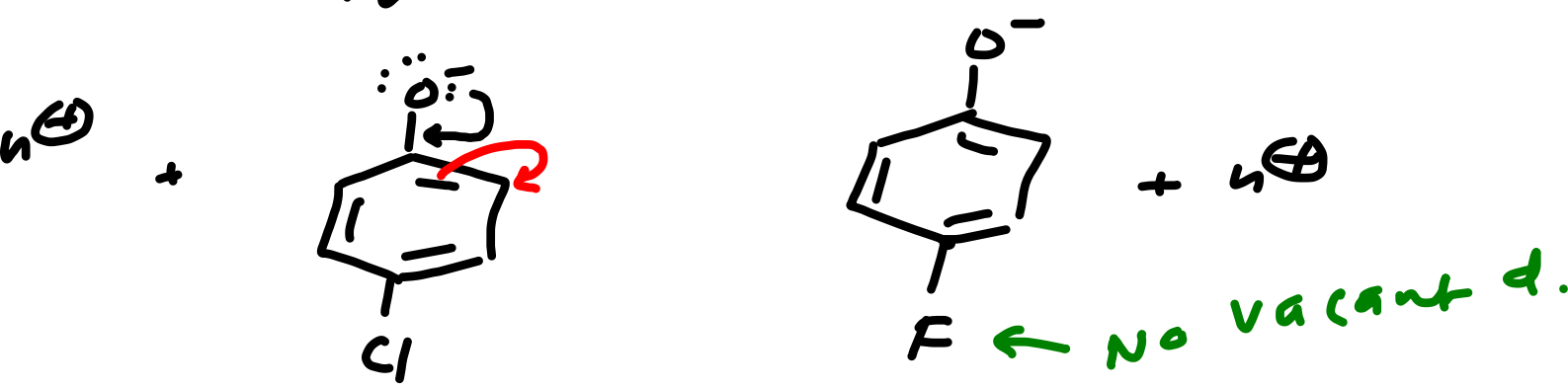
NO vacant d.

$CCl_3^- > CF_3^-$

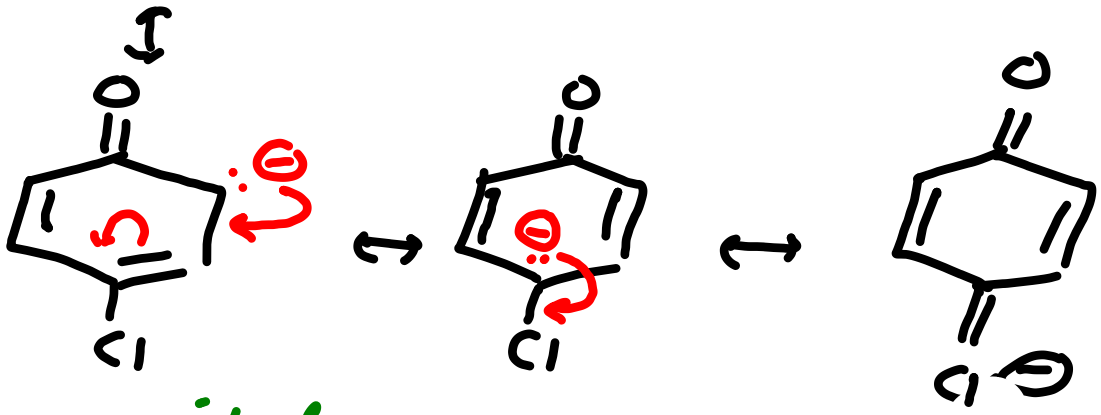
acidic Strength (1) > (2)

(27) Compare Acidic strength





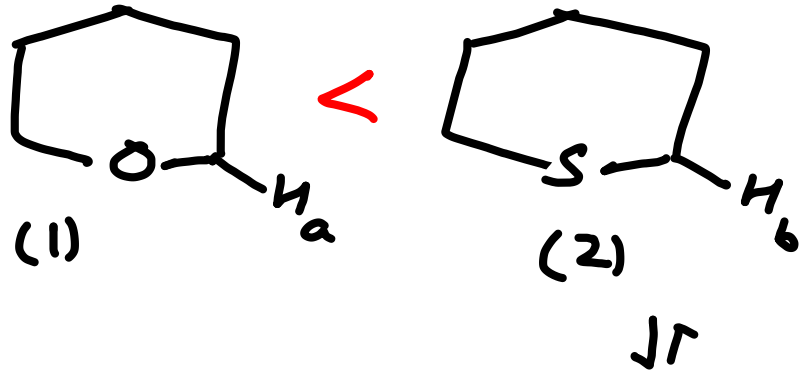
F ← No vacant d.



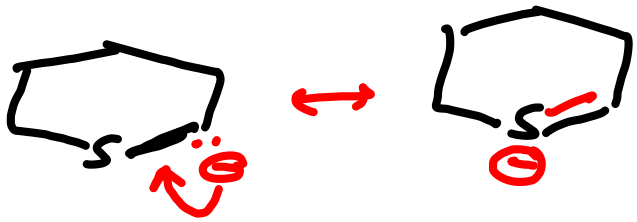
d-orbital Resonance

Acidic Strength (1) > (2)

(28) Compare Acidic strength

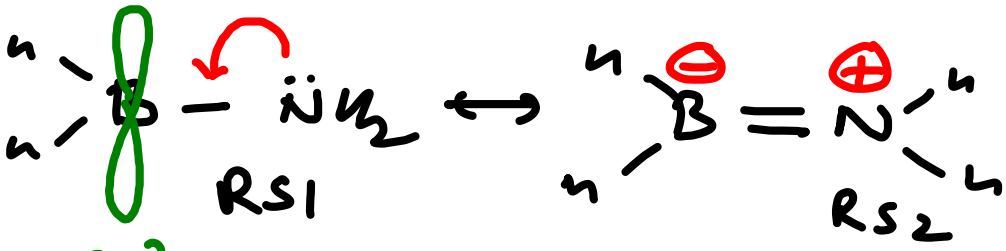


$\downarrow\pi$



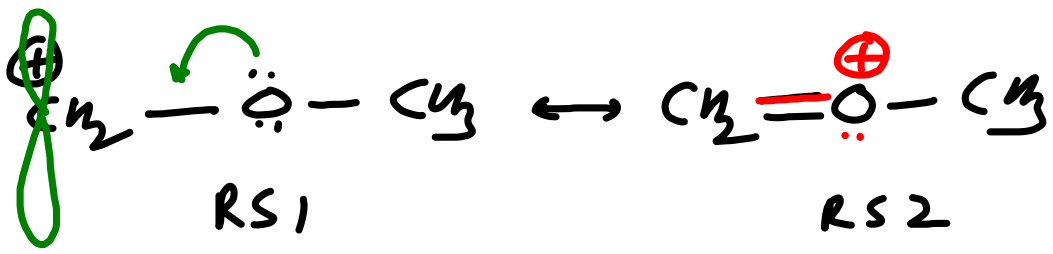
(6) Vacant p-orbital conjugated with Lone pair

(29)



RS1 < RS2

Complete octate for all atoms

(30) 

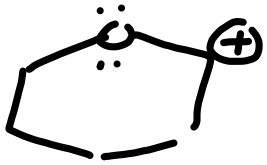
Stability

RS1 < RS2

Complete octate for all atom

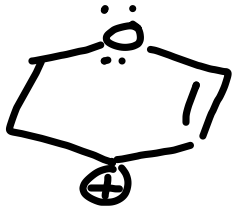
HW

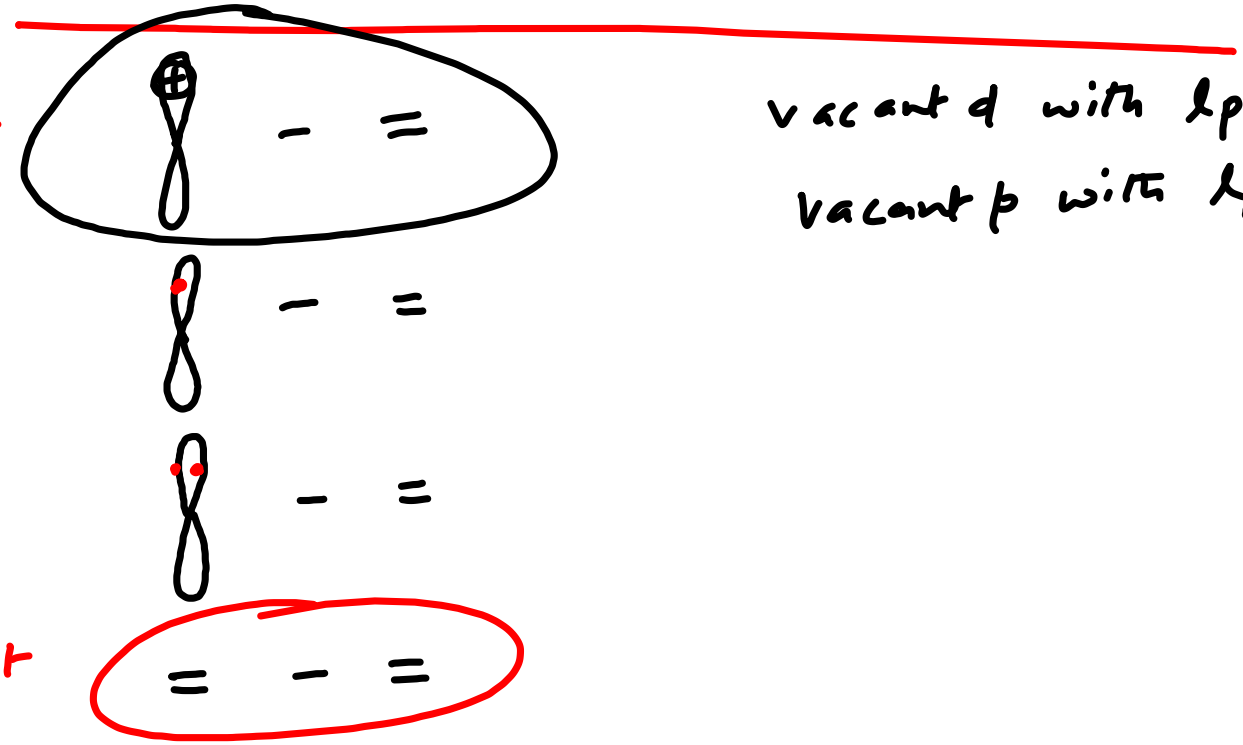
(31) Draw RS & RH



HW

(32)





vacant d with lp

vacant p with lp.

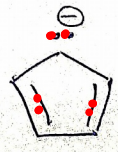
AROMATICITY

	AROMATIC COMP.	ANTIAROMATIC COMP.	Non Aromatic comps
H 1.	Compound must follow Huckel's rule i.e must have $(4n+2)\pi$ electrons $n = 0 \quad 1 \quad 2 \quad 3 \ldots$ $\pi\bar{e} = 2 \quad 6 \quad 10 \quad 14 \ldots$	⇒ Compound must have $4n\pi$ electrons $n = 1 \quad 2 \quad 3 \ldots$ $\pi\bar{e} = 4 \quad 8 \quad 12 \ldots$	Compound which is not Aromatic or also not Anti Aromatic is classified into Non Aromatic compound.
P 2.	Compound must be Planer	⇒ Compound must be planer	
C 3.	Compound must be cyclic	⇒ Compound must be cyclic	
CD 4. OR CC	Compound must have cyclic delocalisation of Huckel's π-electrons # Cyclic Delocalisation in R.H # Cyclic conjugation in R.S	⇒ Compound must have cyclic conjugation of $4n\pi$ electrons.	[Note: If any cyclic atom is sp^3, Then compound will be non Aromatic eg. Non Aromatic

Compound must follow Above 4 Rules for AROMATIC.

Compound must follow Above 4 Rules for Anti Aromatic

eg. (33)



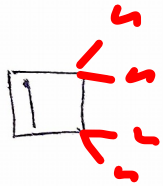
ARO

6 $\bar{e}$ H P C CD

(34)

4 π $\bar{e}$ P C CC

(35)



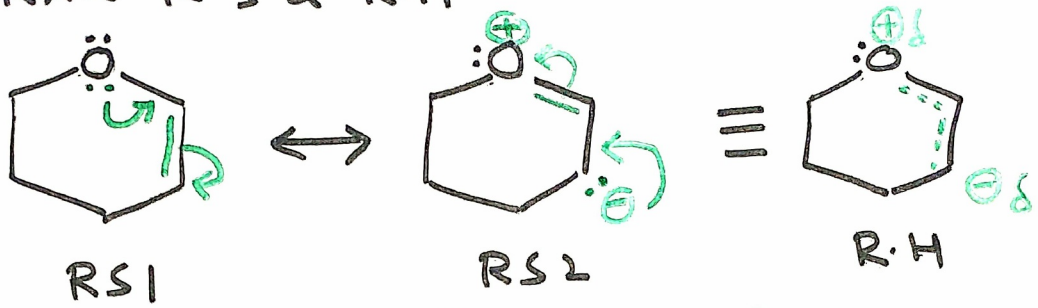
SP3

Non Aromatic

Solution of Homework

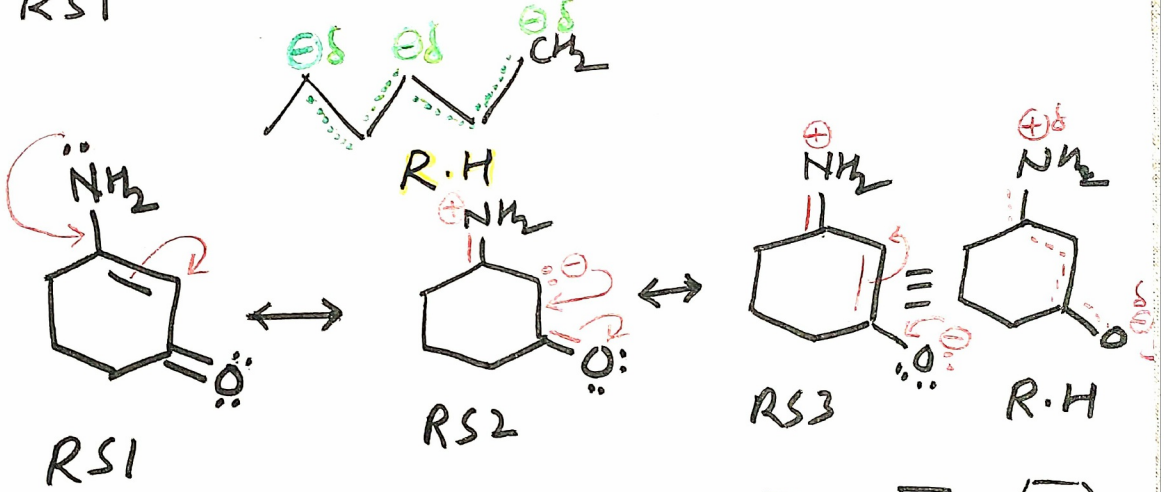
DRAW R.S & R.H

Q⑥

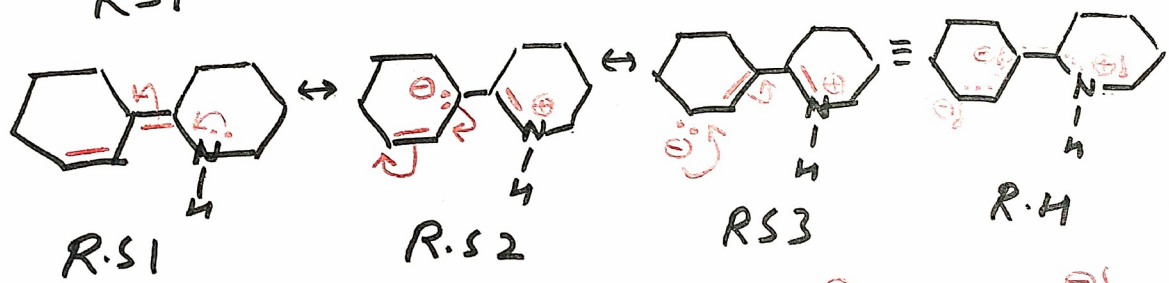


Q.7

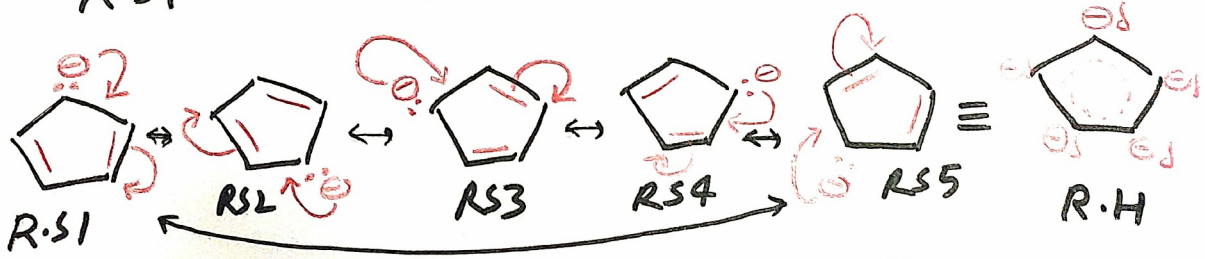
Q.8



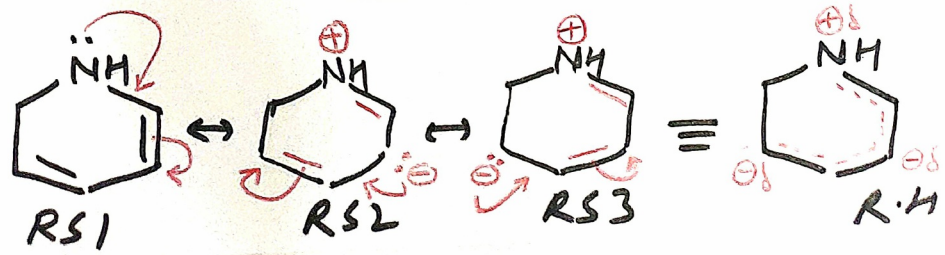
Q.9



Q⑩



Q⑪

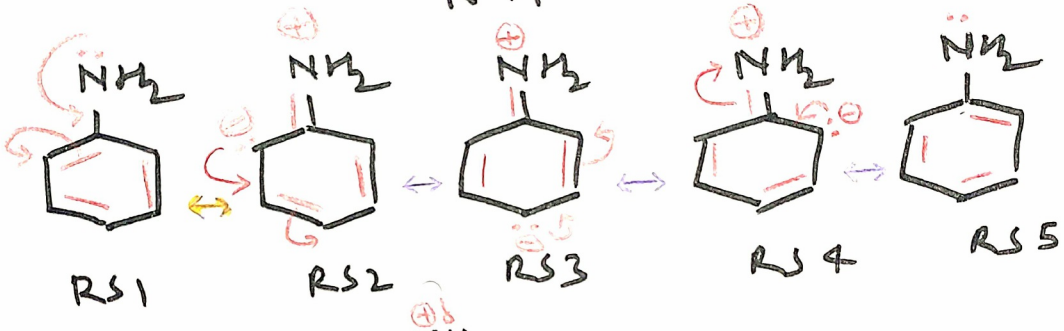


Q12

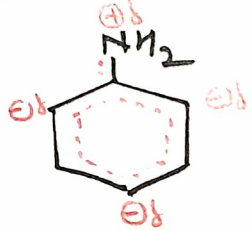
≡

R·H

Q13

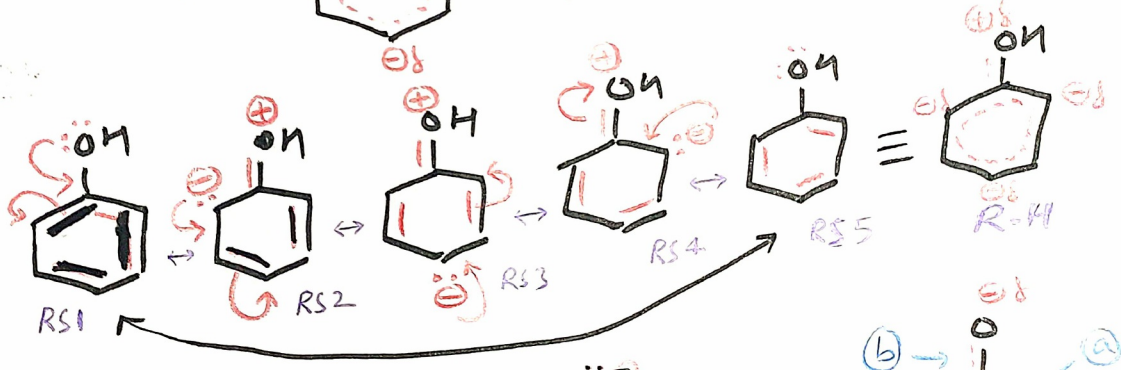


≡



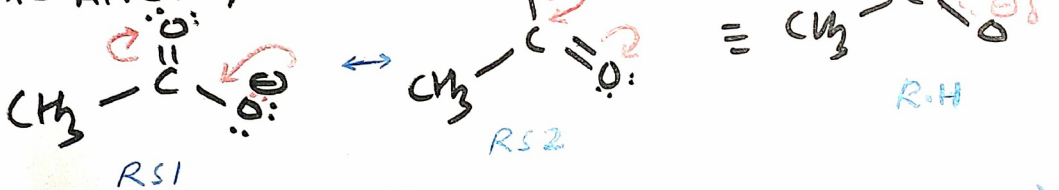
R·H

Q14



Q.15 Done Already

Q.16



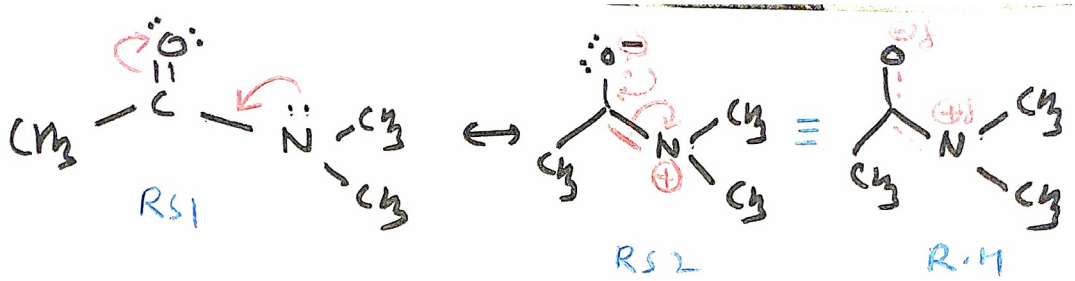
Stability RS1 = RS2 (They are equivalent type RS)

So their contribution is same to R·H.

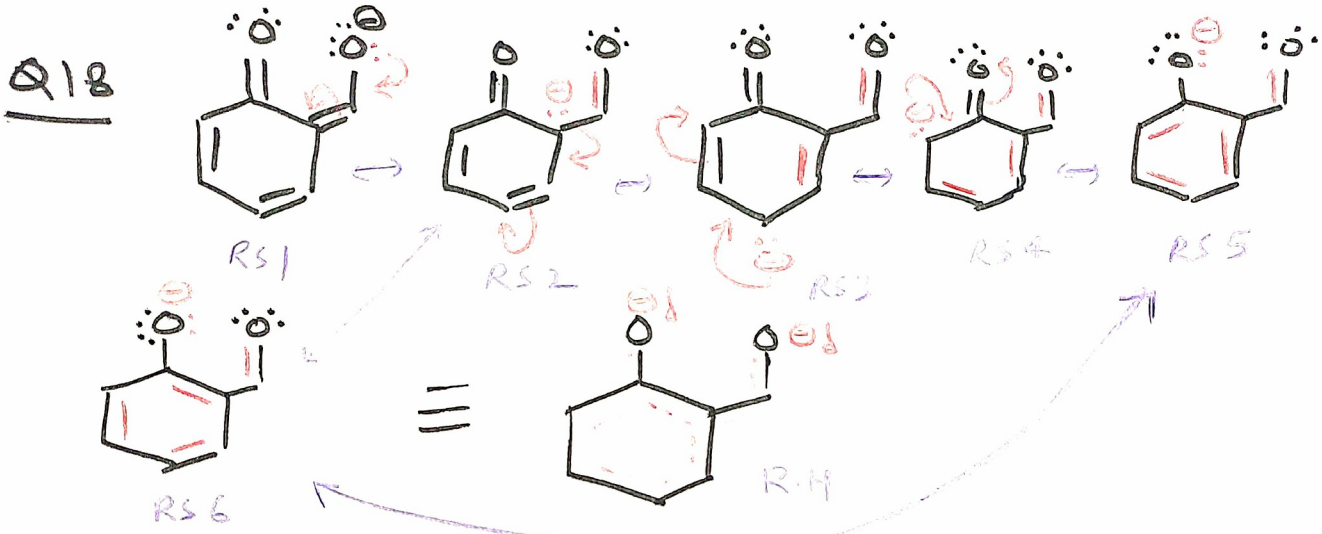
∴ order of B.O ⇒ a = b ✓

∴ order of B.L ⇒ a = b ✓

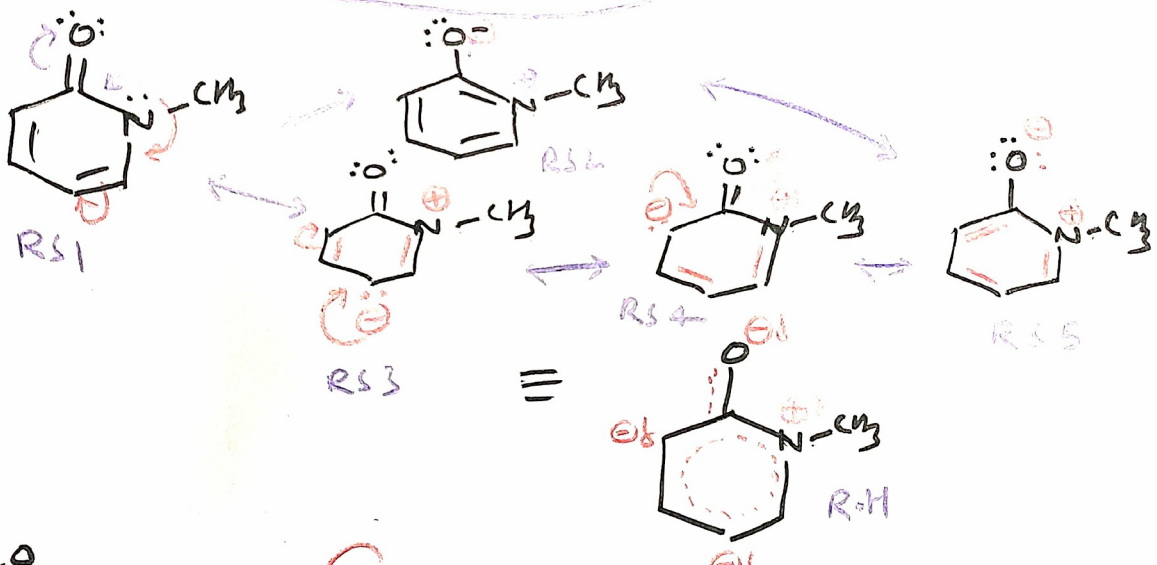
Q 17



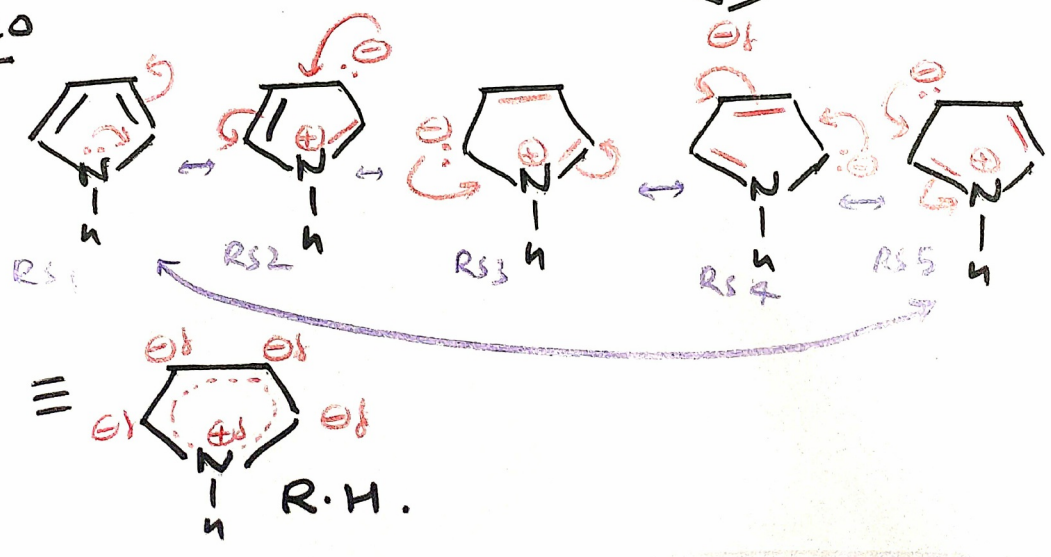
Q18



Q19

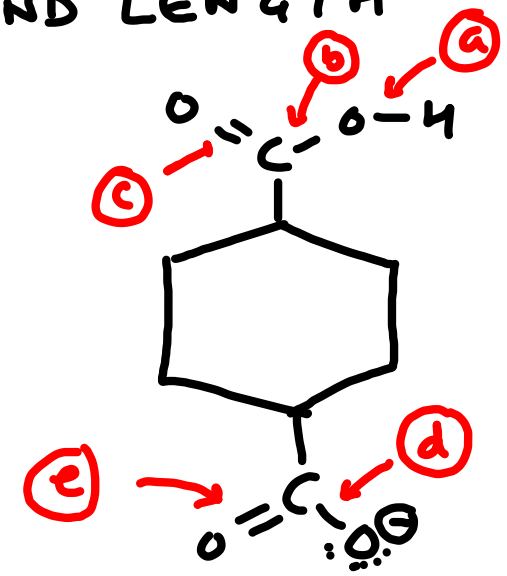


Que 20

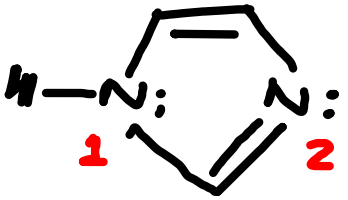


Quiz - 1

Que.1 FIND OUT THE CORRECT ORDER OF BOND LENGTH



Que.2

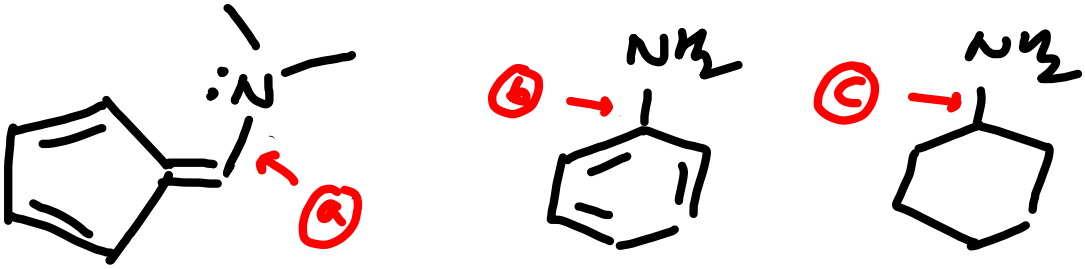


Which is more Basic

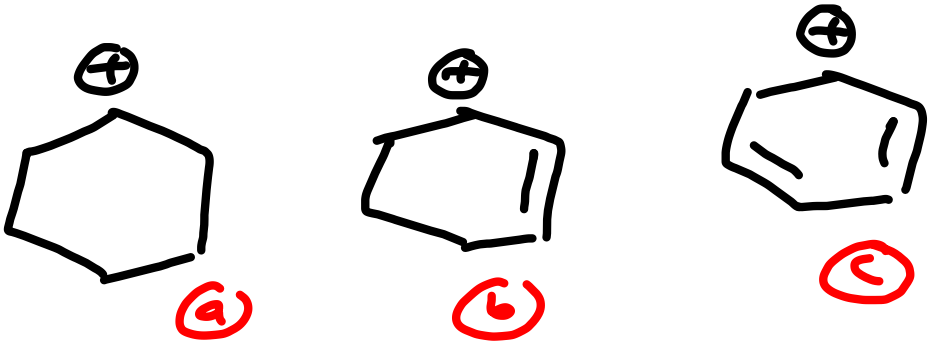
N1 or N2

and explain why?

Que.3 compare Bond energy



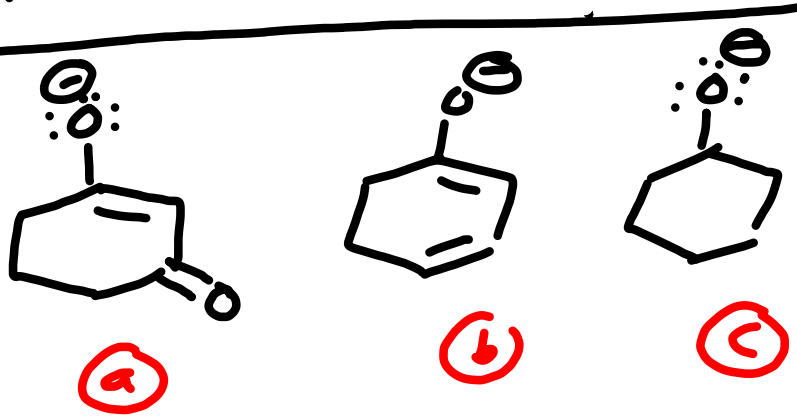
Que 4 compare stability of carbocation



Que. 5 Compare acidic strength

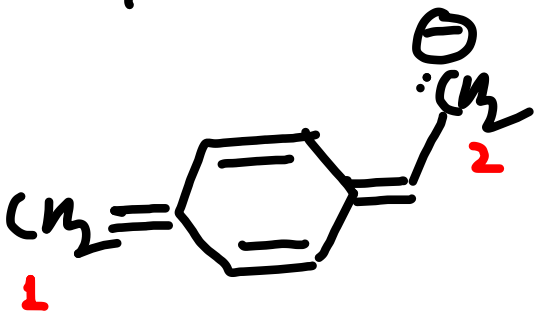
CH_3-CH_2-O-H (a) $CH_2=CH-O-H$ (b)

Que. 6 Compare stability of anion.

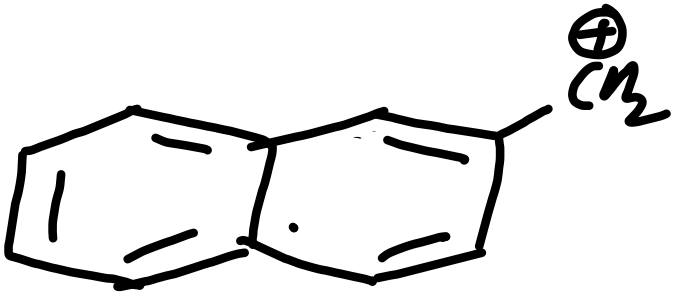


(a) (b) (c)

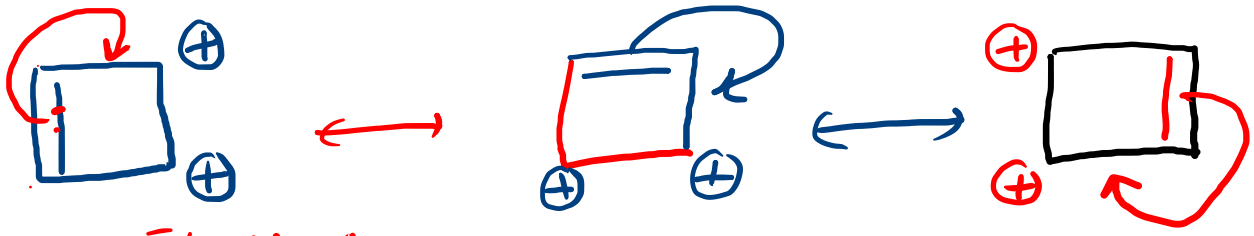
Que. 7 Compare Basic strength b/w C1 & C2



Que 8 How many R.S are possible with



36



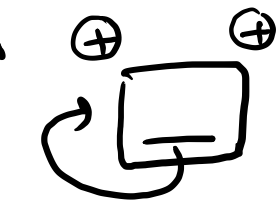
2π ēle H✓

All 4C are sp^2

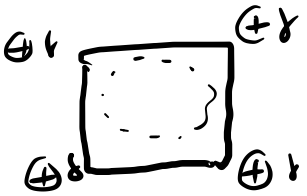
P✓

C✓

CD

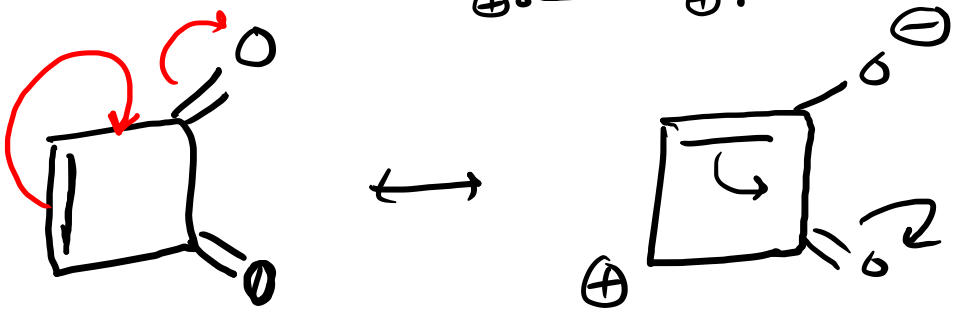


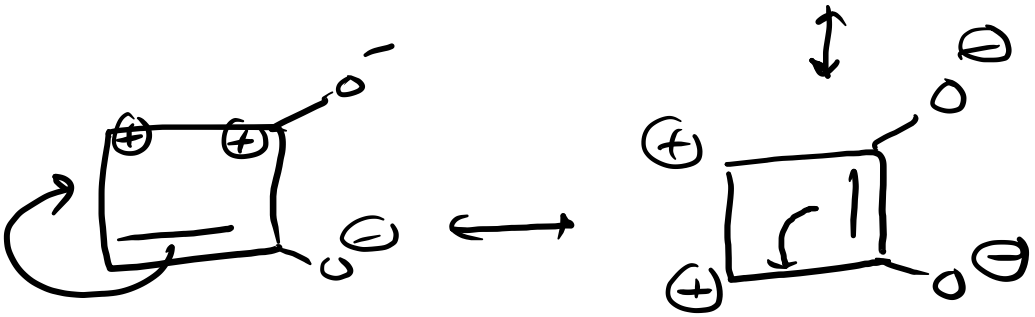
ARO



⊕δ = +½

37



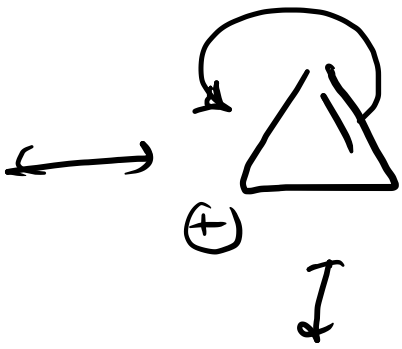


2πe H✓ P✓ C✓ CD✓

ARO.

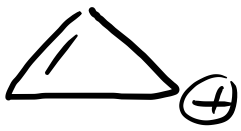
38

2π ē P✓ C✓ CD✓

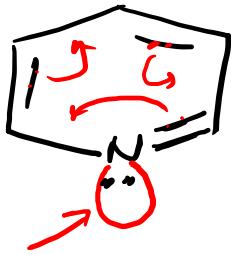


ARO.

R. U.



(39)



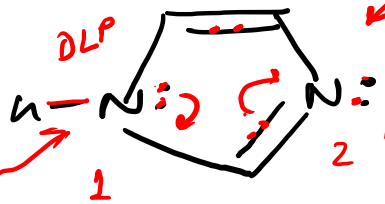
Localised LP.

$2+1 = SP^2$

6πe⁻ ✓
PL ✓
C ✓

ARO

(40)



DLP

SP^2

1

$3+0$

SP^2

2 LLP

6πe⁻
PL ✓
C ✓
CD ✓

ARO.

(41) ✓

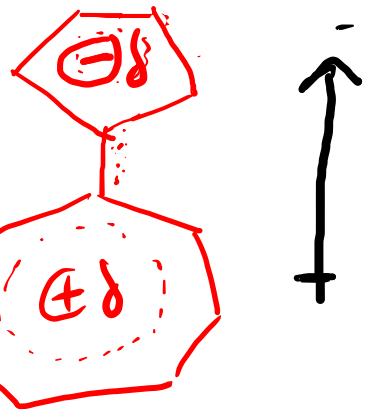
ARO

6πe⁻

6πe⁻

ARO

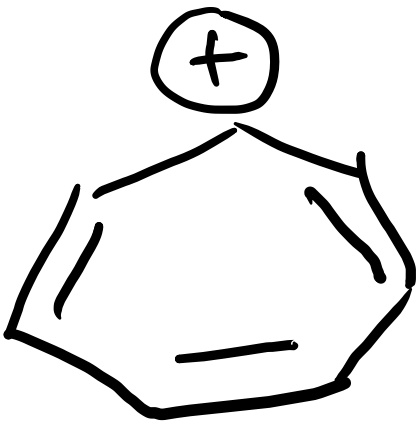
Direction of $\vec{\mu}$



5 mem

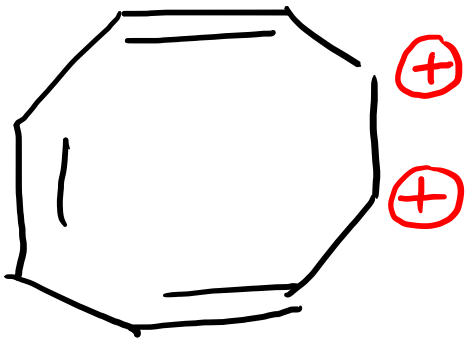
7 mem

(41)



6πe⁻ H ✓
PL ✓
ARO
C ✓
CD ✓

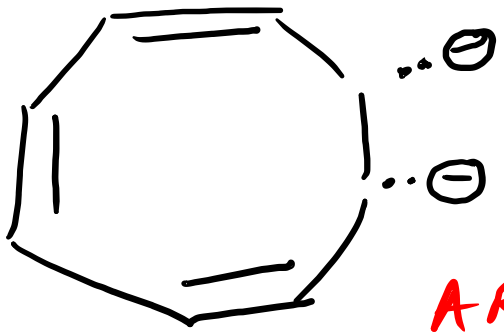
(43)



6 π e⁻ ✓
P ✓
C ✓
CD ✓

ARO

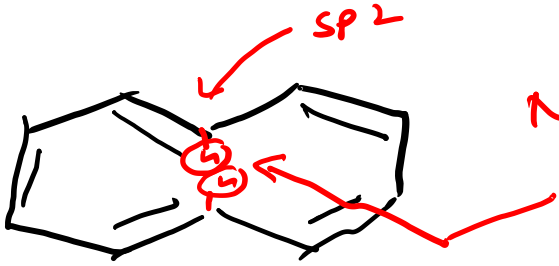
(44)



10 π ele ✓ ✓
P ✓
C ✓
CD ✓

ARO.

(45)

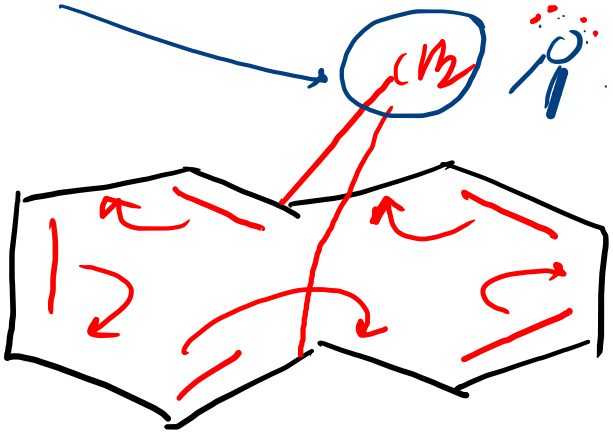


10 π e⁻ ele.
Non planer
due to repulsion
b/w Hydrogens
∴ NON Aromatic

Out of plane CDX

Annulene-10

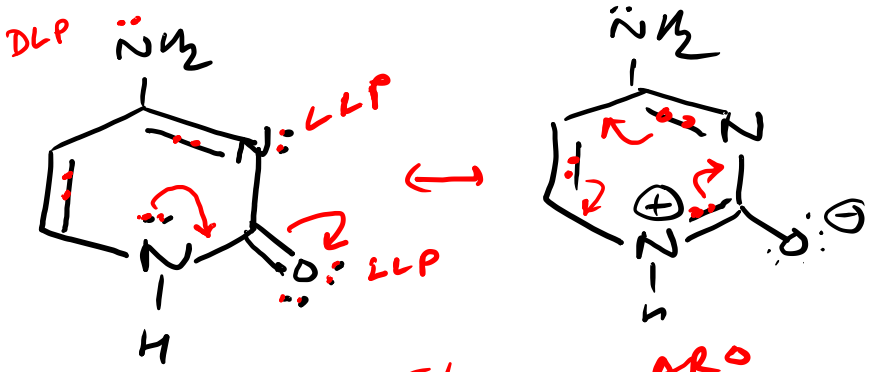
(46)
(47)



10 π e⁻ ele ✓ ✓
P ✓
C ✓
CD ✓

Aromatic.

48



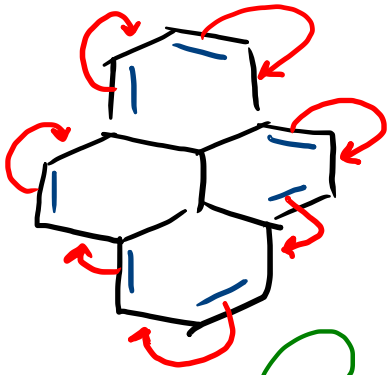
AROMATIC

RS1 — 6 π ele — HL ✓ P ✓ C ✓ CD ✓ — ARO — RS2

LLP = 3
DLP = 2

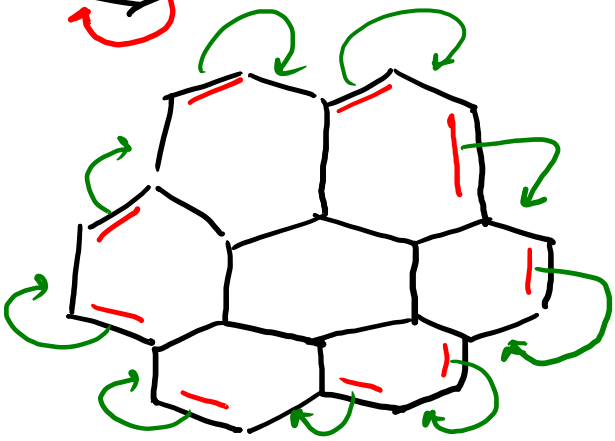
Periferal Resonance (or circumference reso)

49



14 π ele — HL ✓ P ✓ C ✓ CD ✓ — ARO

50

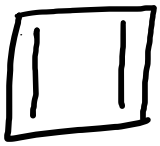


18 π ele — HL ✓ P ✓ C ✓ CD ✓

Annulene :-

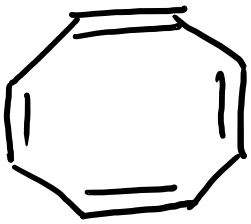
Mono cyclic hydrocarbons having alternate double bonds are called Annulene.

eg (S1)



Annulene-4

(S2)



Annulene.-8

e.g. (S3) Annulene.-10

Note: ①

Monocyclic compounds having 7 or more than 7 membered ring will not exist in Anti Aromatic, because the structure converted into Non-planer structure.
(∴ Non Aromatic)

Stability order

Aromatic compounds > Non Aromatic compounds > Anti Aromatic compounds

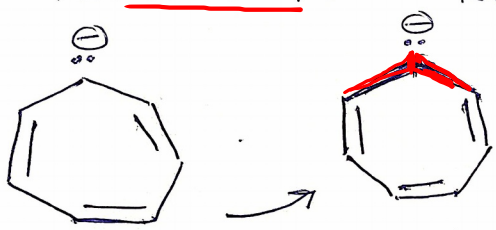
Anti Aromatic compounds are unstable at room temp.

Annulene-8

(S3)

It exist into tub shape → ∴ Nonplaner Non Aromatic

(S4)

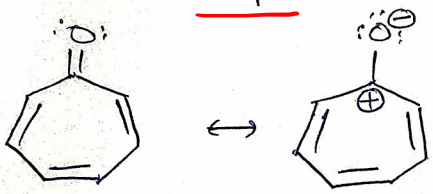


envelope like structure ∴ Non planer Non Aromatic

Note ② If any R.S of given compound is Aromatic then compound will be Aromatic.

CC X

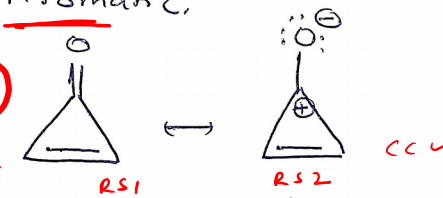
(S5)



Non Aromatic RS1 ↔ Aromatic RS2

compound is AROMATIC

(S6)



CCX Non Aromatic RS1 ↔ Aromatic RS2 CC✓

compound is AROMATIC.

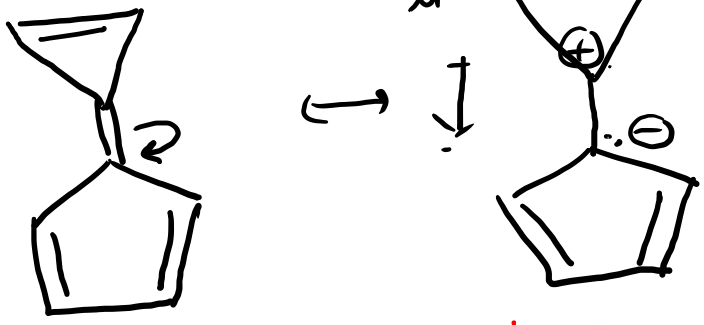
polar Quasi Aro

Quasi Aromatic :-

The compound which becomes polar to attain aromaticity or the compound in which charge also undergoes cyclic delocalisation along with Huckel's π-eles is called Quasi Aromatic compound.

eg (55) & (56)

(57)



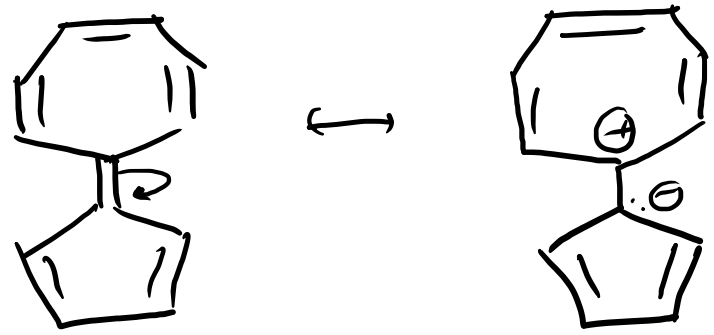
$\vec{\mu}$

$2\pi\ \bar{e}$ H–
R–
C–
COR

6π eles H–
R–
C–
COR

Quasi ARO.

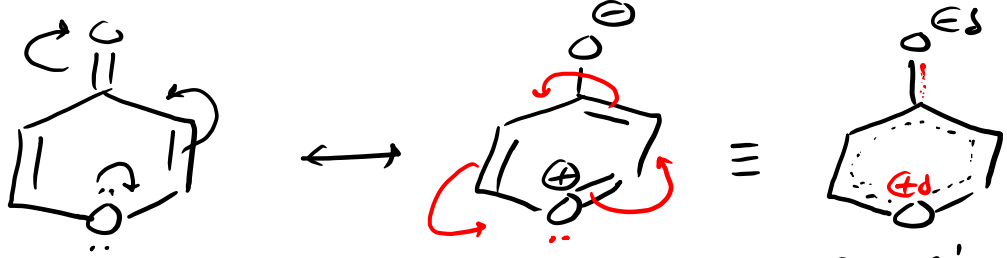
(58)



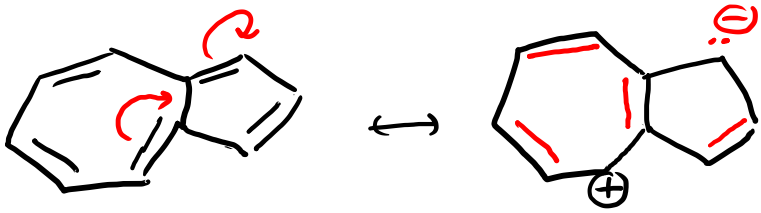
$6\pi e$

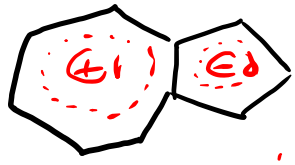
$6\pi e^-$

59



60



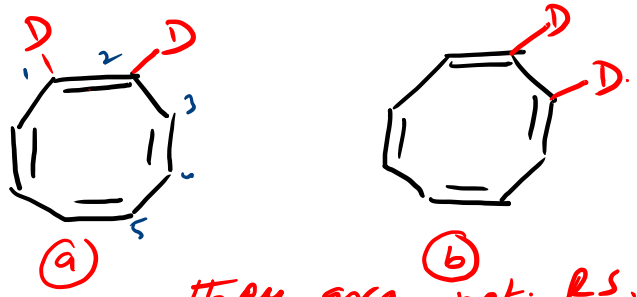


Note

Quasi Aromatic compounds are polar,
soluble in polar solvent
They are highly stable..

Prob. Identify the compounds i.e same or different.

61

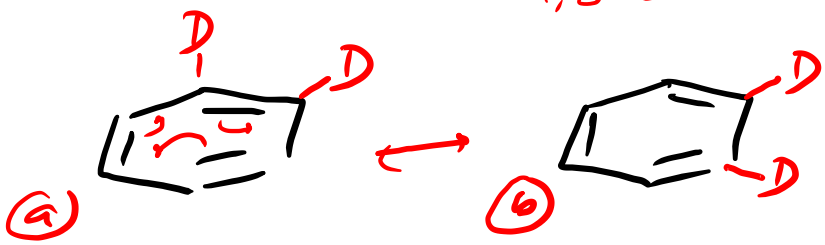


1,2-Di Deutero cyclo octa-1,3,5,7-tetra ene

they are not RS.
∴ they are different compound

1,8-Di deutero " " " "

62

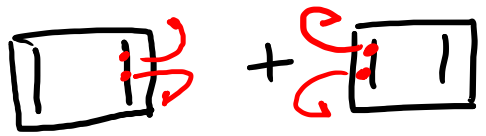


same

Dimerisation :-

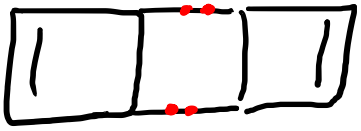
Formation of Dimer is known as Dimerisation.

(63)



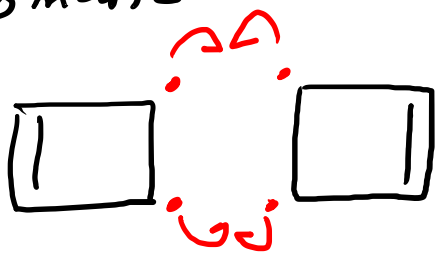
Dimerisation

Dimer

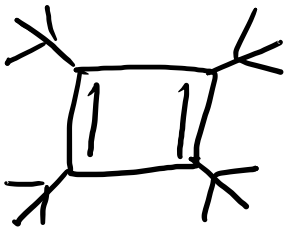


Non Aromatic

Anti Aromatic



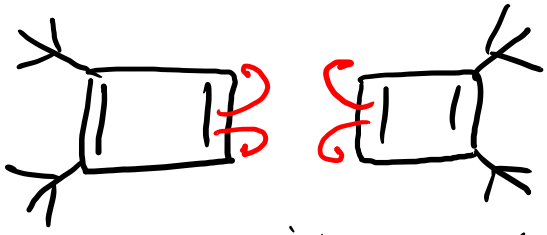
(64)

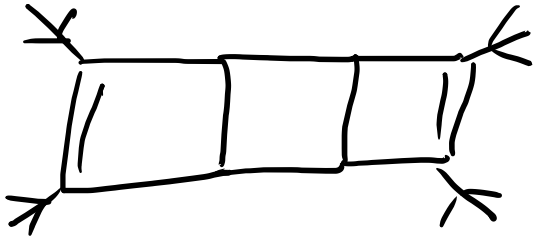


will it form dimer?

Ans NO (steric hindrance)

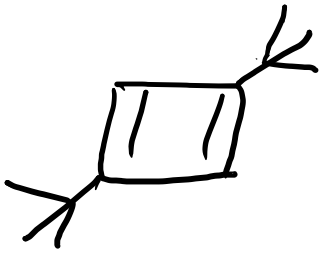
(65)





yes it can form Dimer.

(66)



→ X NO Dimer form.

Relative stability of R.S

QA:

C = complete octate for all atoms (67)

A = Aromaticity (68)

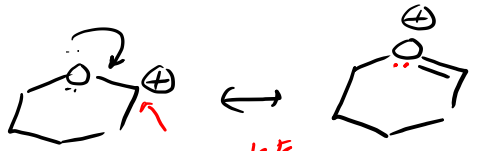
C = covalent bonds (69) (Neutral RS > charged RS) (70) (71) (72)

EN = Electronegativity $\overset{\ominus}{C} < \overset{\ominus}{O}$ $\equiv \overset{\oplus}{N}- > =\overset{\oplus}{O}-$ (73) (74)

EC = Extended conjugation > cross conjugation

DC = Distance of charge. (75) (76)

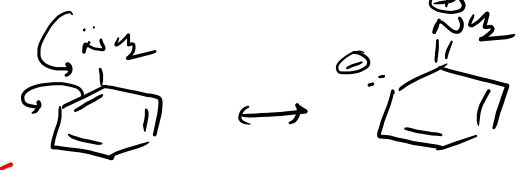
(67)



RS1 Incomplete octate ↔ RS2

RS1 < RS2

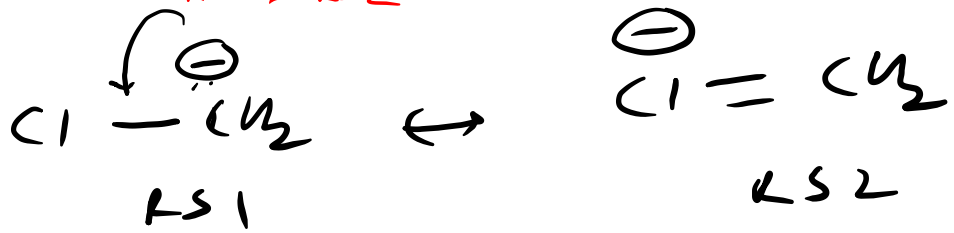
(68)



CC ✓ RS1 ARO ↔ RS2 Non Aro. CC ✗

RS1 > RS2

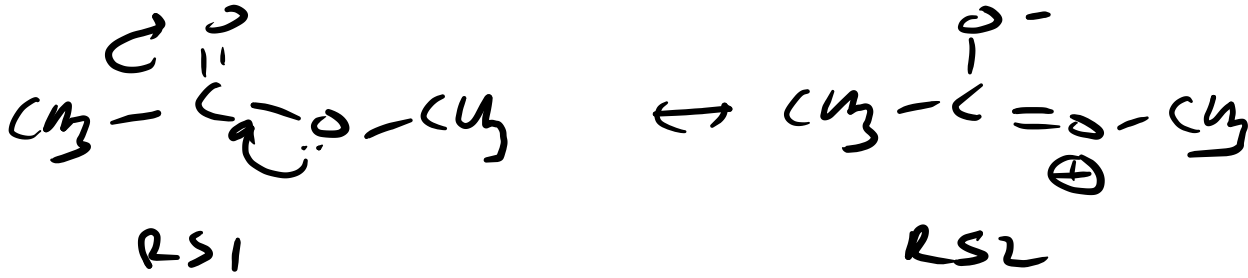
(69)



RS1 ↔ RS2

RS1 < RS2

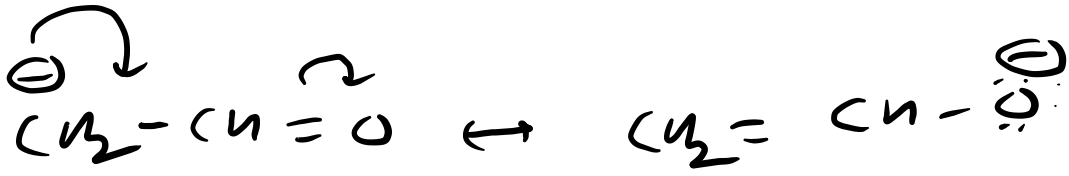
e. (70)



RS1 RS2

RS1 > RS2

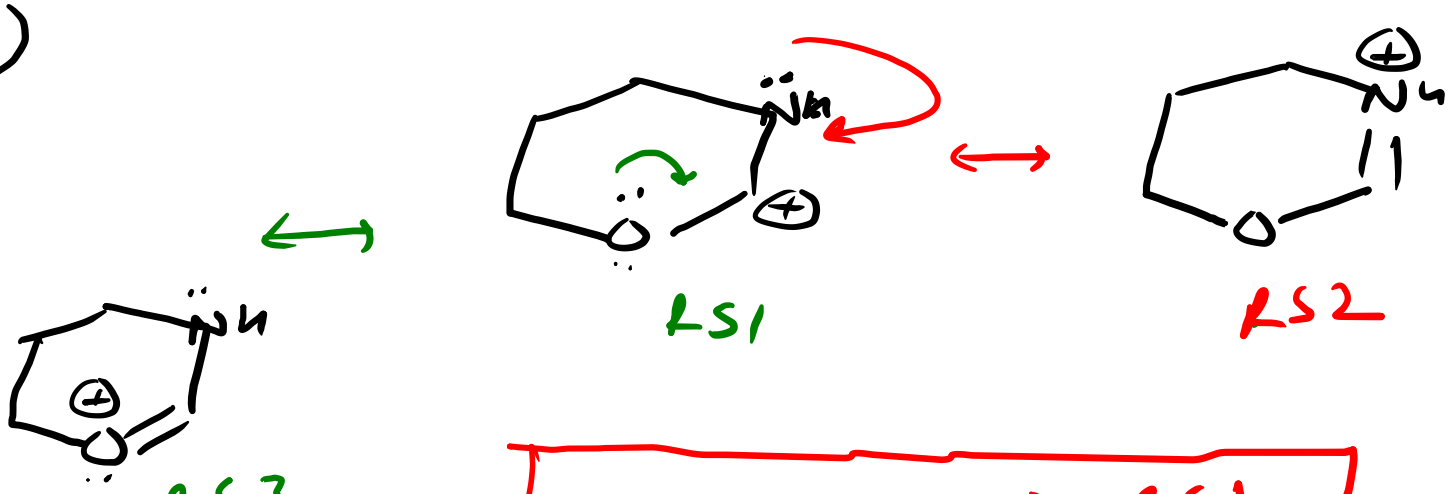
(71)



RS1 RS2

RS1 < RS2

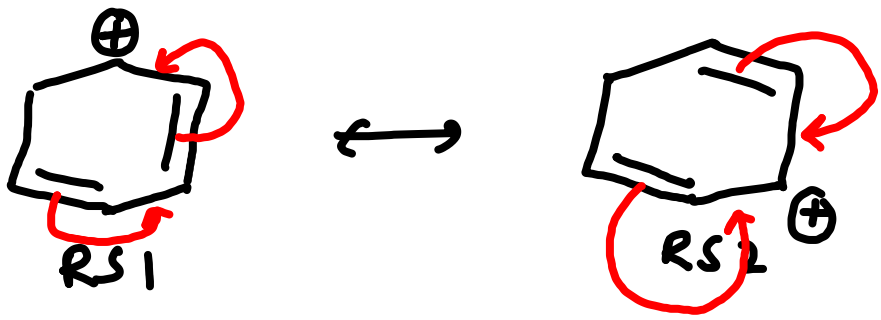
(72)



RS1 RS2 RS3

RS2 > RS3 > RS1

73

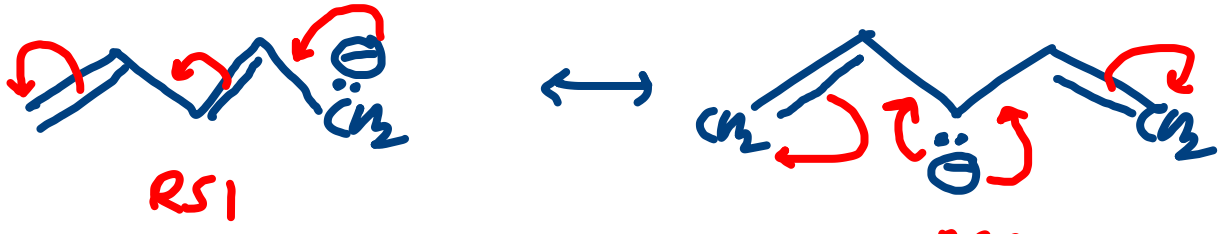


Extended conjugation

cross conjugation

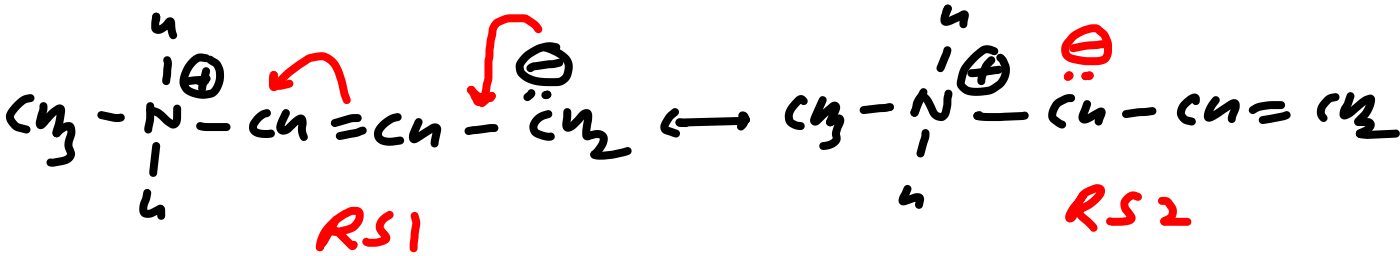
RS1 > RS2

74



Extended conjugation > Cross conjugation

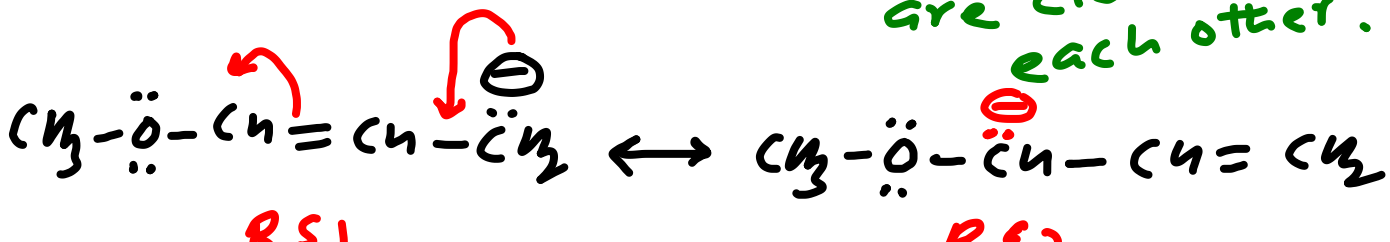
75



RS1 < RS2

opposite charges are close to each other.

76



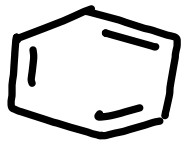
RS1 > RS2

Lone pair -ve repulsion.

Mesomeric Effect:

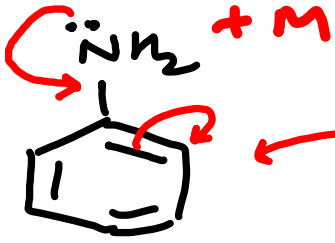
"Participation of mesomeric group in Resonance is called Mesomeric effect.

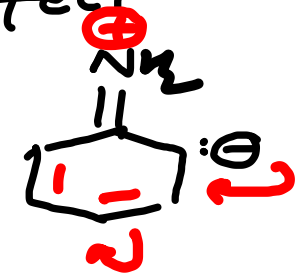
eg. (1)

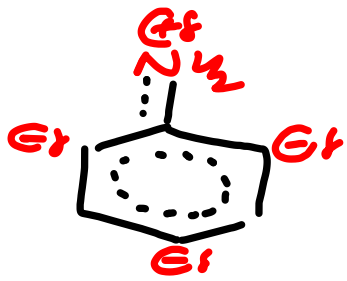


R ✓
M. Group X
M. Effect X

(2)







Resonance ✓
M. Group ✓
M. Effect ✓

es (3)

Mesomeric group

Reso ✓

Mesomeric groups are of two types

(A) +M or +R Group. (EDG or ERG)

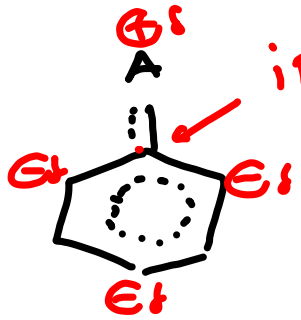
(B) −M or −R Group (EWG)

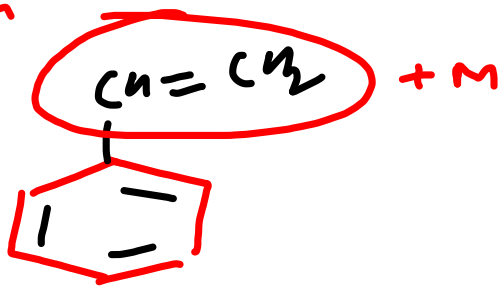
(A) +M or +R Group (ERG or EDG)

The group which release the Electron Through resonance is called +M/+R Group

Identification

eg. (1) $-\ddot{\underset{\ddot{}}{A}}$





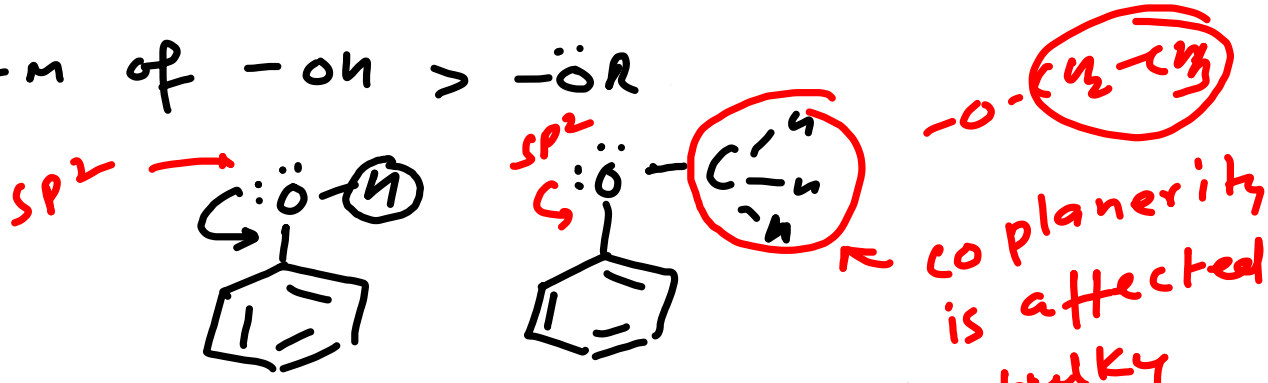
order of strongness of +M / +R Group.

$-\ddot{C}H_2 > -\ddot{N}H > -\ddot{O}: > -\ddot{N}R_2 > -\ddot{N}H_2 > -\ddot{O}H > -\ddot{O}R$

$> -\ddot{N}H-\overset{\overset{O}{\|}}{C}-R > -\ddot{O}-\overset{\overset{O}{\|}}{C}-R > -\ddot{F}: > -\ddot{Cl}: > -\ddot{Br}: > \ddot{I}:$ etc

$2p \leftarrow 2p$ $\quad 2p \leftarrow 3p$ $\quad 2p \leftarrow 4p$

One. +M of $-OH > -\ddot{O}R$



+M order.

$-OH > -O-CH_3 > -OEt$

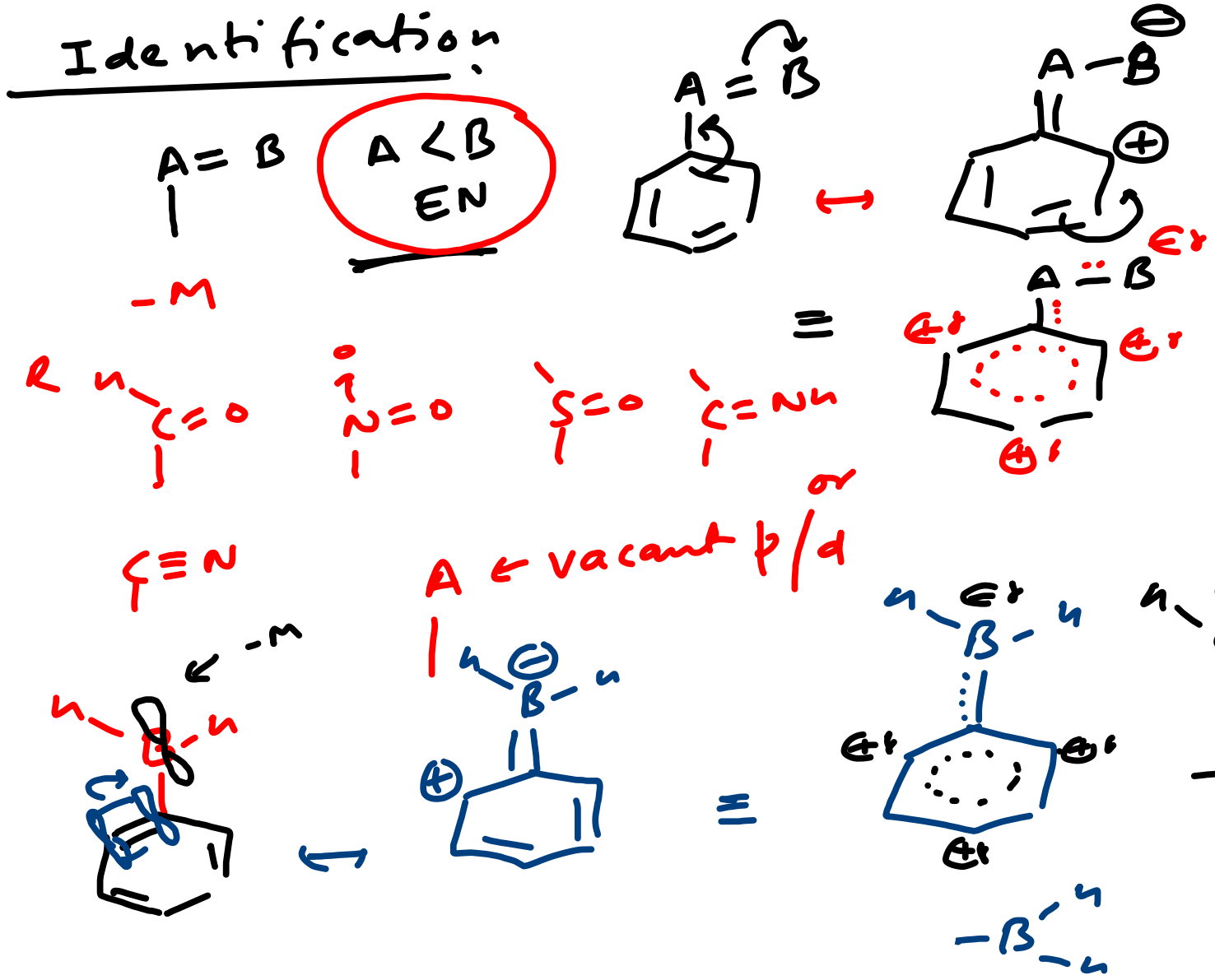
electron density in Benzene ring

a > b > c

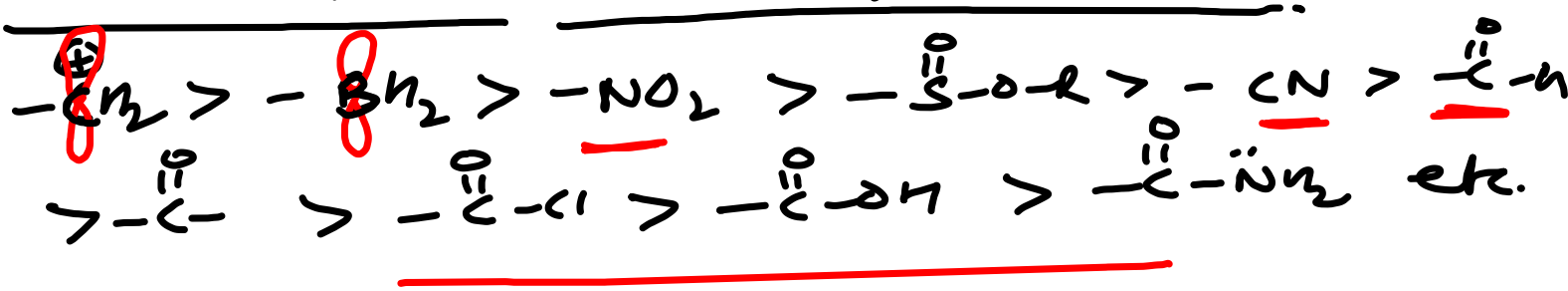
-M or -R Group (EWG)

The group which withdraw the electrons through resonance is called -M group.

Identification:



Order of strongness of -M groups



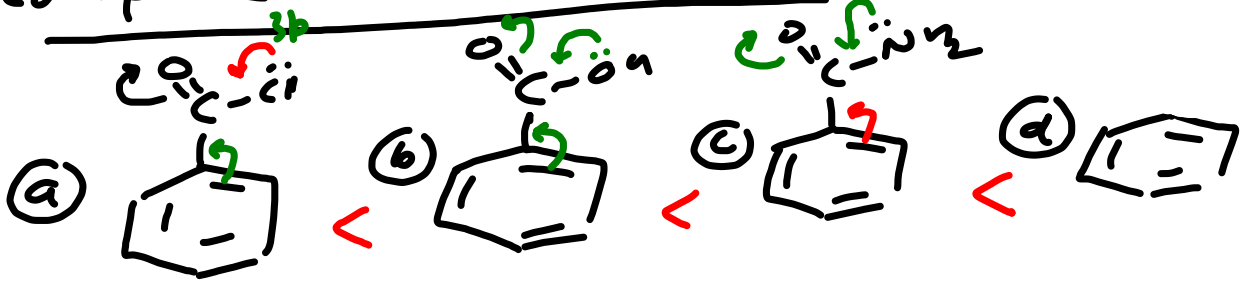
Acid derivative

N C C

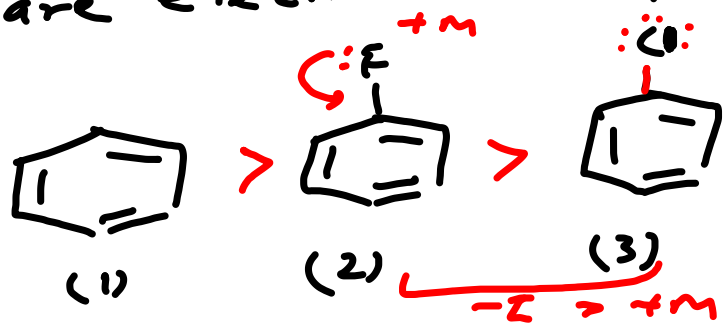
↓ ↓ ↓

NO_2 -CN $-\overset{O}{\overset{\|}{C}}-H$

compare electron density in benzene ring



compare electron density in Benzene ring.



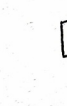
(1) > (2) > (3)

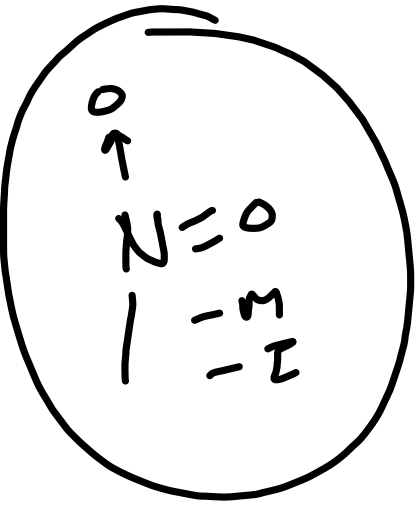
Halide

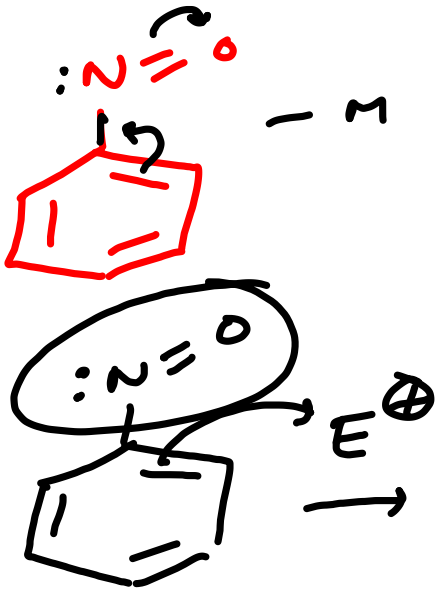
$-I > +M$

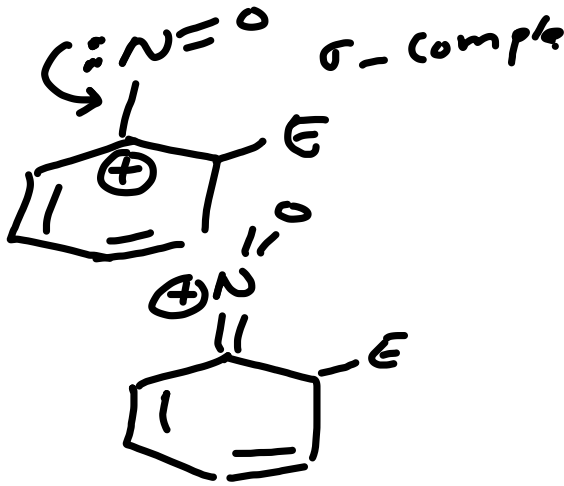
(2) $-I > +M$ (3)

Prob. Identify the nature of groups, i.e +M, −M, +I, −I, +H, −H

S.NO		+M	−M	+I	−I	+H	−H
①		✓			✓		
②					✓		
③			✓		✓		
④		✓	✓		✓		
⑤		✓	✓		✓		
⑥		✓	✓		✓		
⑦				✓		✓	
⑧					✓		✓
⑨			✓		✓		
⑩		✓	✓		✓		
⑪				✓			
⑫		✓			✓		
⑬			✓		✓		
⑭					✓		
⑮			✓		✓		
⑯		✓			✓		
⑰		✓			✓		
⑱					✓		
⑲					✓		✓
⑳					✓		

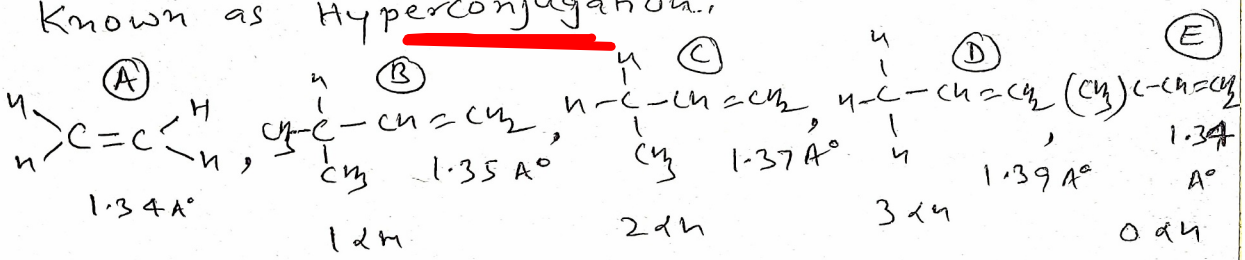

O
N=O
-M
-I


:N=O
-M
:N=O
E⊕

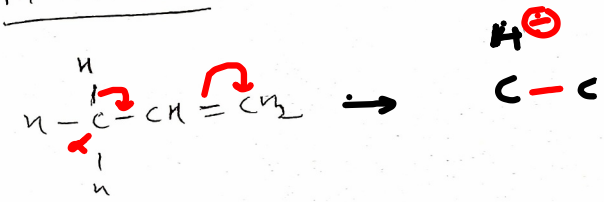

:N=O
σ-complex
E
O
N
E

Hyperconjugation / NO bond resonance / σ-π conjugation effect / or Baker Nathan effect

Permanent delocalisation of σ-eles of α C-H bond into π-molecular orbital is known as Hyperconjugation.



Hyperconjugation in Alkene

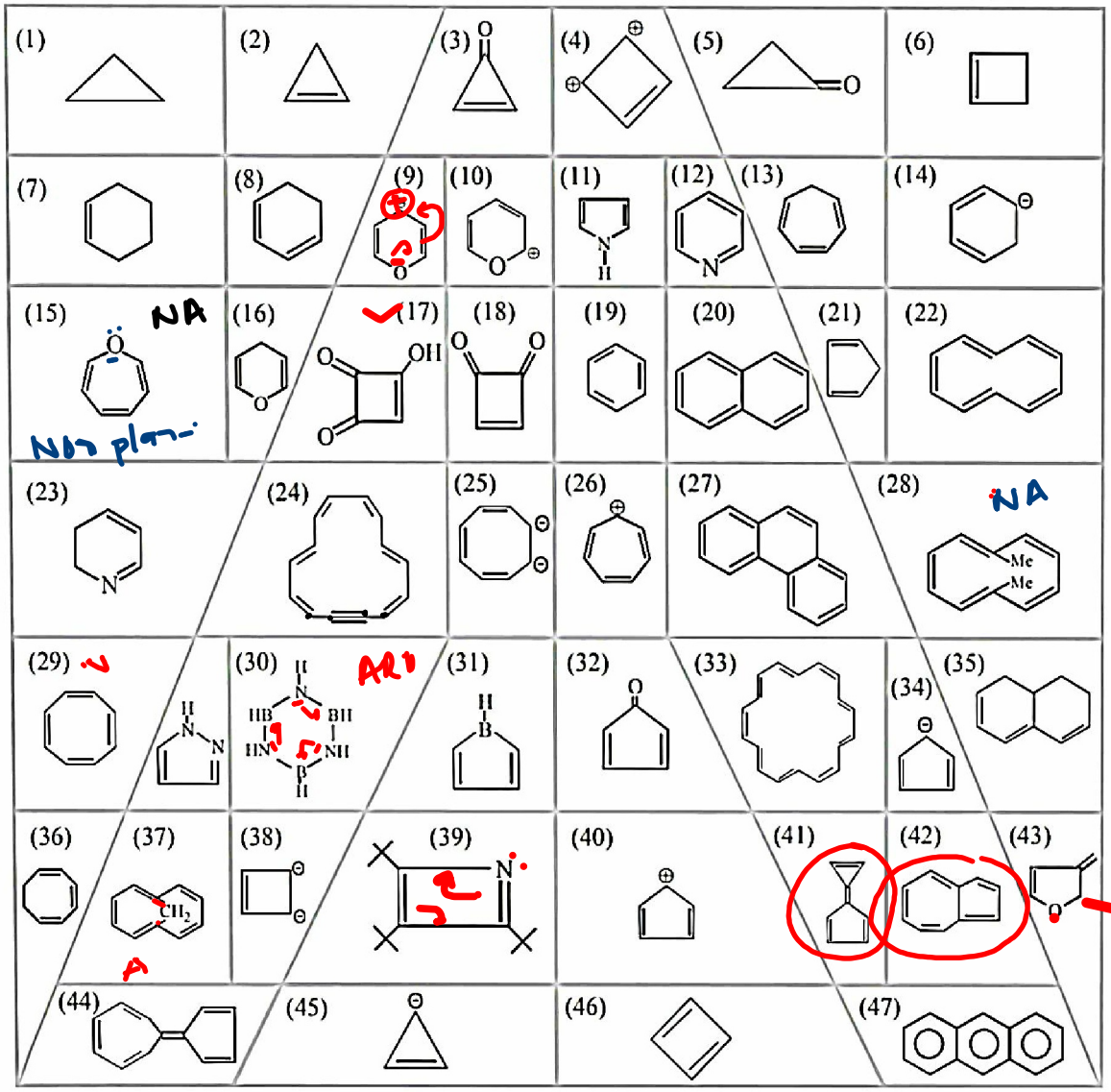


Important points related to Hyperconjugation

1. Carbon attached to double bonded carbon or $C^{\oplus}$ or $C^{\bullet}$ is known as α-carbon and Hydrogens attached to α carbon are known as α-Hydrogens.
2. Hybridisation state of α-carbon converted from sp^3 to sp^2 in Hyperconjugation.
3. More the number of α-hydrogens, more will be the Hyperconjugation structures, which results more stability of alkene, or carbocation or Free Radical.

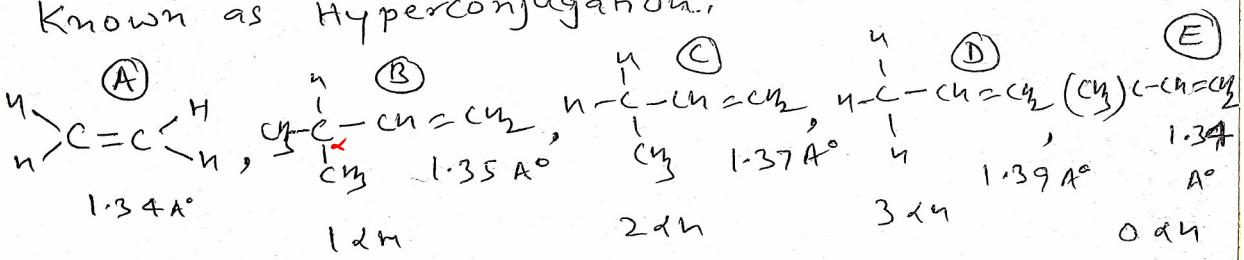
3. Darken the complete area by pencil, having aromatic compounds in it. [25, 0]

3. एरोमेटिक वाले पूरे भाग को पेंसिल से गहरा कीजिए। [25, 0]

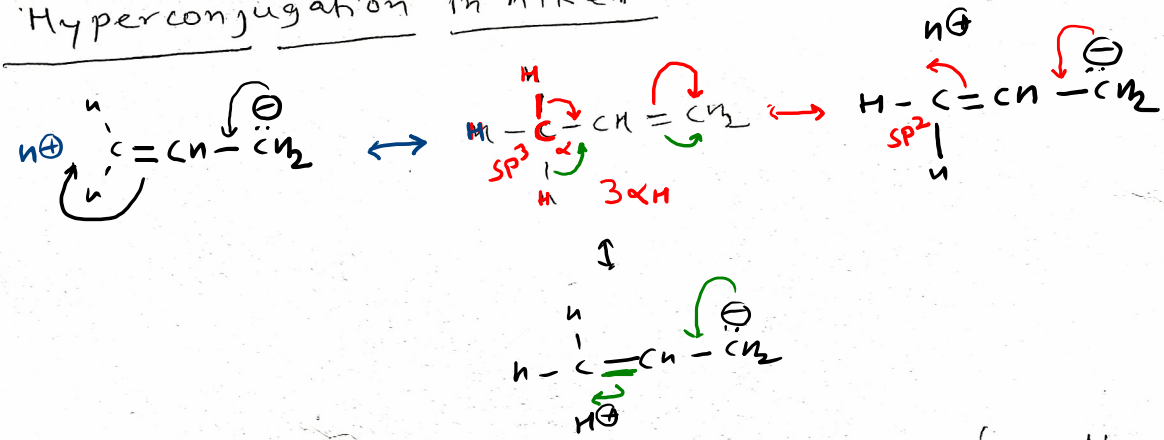


Hyperconjugation / NO bond resonance / σ-π conjugation effect / or Baker Nathan effect

Permanent delocalisation of σ-eles of α C-H bond into π-molecular orbital is known as Hyperconjugation.



Hyperconjugation in Alkene



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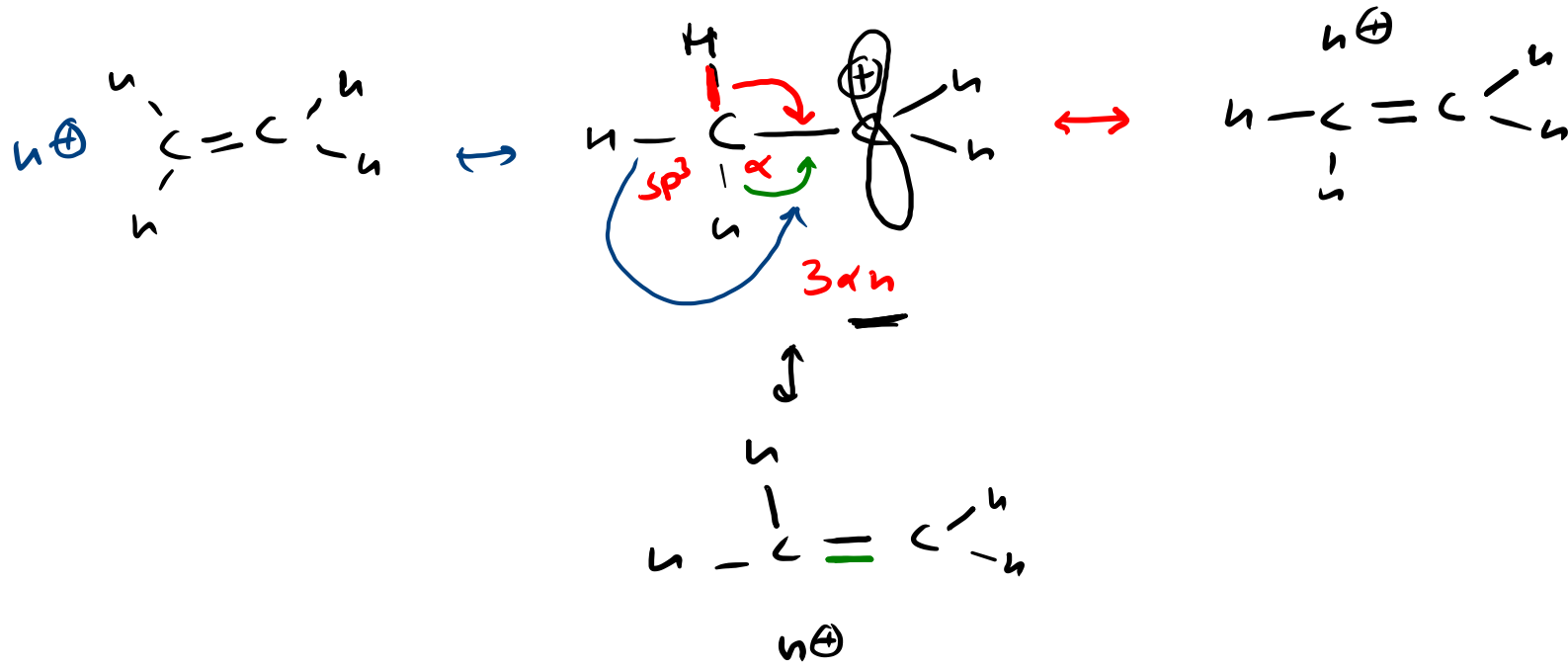
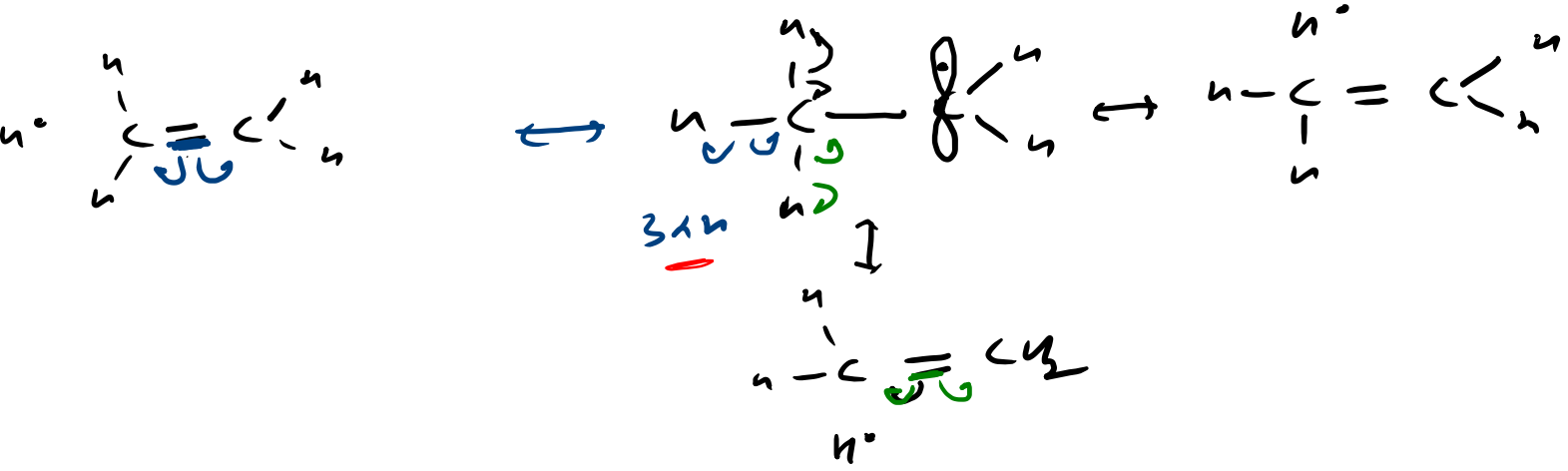
2. Hybridisation state of α-carbon converted from sp^3 to sp^2 in Hyperconjugation.
3. More the number of α-hydrogens, more will be the Hyperconjugation structures, which results more stability of alkene, or carbocation or Free Radical.

Hyperconjugation in carbocation

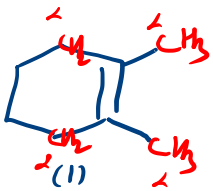


Hyperconjugation in FR

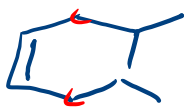


Compare the stability of Alkene

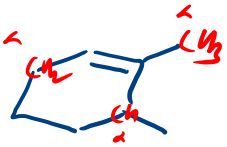
④



10 α-H



4 α-H



6 α-H

i > iii > ii.

Non ii > iii > i

⑤

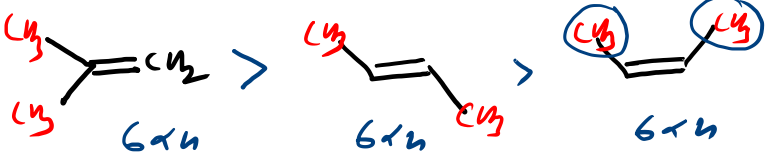
i > ii > iii

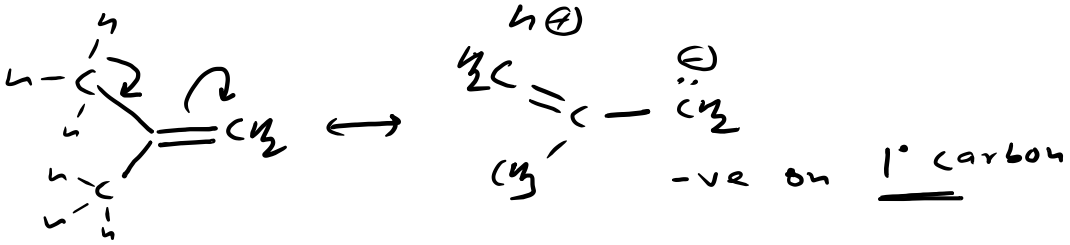
Non

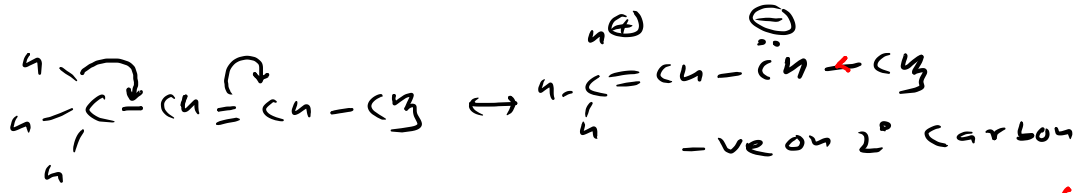
iii > ii > i

⑥

⑦







-ve on 1°C is more stable than -ve on 2° carbon

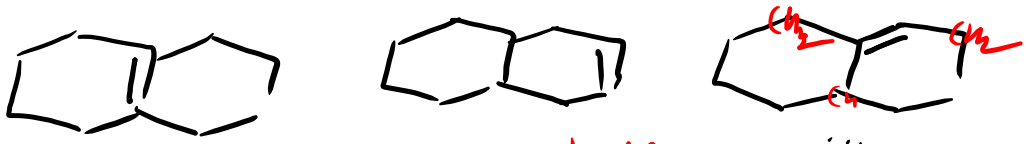
∴ Iso butene (i) is more stable than 2-butene

(8)

1- hexene　　2- hexene　　3- hexene

2 < 4　　5 < 4　　4 < 4

ii > iii > i.

(9) 

(i) 8 α H　　(ii) 4 α H　　(iii) 5 < H

(i) > (iii) > (ii).

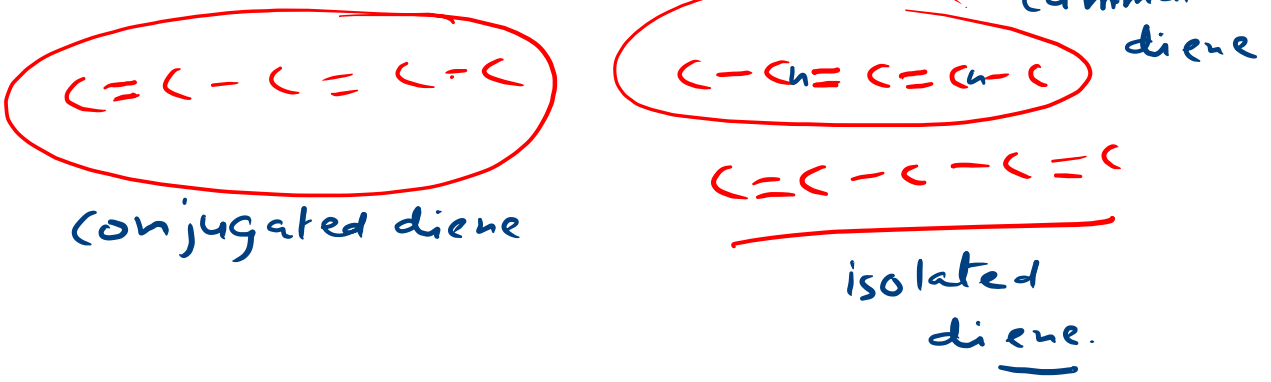
(10) BST

(i)　(ii)　(iii)　(IV)

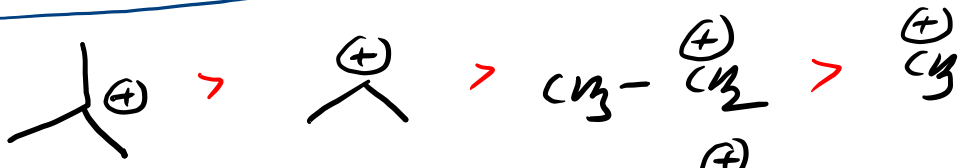
(11) $CH_3-C\equiv CH$ > $CH_3-\overset{\alpha}{C}H_2-C\equiv CH$

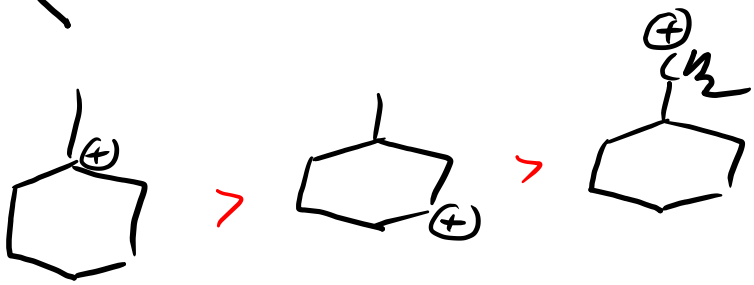
3 < H (i)　　(ii) 2 < H

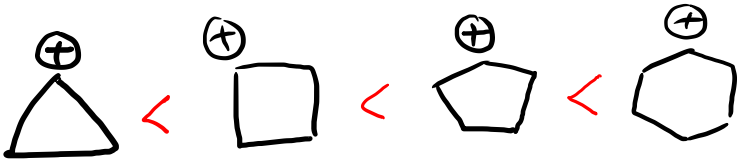
alkene > Alkyne　　(hyperconjugation)

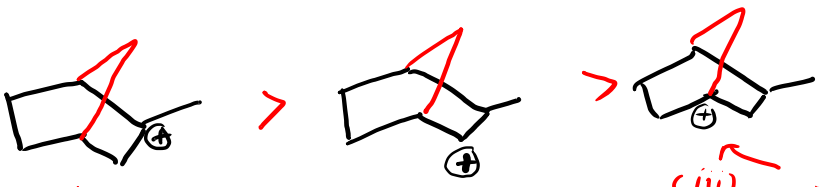


Compare the stability of carbocation

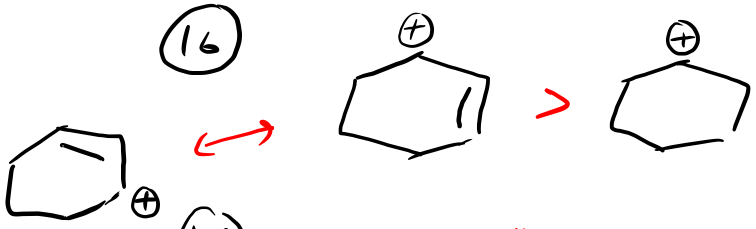
(12) 

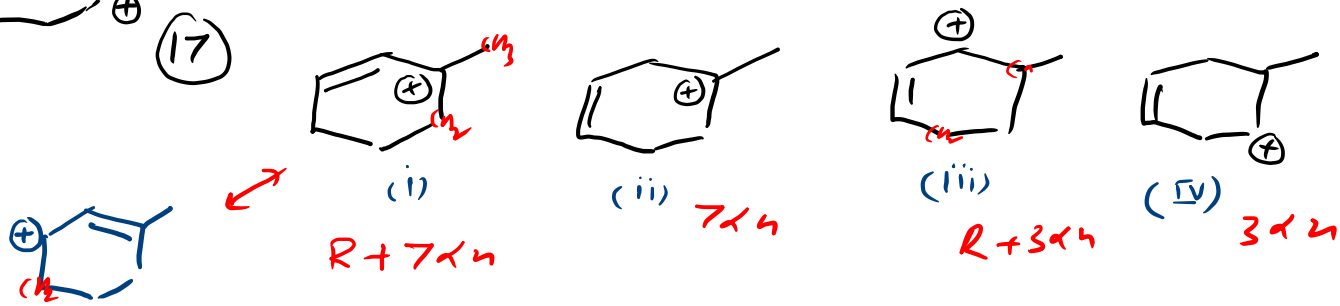
(13) 

BST (14) 

Bredt's rule (15) 

(i) (ii) (iii) sp^2 unstable

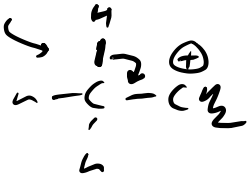
(16) 

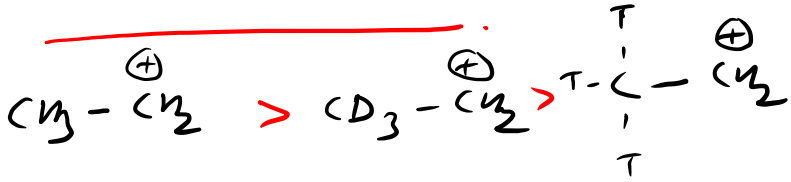
(17) 

(i) R + 7 α H (ii) 7 α H (iii) R + 3 α H (iv) 3 α H

I > III > II > IV.

Less B.E

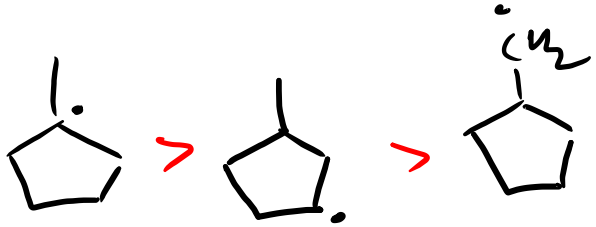


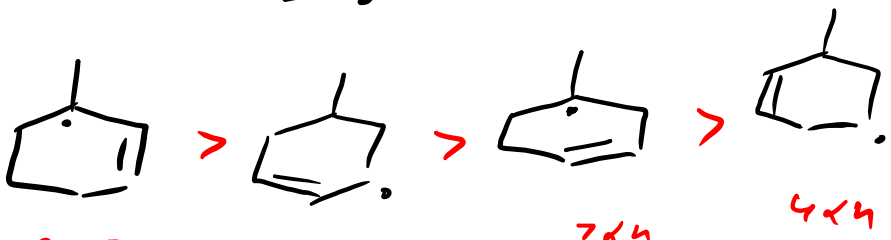
(18) 

+ H order $-CH_3 > -CD_3 > -CT_3$.

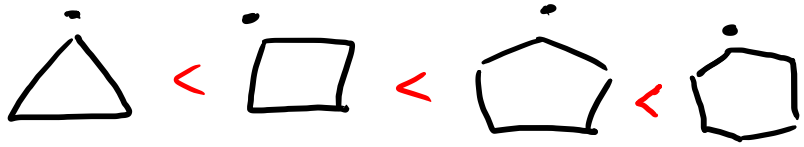
Compare stability of FR

(19)

(20) 

(21) 

R+2×h R+4×h 2×h 4×h

BST (22) 

R > H > I

Baeyer's strain theory (BST)

More the deviation from original bond angle, more will be the angle strain, which results more unstability.

(23) Compare stability

(i) $109^\circ 28' - 60^\circ = 49^\circ 28' \times 3$

(ii) $109^\circ 28' - 90^\circ = 19^\circ 28' \times 4$

(iii) $109^\circ 28' - 108^\circ = 1^\circ 28' \times 5$

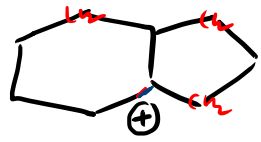
(iv) ↑ 9r is strain free 9r exist in chair form.

angle strain

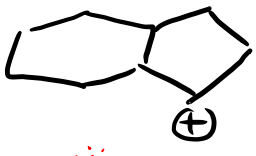
eg. (14), (22)

(24) Compare stability

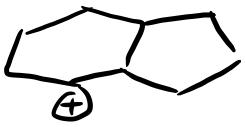
BST.



(i) 5 membered Ring



(ii) 3 membered



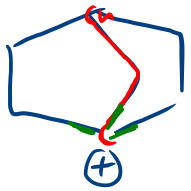
(iii) 3 membered

1 > 3 > 2

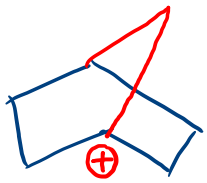
Bredt's Rule

Bridge headed sp^2 atom is unstable in bicyclo compounds until one of the ring contain min 8 atoms.

(25)

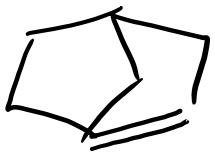


or



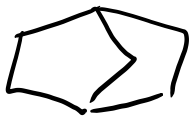
sp^2 unstable

(26)



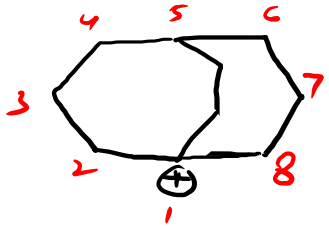
sp^2 unstable

(27)



sp^2 unstable.

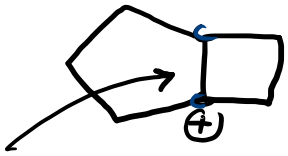
(28)



No Bredt's rule.

Note Bredt's rule is not applicable for those bicyclo compounds which has Zero carbon Bridge is present.

(28)

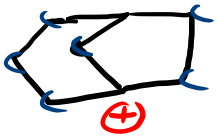


Zero carbon bridge

↑

Bredt's rule is not applicable.

(29)

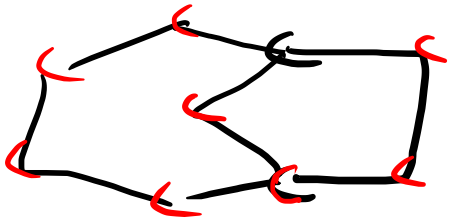


↑

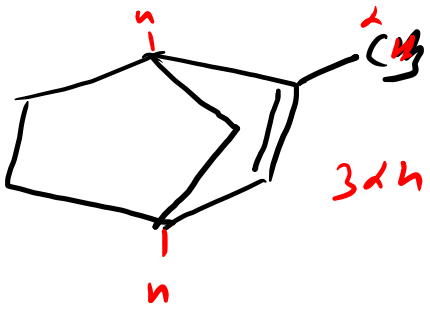
Unstable

↑

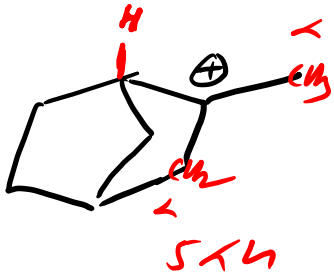
Bredt's rule is applicable.

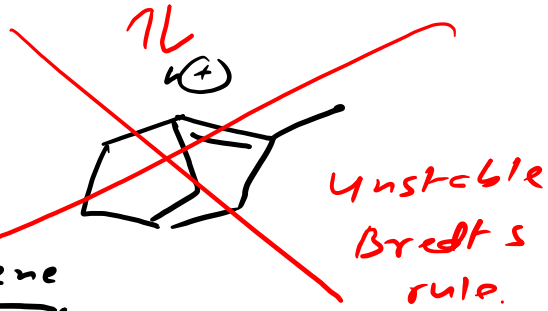


Que 30 How many hyperconjugable α-H are present in following compound.

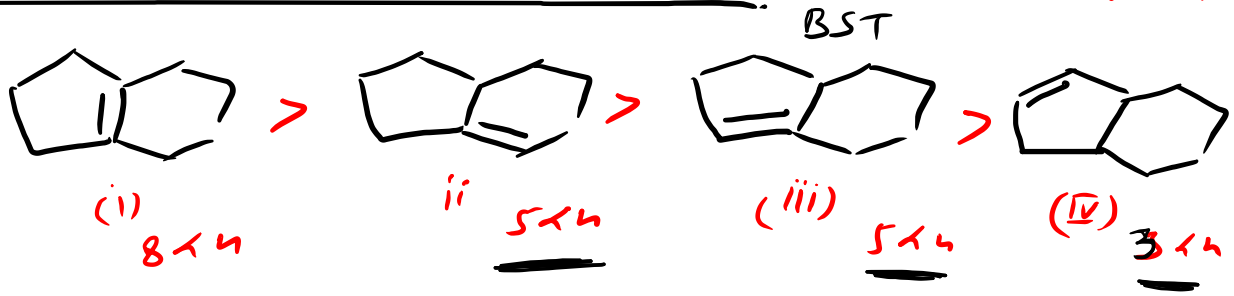


(31)





(32) Compare stability of Alkene



Heat of Hydrogenation (HOH)

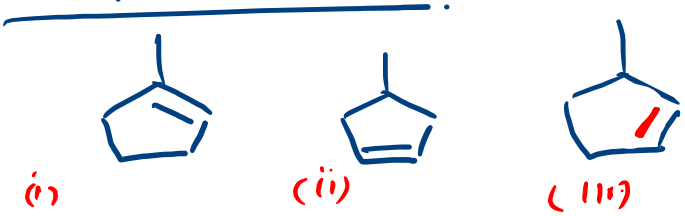
The amount of energy released during hydrogenation (i.e Addition of H_2) is called HOH.

eg. (33) + $\Delta H = -120\ KJ/mol$

Case (1) Alkenes having same no. of π bonds.

Heat of hydrogenation $\propto \frac{1}{\text{stability of alkene}}$

(34) **Compare HOH**



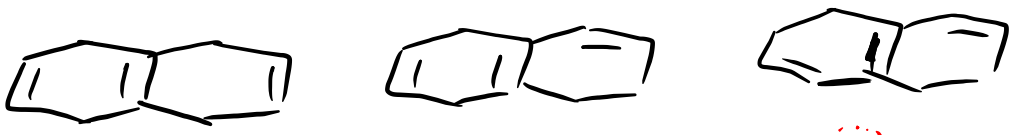
(i) (ii) (iii)

stability i > ii > iii

order of HOH iii > ii > i.

eg. 4 to 10

(35) **compare HOH**



(i) (ii) (iii)

stability iii > ii > i

HOH i > ii > iii

Case (2) If No of π bonds are different

eg (36)

(i) (ii) (iii)

non
iii > ii > i

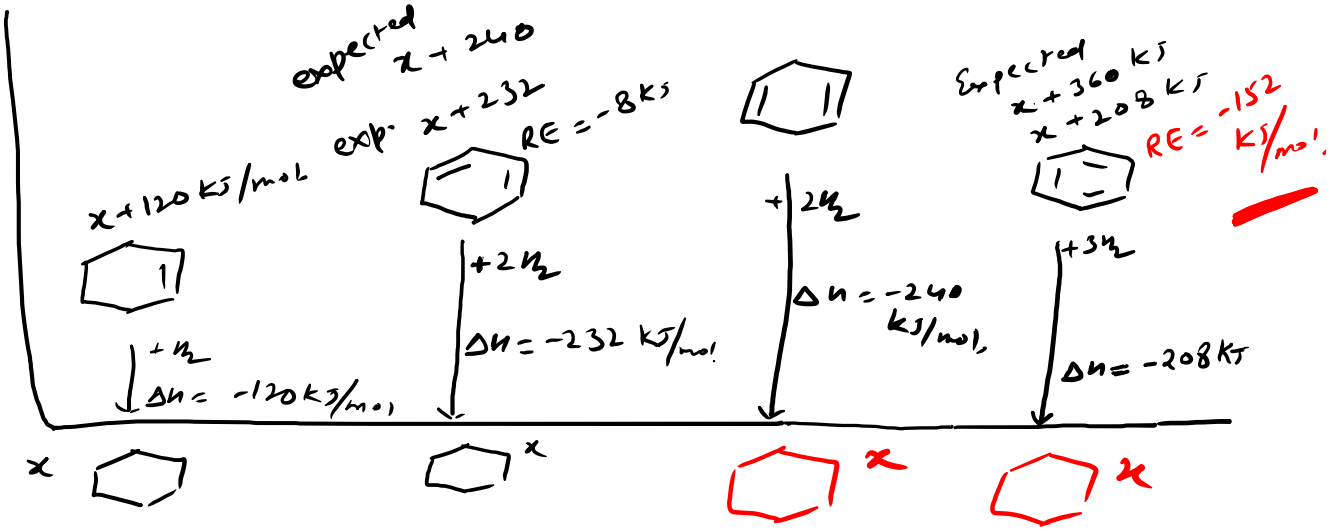
Non ∝ No. of bonds

eg (37) compare Hon

order of non
iii > ii > iv > i

(i)

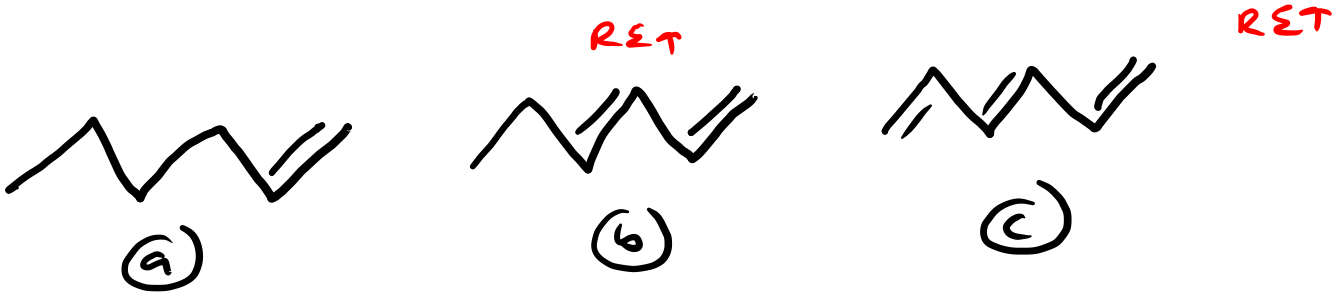
(ii) (iii) (iv)



case (3) HOH per π bond

$$HOH \propto \frac{1}{\text{Stability of each } \pi \text{ bond}}$$

eg (38) Compare HOH per π bond



$$\frac{HOH_a}{1} > \frac{HOH_b}{2} > \frac{HOH_c}{3}$$

eg (39) Compare HOH per π bond

$$\frac{HOH_a}{1} > \frac{HOH_b}{2}$$

↑ (120)

$\frac{232}{2} = 116$

Heat of Combustion (HOC)

The amount of energy released due to burning (complete oxidation) of organic compound is called Heat of combustion (HOC).

eq (1) + $6O_2$ ⟶ $4CO_2 + 4H_2O + \Delta H = -2719$ kJ/mol

1-butene

ΔH ↑ HOC

Case-1 If No. of carbons are different.

> HOC ∝ No. of carbons

eq. (1) Compare HOC

eq. (2)

(i) (ii) (iii) $CH_2 = CH - CH_3$ (IV)

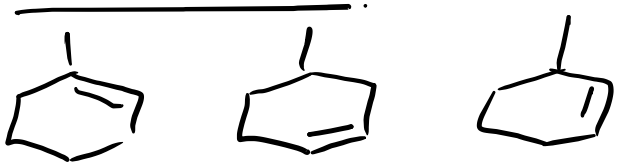
order of HOC i > ii > iii > IV

order of HON IV > III > II > I

Case (2) HOC for isomeric alkene (same MF)

$$HOC \propto \frac{1}{\text{Stability of alkene}}$$

eg. (3) Compare HOC ?



(i) 7 ring (ii) 4 ring (iii) 3 ring

HOC iii > ii > i.
or HOH

example

Case (3) HOC per carbon (or per $-CH_2$) (last lecture)

$$HOC \propto \frac{1}{\text{Stability of each carbon (or per } CH_2)}$$

eg. (4) compare HOC per $-CH_2$ (or per C)

angle strain →

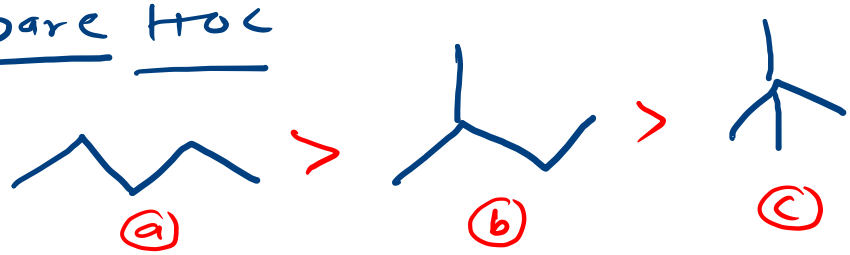
(a) (b) (c) (d)

$$\frac{HOC_{(a)}}{3} > \frac{HOC_{(b)}}{4} > \frac{HOC_{(c)}}{5} > \frac{HOC_{(d)}}{6}$$

Case 4 HOC for isomeric alkane

HOC ∝ Surface area of isomeric alkane ∝ $\frac{1}{\text{Branching}}$.

eg (5) Compare HOC



(a) (b) (c)

Resonance Energy (RE)

Def: the amount of energy released due to resonance is called RE.

OR

PE difference b/w RH and it most stable RS is known as RE.

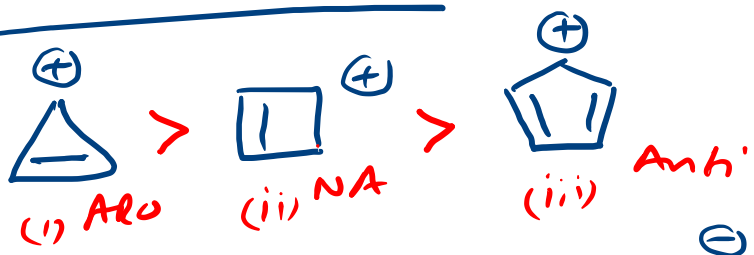
$$RE = PE \text{ of } R.H - PE \text{ of most stable } R.S$$

RE = -ve ↑

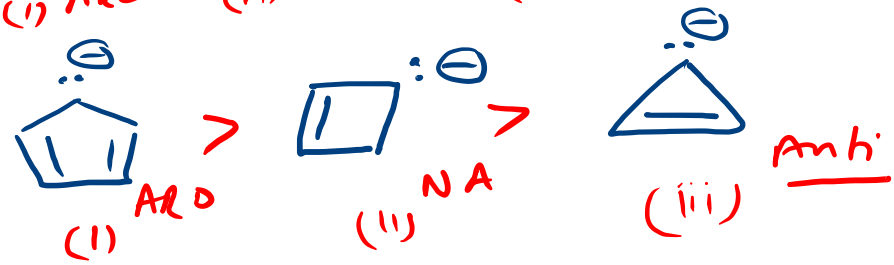
case (1) faster the resonance, more RE

(a) Aromatic > Non Aromatic > Anti Aromatic.

eg. (1) Compare R.E

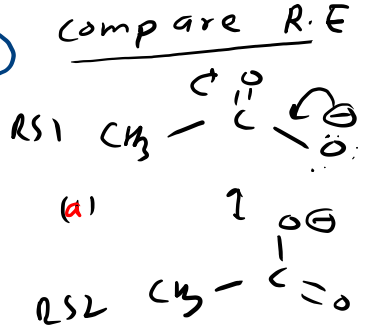


eg. (2)

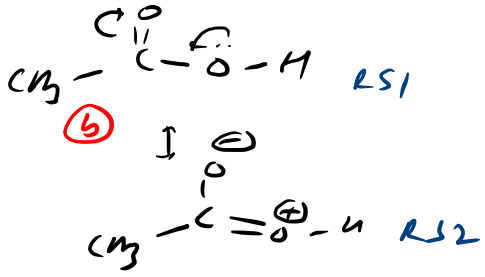


(b) Equivalent type RS has faster Resonance (∴ more R.E) Than non equivalent type RS.

eg. (3) compare R.E.

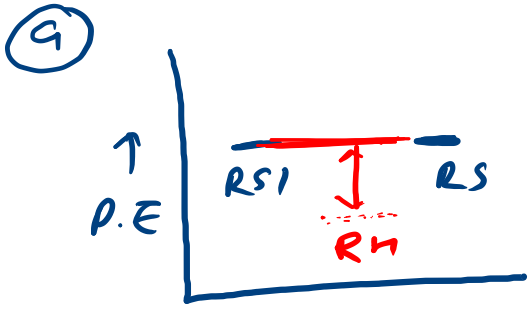


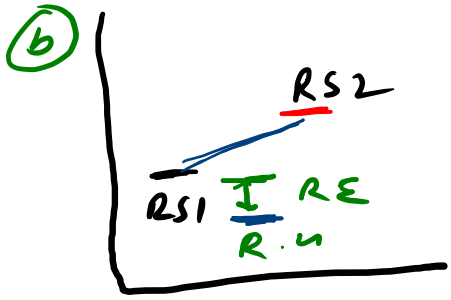
RS1 = RS2



Stability

RS1 > RS2

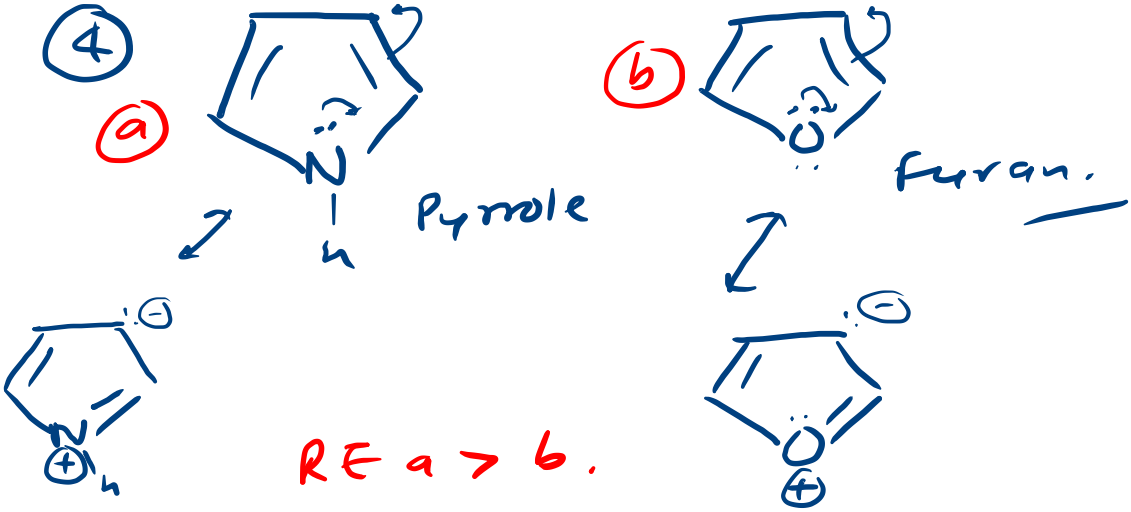




R.E a > b

(c) RE of compound having stable RS has more RE than the compound having less stable RS.

eg. (4)



(d) RE of extended conjugation is more than cross conjugation.

eg (5) compare RE

(a) Extended conjugation

(b) cross conjugation

R.E a > b.

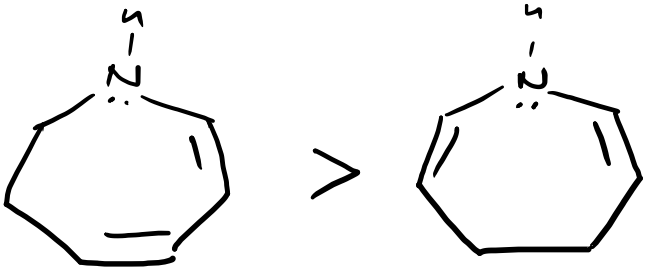
(6)

(i) Cross

(ii) Extended

RE I < II

(7) compare RE



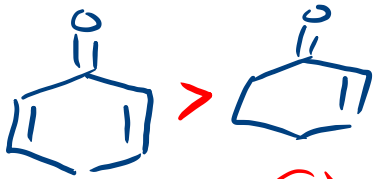
(a) > (b)

case (2) RE of long conjugation is more than short conjugation

eg (8) compare RE

(a) long > (b) short

(9)

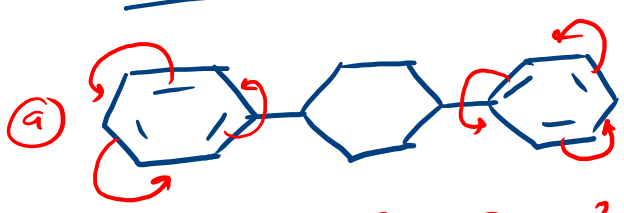


(a) > (b) RE.

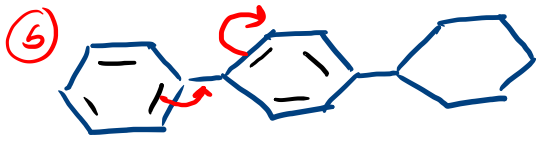
case (3) Fries Rule

More the benzenoid (Benzene like) structure, more will be the R.E

eg. (10) Compare R.E

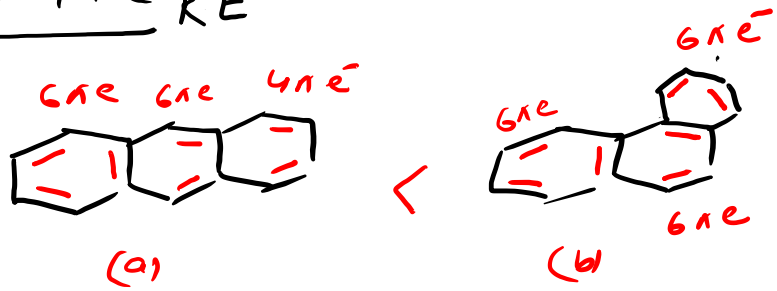


RE = -152 × 2 = -304 kJ/mol



RE = -297 kJ/mole

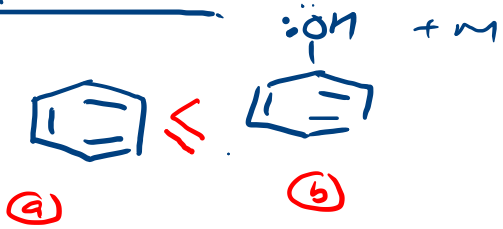
eg (11) Compare RE



RE $\boxed{a < b}$

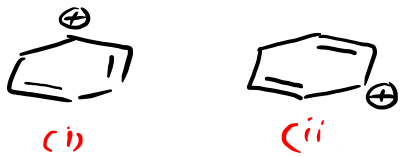
Note (1) +M slightly increases R.E.

eg. (12) Compare R.E



Note: (2) RE is compared b/w different compounds.
(NOT b/w different Rs of same compound)

eg. (13) Compare RE



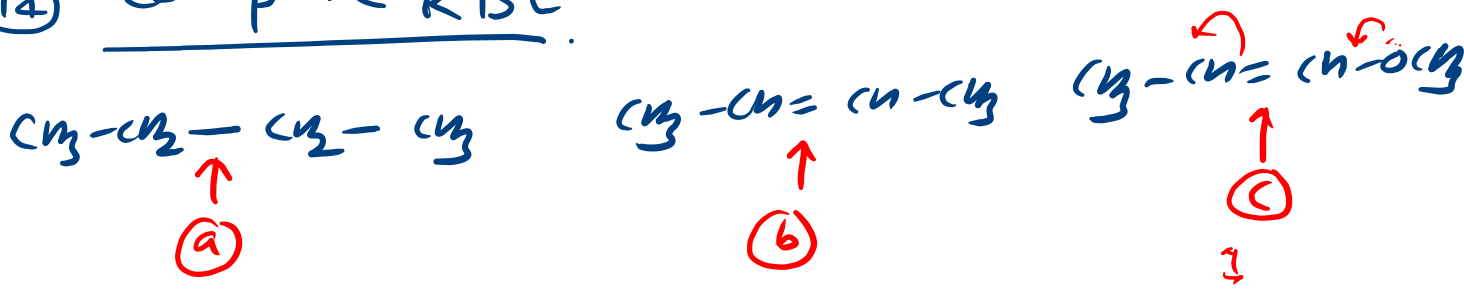
(I) = (II)
RS1 RS2

Rotational Barrier Energy (RBE)

The min amount of energy required for rotation across σ-bond is called RBE

$$RBE \propto B.O$$

eg. (14) Compare RBE

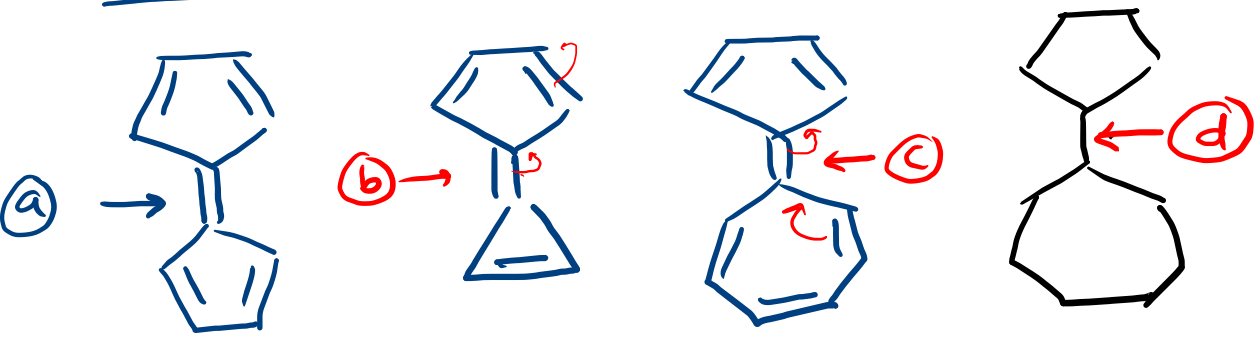


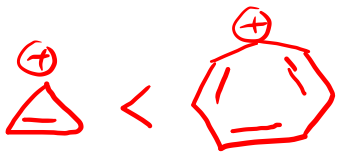
B.O $b > c > a$

RBE $b > c > a$

$1 < B.O > 1$

(15) Compare RBE





Bond order

$a > b > c > d$

RBE $a > b > c > d$.

B.O / B.E / B.D.E / B.L

B.E $\propto$ B.O

B.D.E $\propto$ B.O

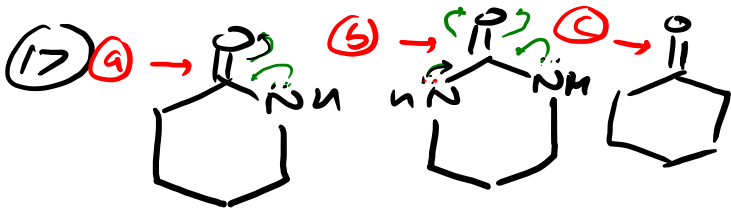
B.L $\propto \frac{1}{\text{B.O}}$

Ex. (16)

B.O a > b

B.L b > a

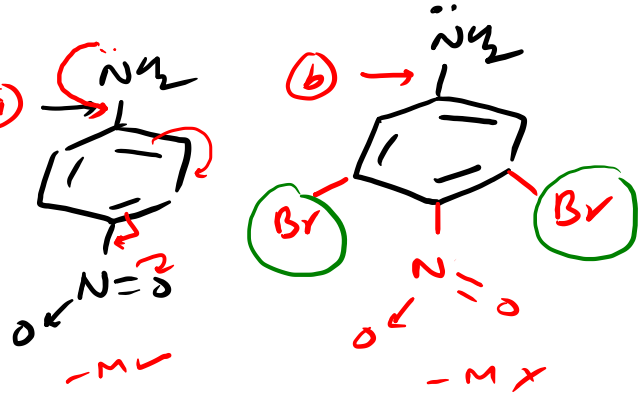
(17)



B.O = c > a > b

B.L = b > a > c

SIR (18)



B.O = b > a

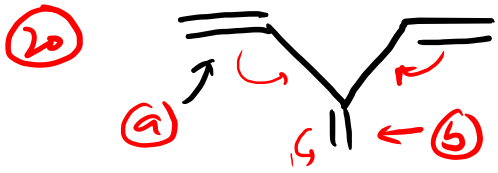
B.L = a > b.

(19)

B.O x > y

B.L y > x

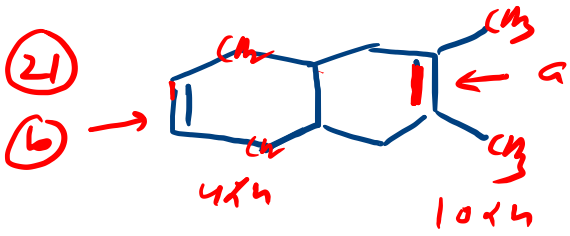
(20)



B.O a > b.

B.L b > a

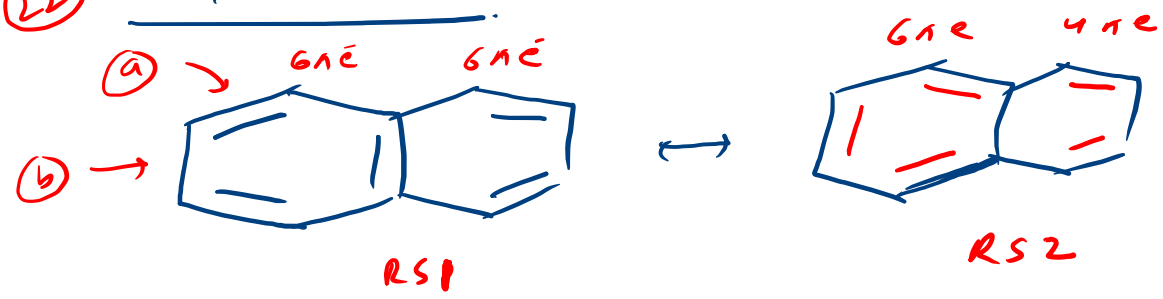
(21)



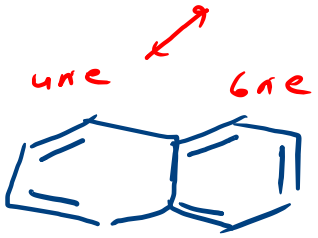
B.O b > a

B.L a > b.

(22) Compare B.L



RS1 ⟷ RS2



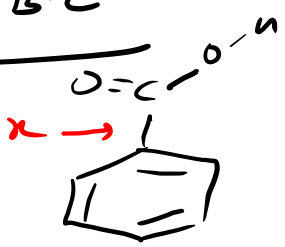
RS3

B.O = a > b

B.L = b > a

RS1 > RS2 = RS3

(23) Compare B.L



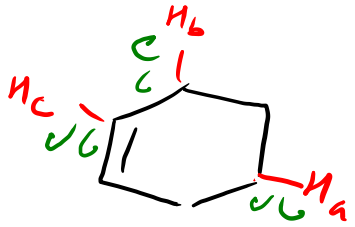
out of plane.

B.O = x > y

B.L = y > x

(24) Compare B.E

$$B.E \propto \frac{1}{\text{stability of } FR}$$



Ha → (i)

Hb → (ii)

Hc → (iii)

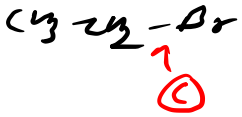
stability

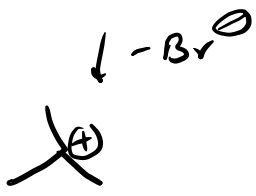
ii > i > iii.

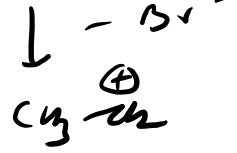
B.E

iii > i > ii

(25) Compare BE







B.E ∝ $\frac{1}{\text{Stability of } C^{\oplus}}$

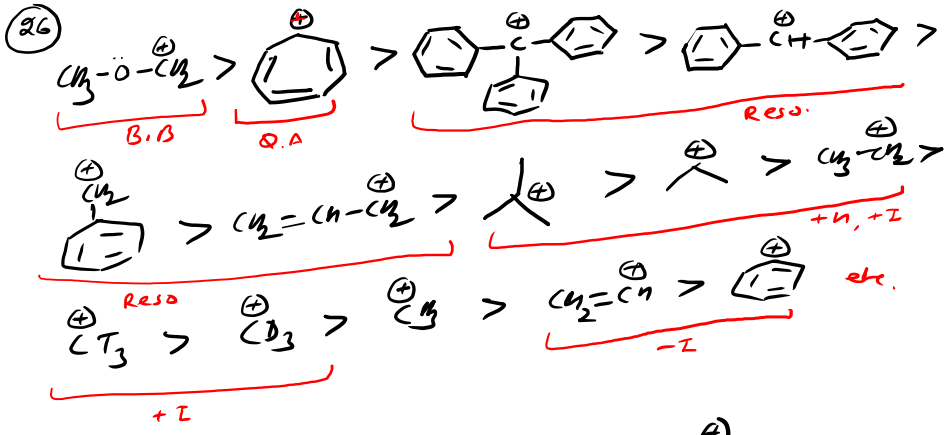
B.E c > b > a

Compare stability of Carbocation

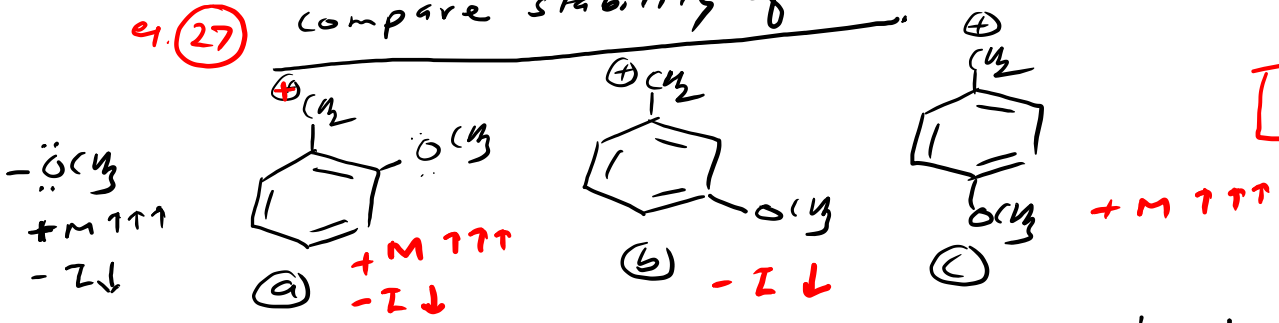
QA > R > H > I

BB

(26)



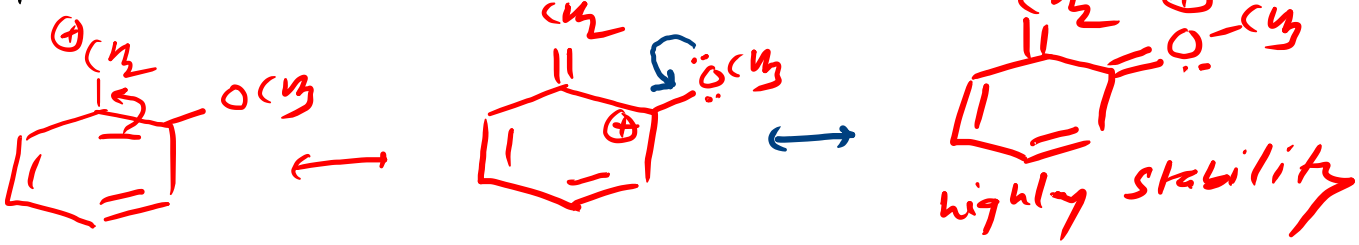
eg. (27) Compare stability of $C^{\oplus}$



C > a > b

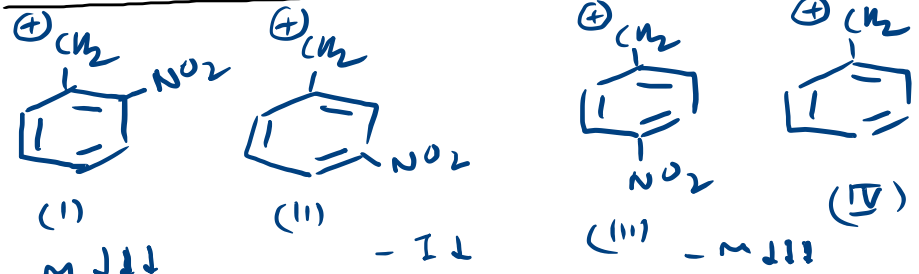
VIP Mesomeric effect (+M, -M) or Hyperconjugation effect (+H or -H) operates from ortho and para positions (NOT from meta).

VIP Inductive effect (+I or -I) operates from ortho > meta and very less from para position. and it is distance dependent.



Compare stability of carbocation

28



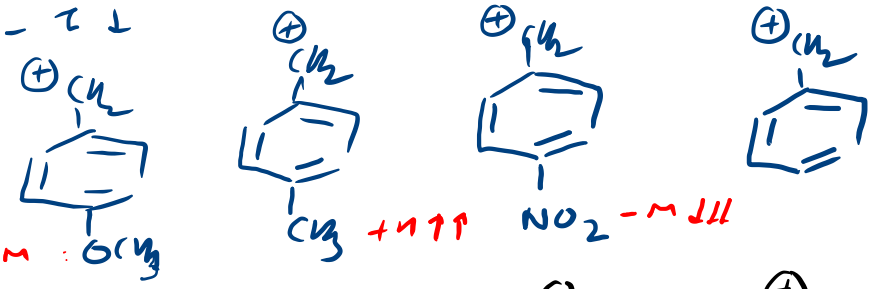
-M -I

-M -I

-I

IV > II > III > I

29



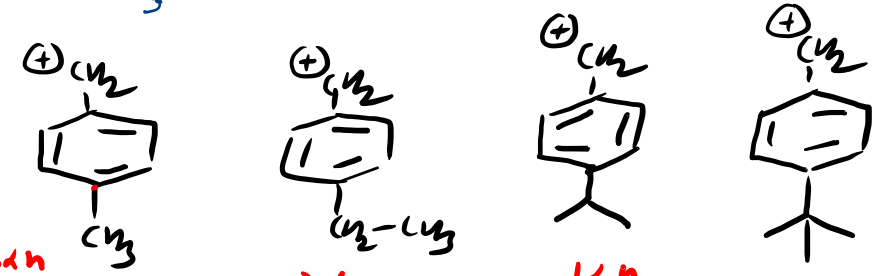
+M : OCH3

+M

-M

I > II > IV > III

30



3°

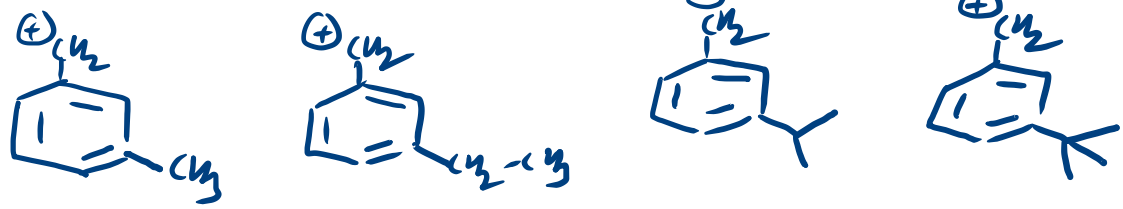
2°

1°

0°

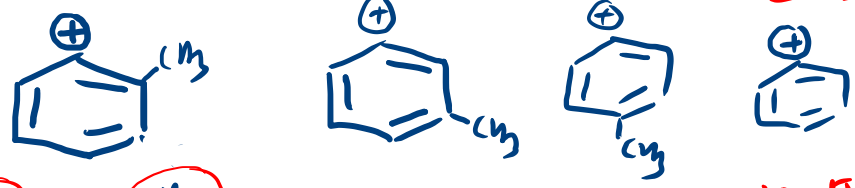
I > II > III > IV

31



IV > III > II > I

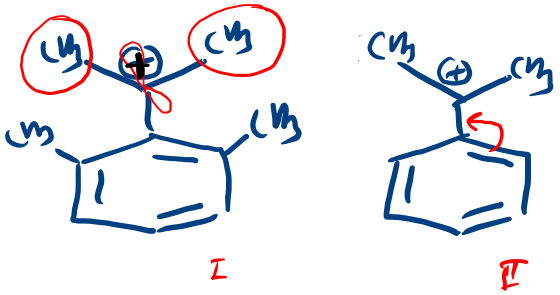
32



I > II > III > IV

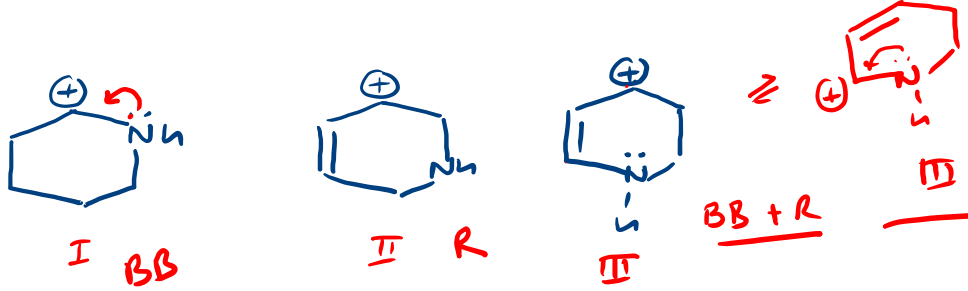
SiR

33



I < II

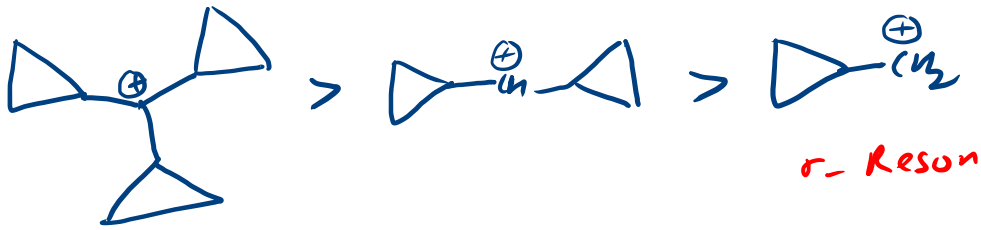
34



I BB II R III BB + R

III > I > II

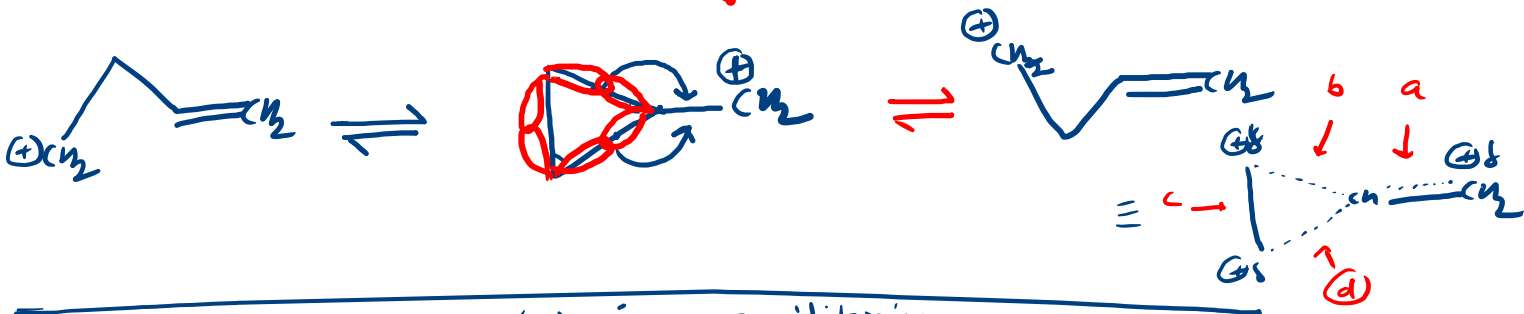
35



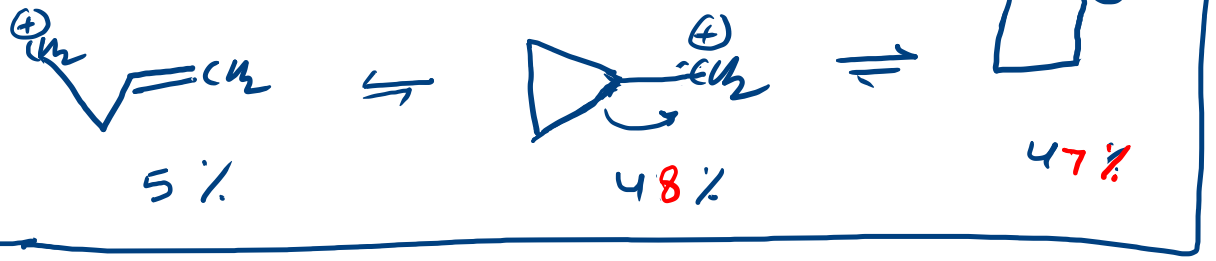
σ- Resonance.

σ- Resonance / or Dancing Resonance

9+ is highly stable ⊕.



9+ exist in equilibrium

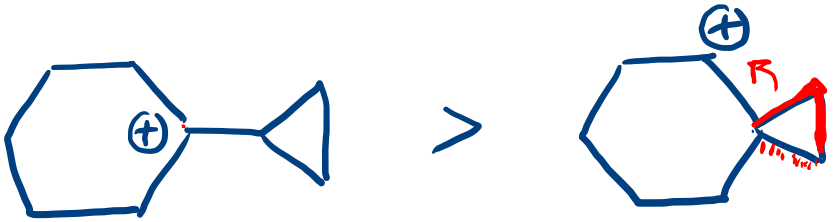


5 % 48 % 47 %

BL =

b = d > c > a

36



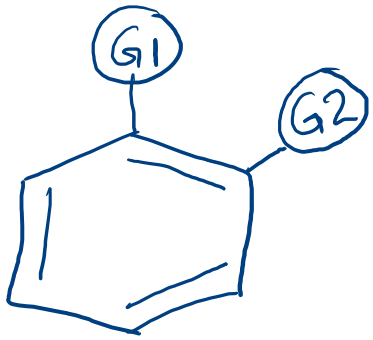
(a) σ-Reso ✓

(b) σ-Reso ✗

SIR Effect (Steric Inhibition of resonance)

Inhibition of Resonance due to steric hindrance is called SIR effect.

eg. (36)



Essential condition for SIR effect.

cond. (1) Both G1 and G2 must bulky.

Smaller groups → H , -F , -OH , -NH_2

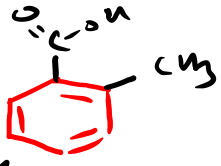
Head F O N

group other than smaller group are Bulky group.

cond (2) Alleast one group (G1 or G2) must be sp^2 when it attached to benzene ring.

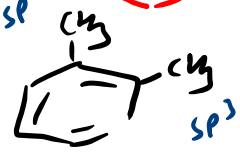
Note: More the bulkier group, more effective SIR effect.

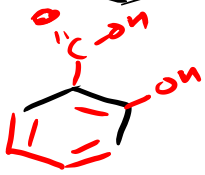
prob. Identify the compound in which SiR effect is present.

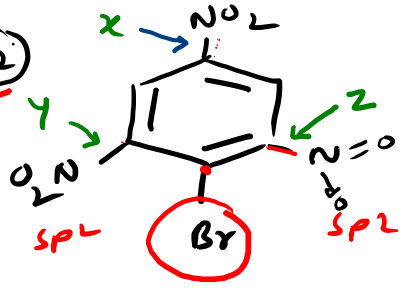
37 ✓ 

38 ✓

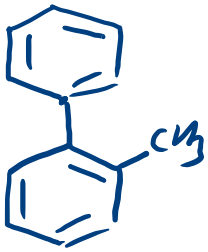
39 ✗

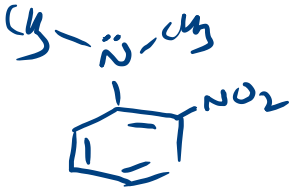
40 ✗ 

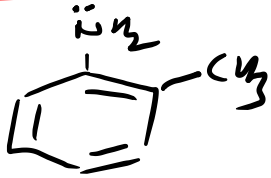
41 ✗ 

42 ✓ 

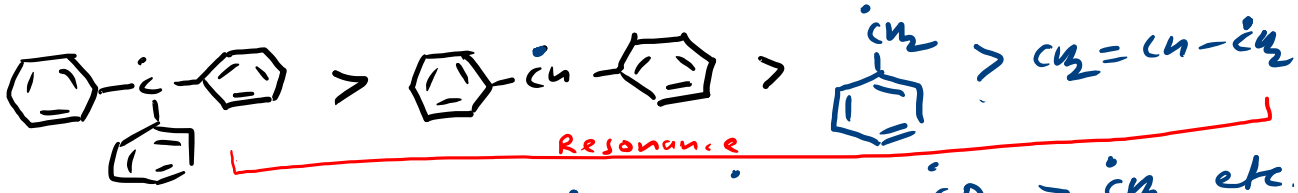
B.L $Y = Z > X$

43 ✓ 

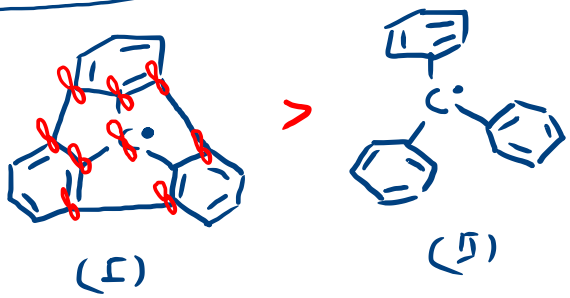
44 ✓ 

45 ✗ 

Relative stability of Free Radical.

46 

Compare stability of F.R

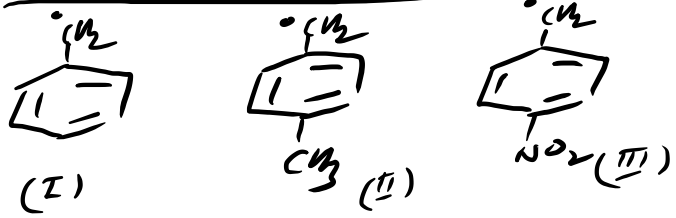
47 

(I) (II)

I > II

↑ more effective Resonance.

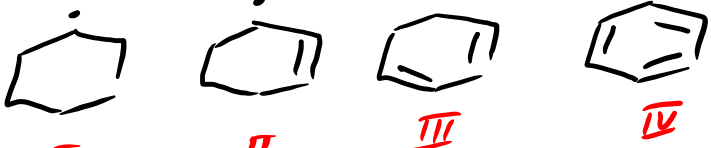
(48) compare stability of FR



(I) (II) (III)

II > I > III

(49)



I II III IV

III > II > I > IV

R R H NO R

NO H

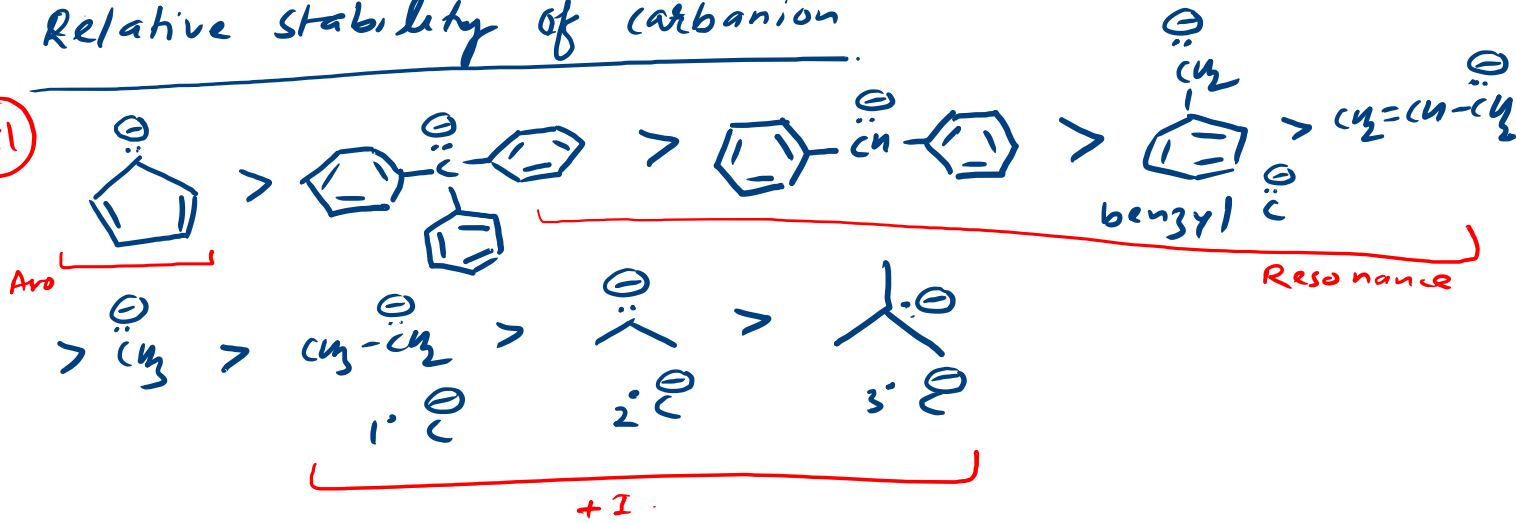
(50)

I II III IV

II > II > I > IV.

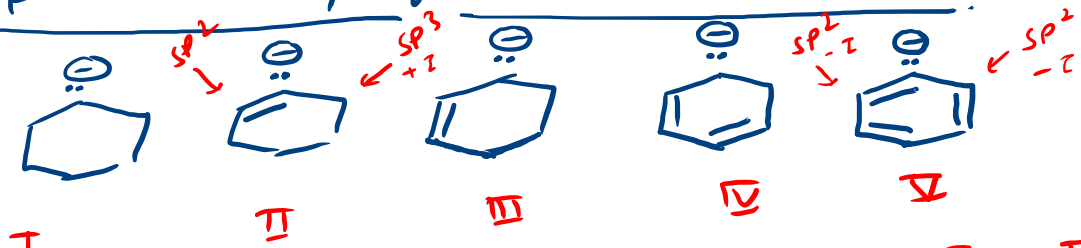
Relative stability of carbanion

(51)



compare stability of carbanion or anion

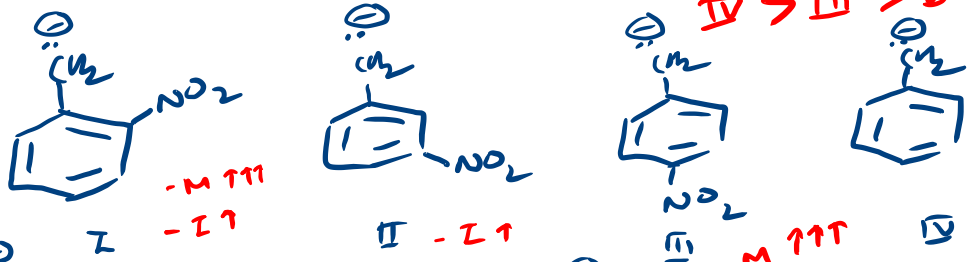
(S2)



I II III IV V

IV > III > V > II > I

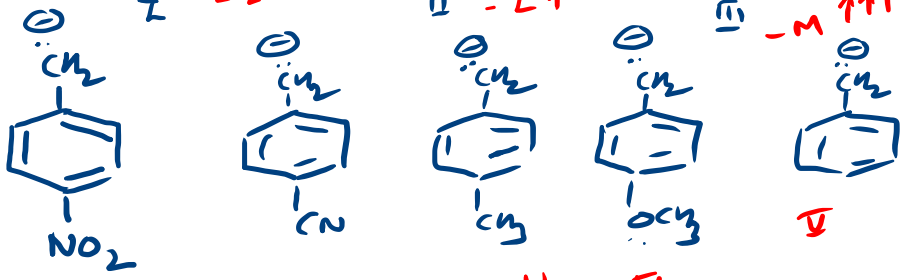
(S3)



I II III IV

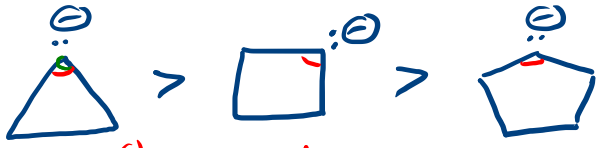
I > III > II > IV

(S4)



I > II > V > III > IV

(S5)

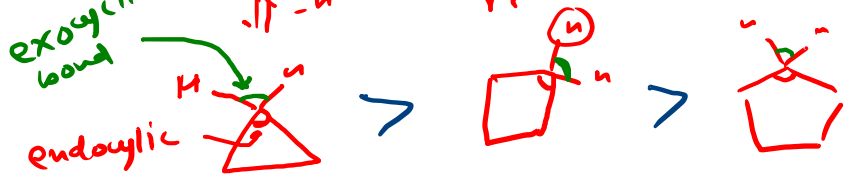


II > I > III

%s character of exocyclic

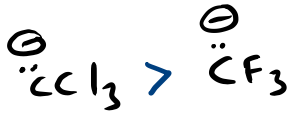
exocyclic bond

endocyclic



order of acidic strength

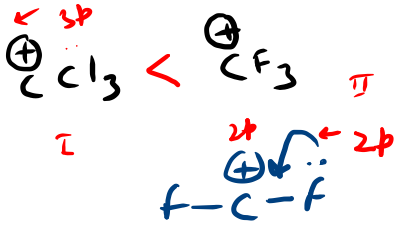
(S6)



I > II

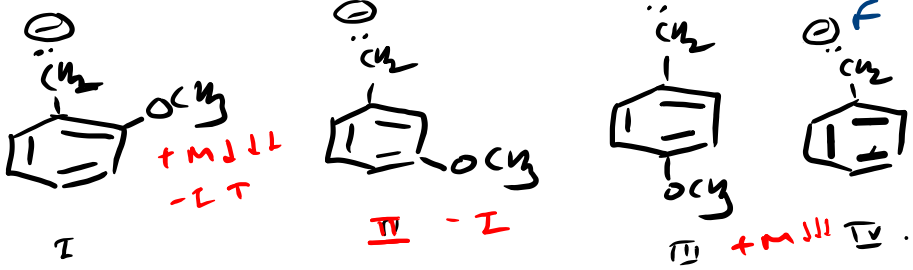
B.B

(S7)



I < II

(S7)

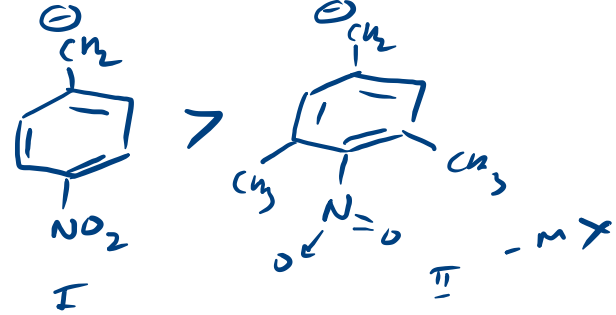


-OCH₃
+M
-I

II > IV > I > III

-I

58

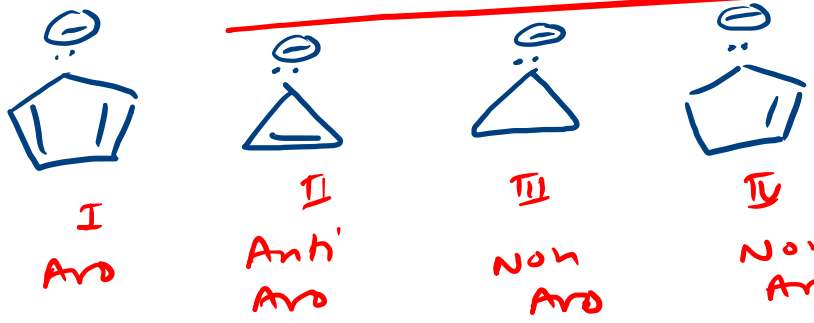


I > II

59

I > III > II > V > IV

60



I > IV > III > II

I: Aro

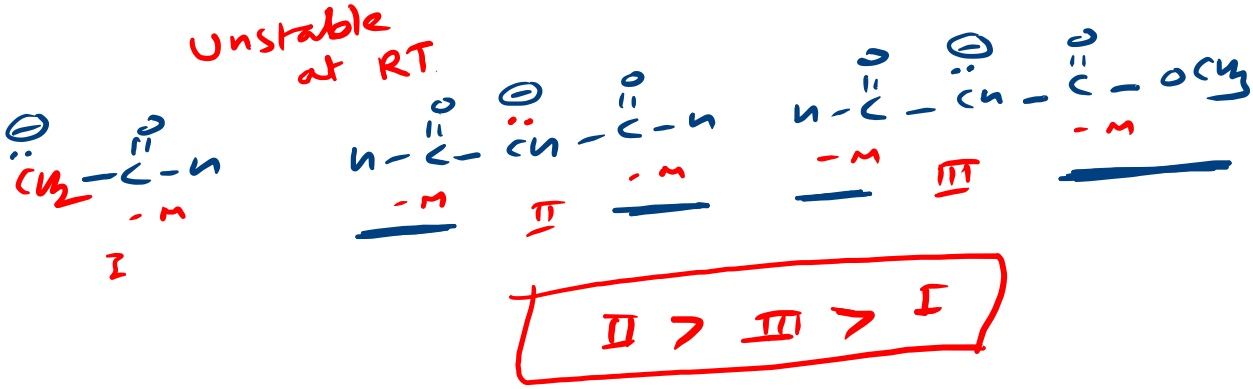
II: Anh' Aro ↑ Unstable at RT

III: Non Aro

IV: Non Aro

II — Anh

61



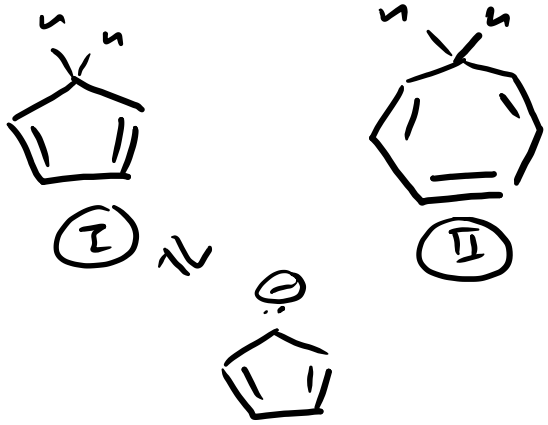
II > III > I

Acidic strength or acidic Nature

$H\text{-}A \rightleftharpoons H^{\oplus} + \overset{\ominus}{A}$ (conjugate Base)

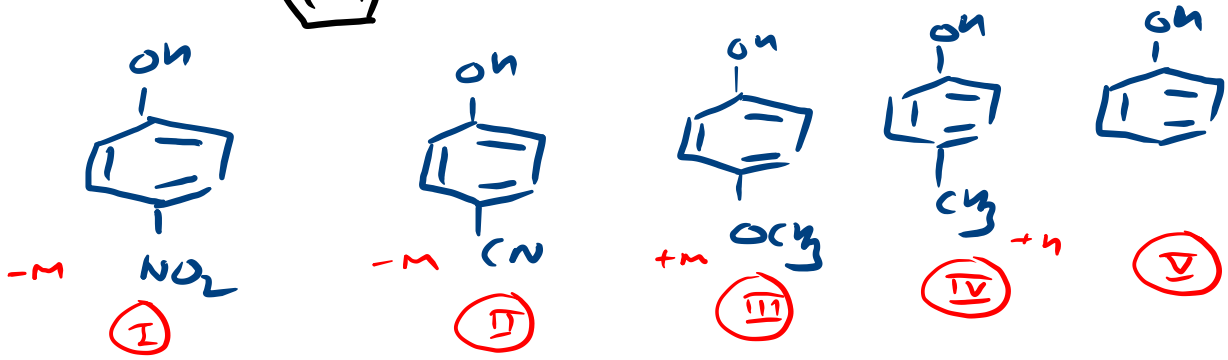
Acidic strength ∝ stability of conjugate

(62) **Compare acidic strength**



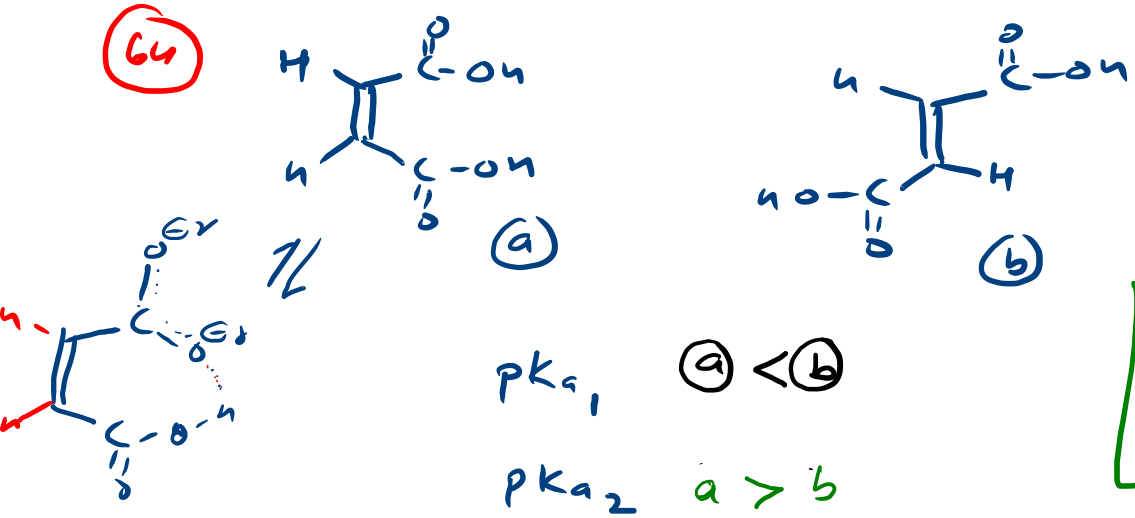
I > II

(63)



I > II > V > IV > III

(64)

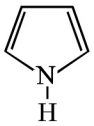


pka₁ (a) < (b)

pka₂ a > b

Acidic strength ∝ 1/pka

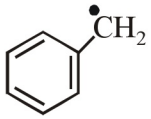
Q.27 In the given pair of compounds select the one in each pair having higher resonance energy :

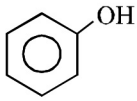
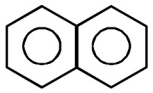
(i) and 

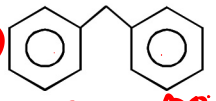
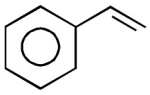
(ii) $CH_2 = CH - \ddot{O} - CH = CH_2$ and $CH_2 = CH - \ddot{N}H - CH = CH_2$

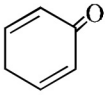
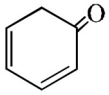
(iii) $CH_2 = CH - \overset{\ominus}{N}H$ and $HN = CH - \overset{\ominus}{N}H$

(iv) $CH_2 = CH - F$ and $CH_2 = CH - Br$

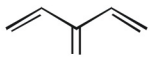
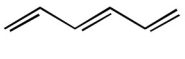
(v)  and $CH_2 = CH - \dot{C}H_2$

(vi)  and 

(vii)  and 

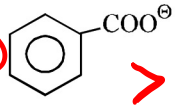
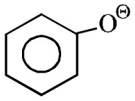
(viii)  and 

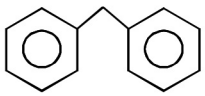
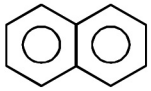
(ix) $CH_2 = CH - OH$ and $CH_2 = CH - CH = CH - OH$

(x)  and 

(xi) CH_3COOH and CH_3COONa

(xii) $CH_2 = CH - \overset{\ominus}{O}$ and $CH_2 = CH - \ddot{O}H$

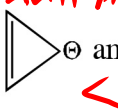
(xiii)  and 

(xiv)  and 

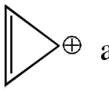
(xv) and $CH_2 = CH - CH = CH - CH = CH_2$

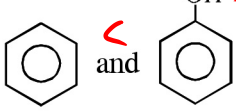
Q.28 In the given pair of compounds select the one in each pair having lesser resonance energy :

(a) CO_3^{2-} and $HCOO^-$

(b)  and $CH_2 = CH - \overset{\ominus}{C}H_2$

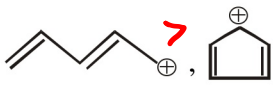
(c) and $CH_2 = CH - CH = CH_2$

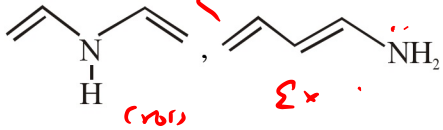
(d)  and $CH_2 = CH - \overset{\oplus}{C}H_2$

(e) 

Q.29 In which of the following pairs first one is having more resonance energy than the second one -

(A)

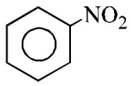
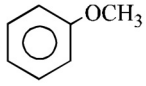
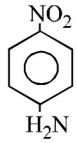
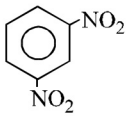
(B) 

(C) 

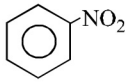
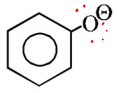
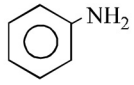
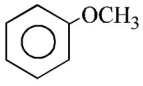
(D) None of these

node06\B0BO-BA\Other center\JEE(Advanced)\Nurture\Physics\Module#01\Eng\ELECTRONIC DISPLACEMENT EFFECT

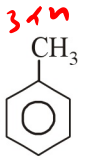
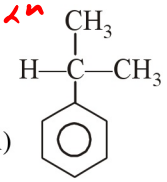
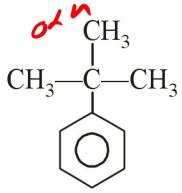
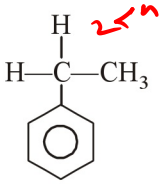
Q.30 In which of the following molecules π – electron density in ring is minimum :

(A)  (B)  (C)  (D) 

Q.31 In which of the following molecules π – electron density in ring is maximum :

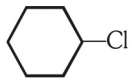
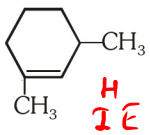
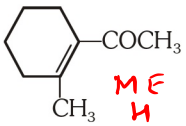
(A)  (B)  (C)  (D) 

Q.32 Arrange following compounds in decreasing order of reactivity of ring towards attack of electron deficient species -

(i)  (ii)  (iii)  (iv) 

(A) i > ii > iii > iv (B) iii > iv > ii > i (C) i > iv > ii > iii (D) i > ii > iv > iii

Q.33 In which of the following molecule all the effect namely inductive, mesomeric & hyperconjugation operate:

(A)  (B)  (C)  (D)

Q.34 Which one of the following molecules has all the effect, namely inductive, mesomeric and hyperconjugative?

(A) CH_3Cl (B) $CH_3–CH = CH_2$

(C) $CH_3 - CH = CH - \underset{\underset{O}{\|}}{C} - CH_3$ (D) $CH_2 = CH - CH = CH_2$

Q.35 Select the correct statement.

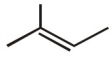
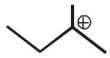
(i) Delocalisation of σ-electron is hyperconjugation.

(ii) Delocalisation of π-electron is resonance.

(iii) Permanent partial displacement of σ-electron is inductive effect.

(A) i & iii (B) ii & iii (C) i & ii (D) i, ii, iii

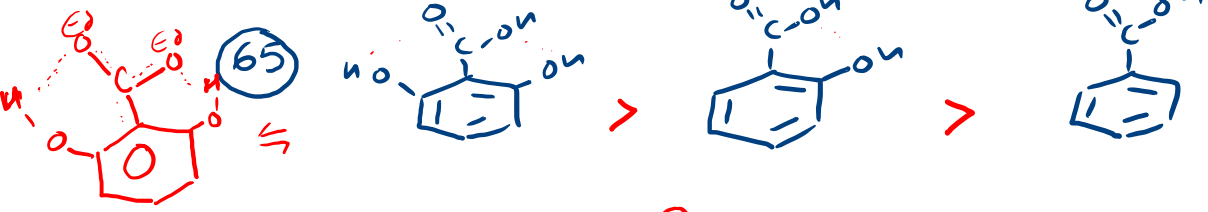
Q.36 Which of the following compound is correctly matched with number of hyperconjugating structures (involving C—H bond) :

(A) (3) (B)  (9) (C)  (8) (D) $CH_3–C\equiv C–CH_3$ (5)

node06\B0B0-BA\Other center\JEE(Advanced)\Nurture\Physics\Module#01\Eng\ELECTRONIC DISPLACEMENT EFFECT

Compare acidic strength

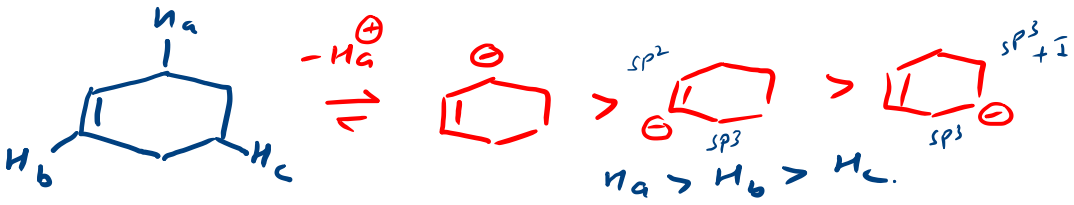
65



66

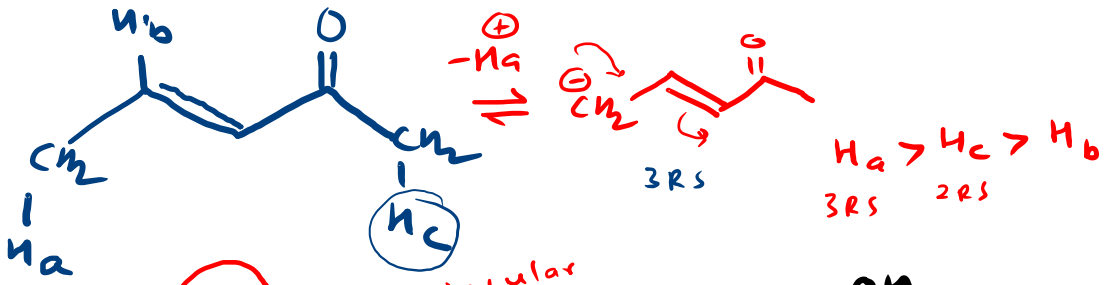
$H_a > H_b > H_c$

67



$H_a > H_b > H_c$.

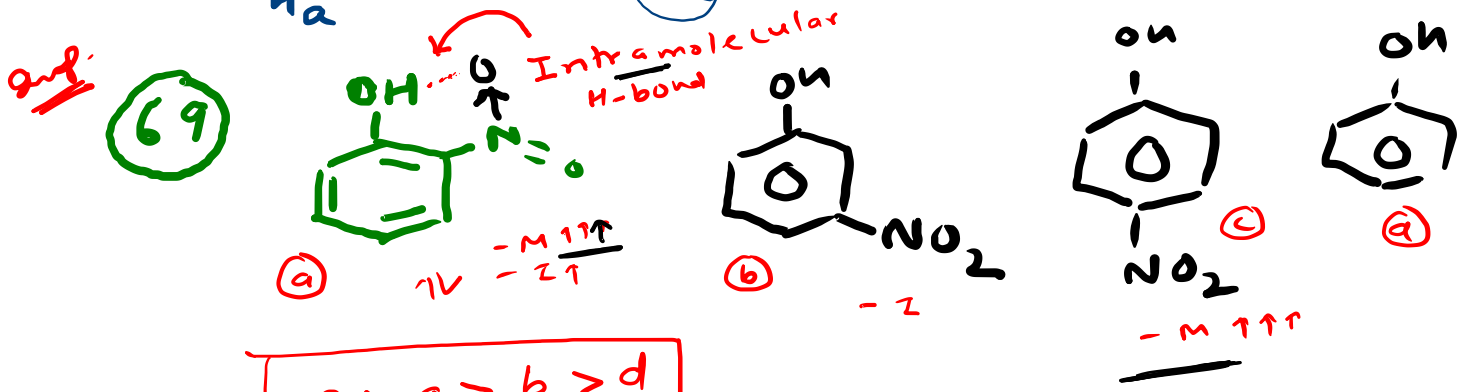
68



$H_a > H_c > H_b$

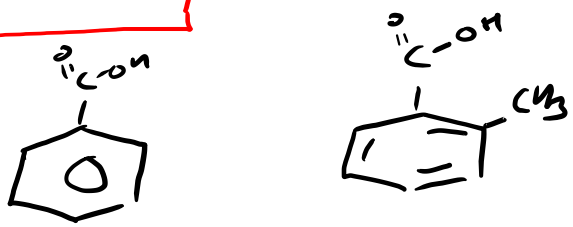
3RS 2RS

69



$c > a > b > d$

ortho effect (SIR) 70

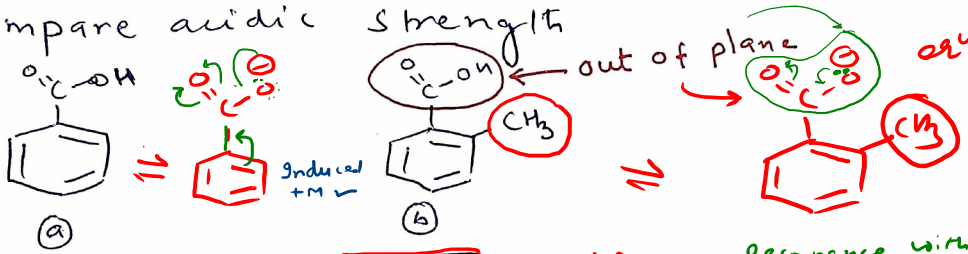


Acce ↑

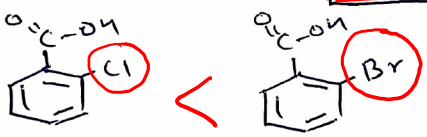
ORTHO EFFECT FOR Acidity (SIR Effect)

Ortho Substituted Benzoic acid (Substituted by bulky group) is more acidic than it's corresponding meta or para substituted benzoic acid, due to SIR (steric inhibition of resonance), which is known as ortho Effect for Acidity.

eg (70) Compare acidic strength

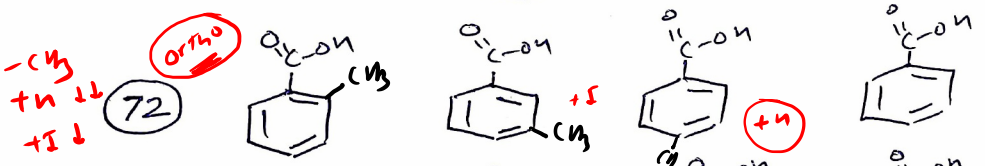


(71)



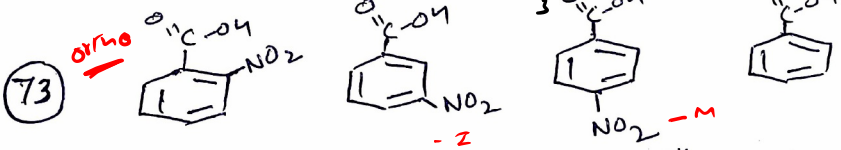
more bulky group, more effective SIR or ortho effect

(72)



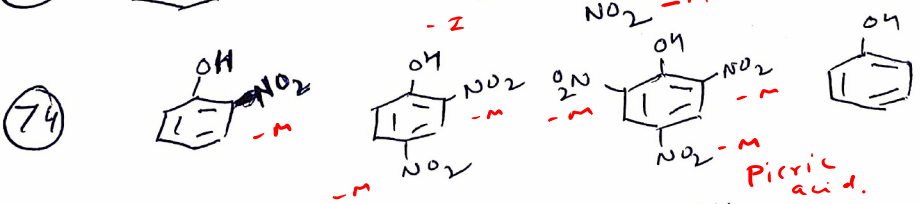
I > IV > II > III

(73)



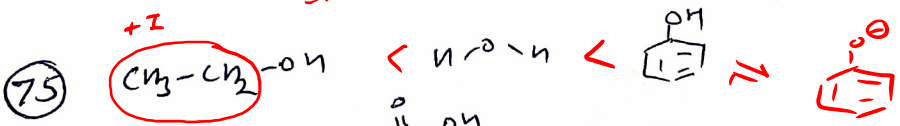
I > III > II > IV

(74)



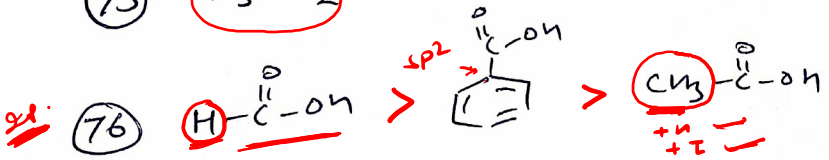
II > II > I > IV

(75)

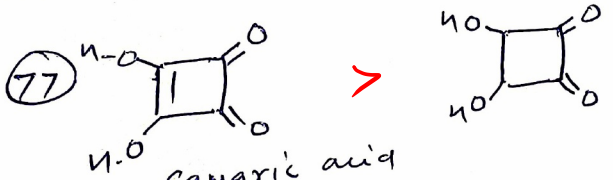


III > I > I

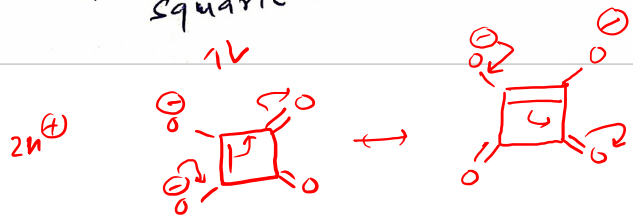
(76)



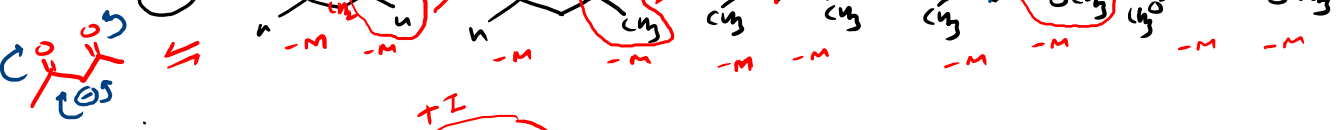
(77)



I > II



(78) Active methylene



(79) H_2O 3.5 > CH_3-CH_2-OH 3.5 > $R-C\equiv CH$ 3.25 > NH_3 3

+I

I > II > III > IV

3.25 3

EN

(80) $CH_4 < NH_3 < H-O-H < H-F$ EN

(81) $HF < HCl < H-Br < HI$ size E

(82) $H-Cl$ > $R-\overset{O}{\overset{\|}{C}}-OH$ > H_2CO_3 >

1V

$Cl^{\ominus}$

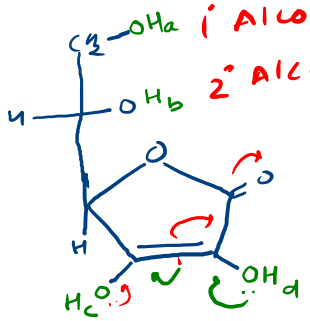
$R-\overset{O}{\overset{\|}{C}}-O^{\ominus}$ > $HO-\overset{O}{\overset{\|}{C}}-O^{\ominus}$

+M

(83) $CH_3-CH_3 < CH_3-\overset{O}{\overset{\|}{C}}-H < CH_3-CN < CH_3-NO_2$

-M -M -M

(84)



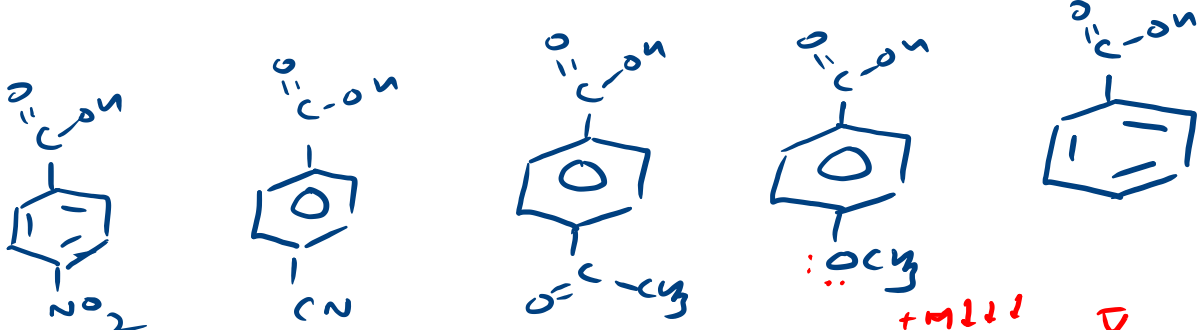
1° Alco.

2° Alco.

$H_c > H_d > H_a > H_b$

1° 2° Alco.

(85)

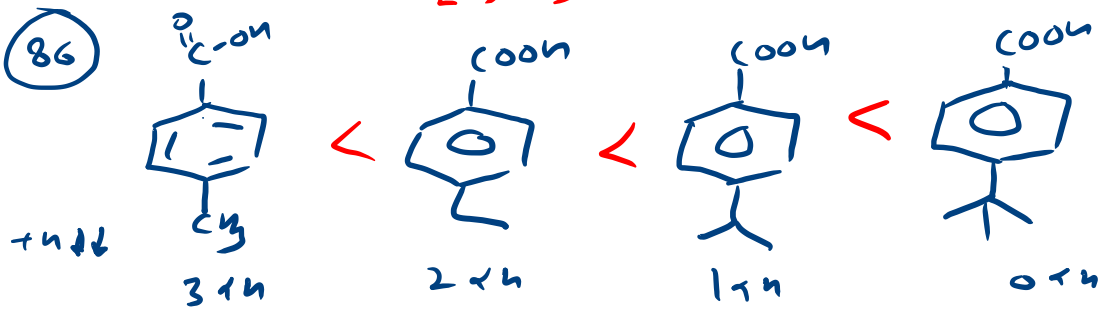


I II III IV V

+M III

I > II > III > V > IV .

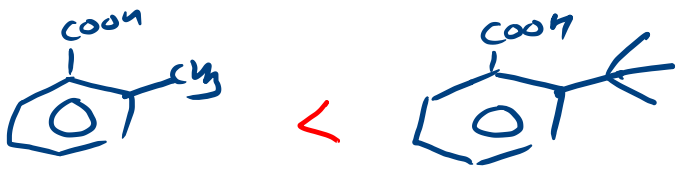
(86)



+M III

3 +M 2 +M 1 +M 0 +M

(87)

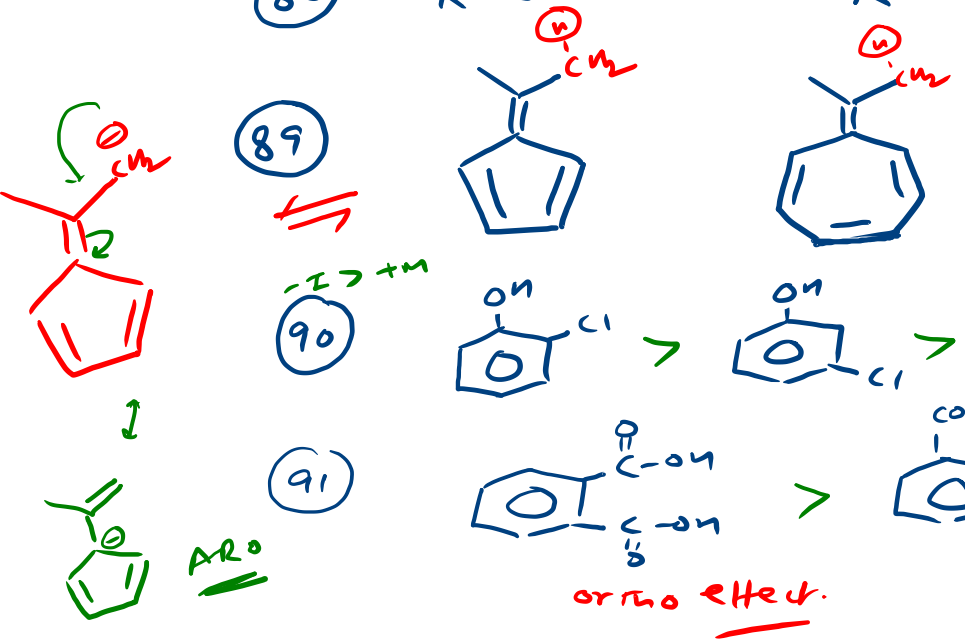


Bulky. effect SIR or ortho

(88)

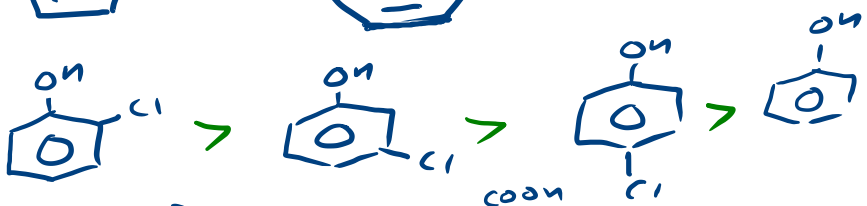
3 or type RS.

(89)



-I > +M

(90)



(91)

I > II .

ortho effect.

Basic strength / Basic Nature / Basicity

The tendency to donate the pair of electrons to $H^{\oplus}$ is known as Basicity (or Basic strength)

Basic strength $\propto$ ēle density on an atom

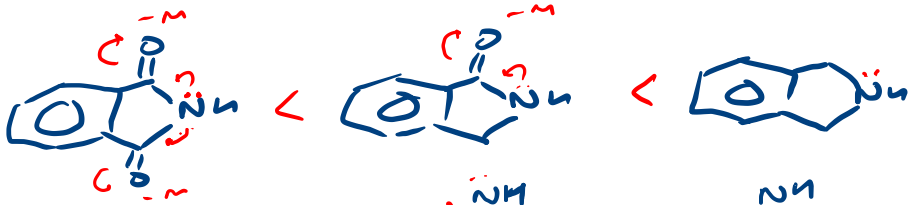
Basic strength $\propto \frac{1}{\text{Stability}}$

Basic strength $\propto K_b \propto \frac{1}{pK_b}$

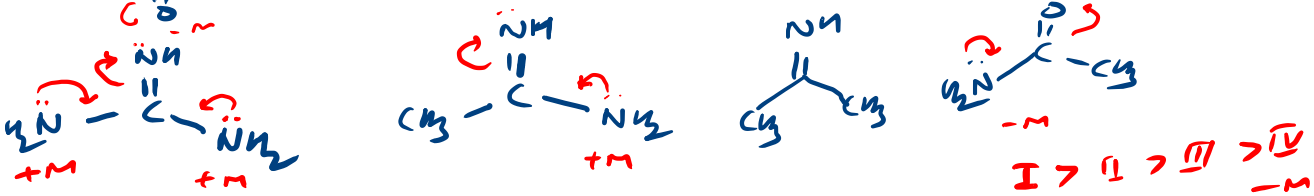
compare Basic strength

(92) $CH_2=CH-\ddot{N}H_2 < CH_3-CH_2-\ddot{N}H_2$

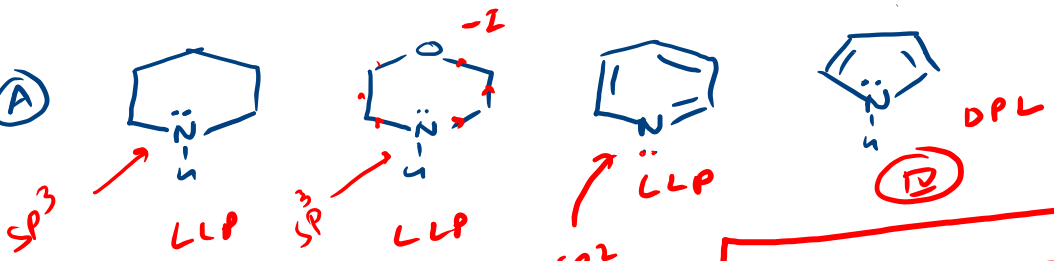
(92 A)



(93)



(93 A)



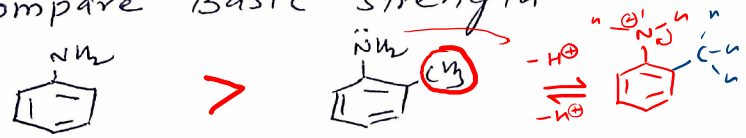
I > II > III > IV

ORTHO EFFECT FOR BASICITY (SIP)

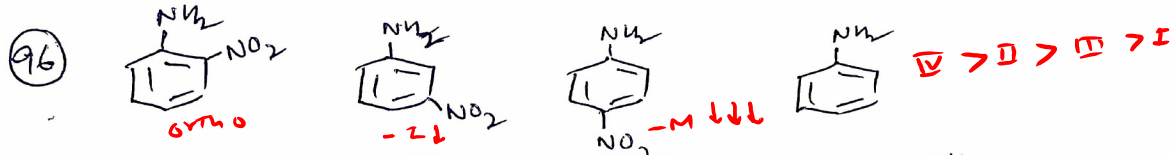
Break

Ortho substituted aniline (substituted by bulky group is less Basic than it's corresponding meta or para substituted aniline, due to SIP (steric inhibition of protonation), which is known as ortho effect for Basicity.

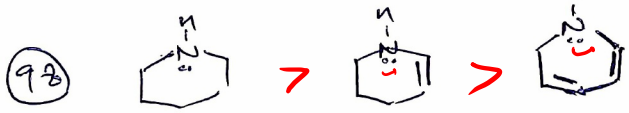
Compare Basic strength

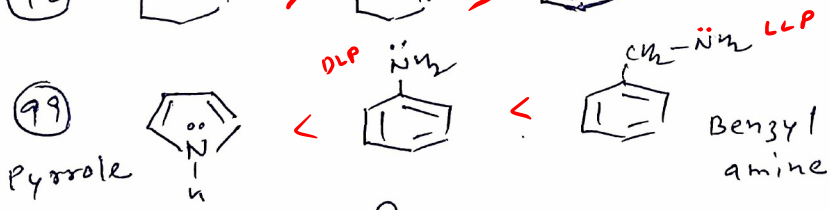
(94) 

(95)

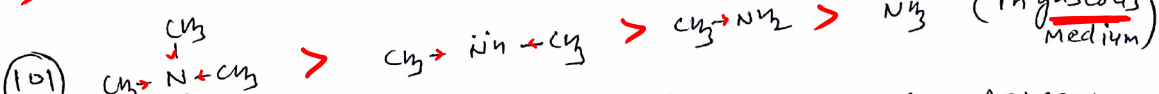
(96) 

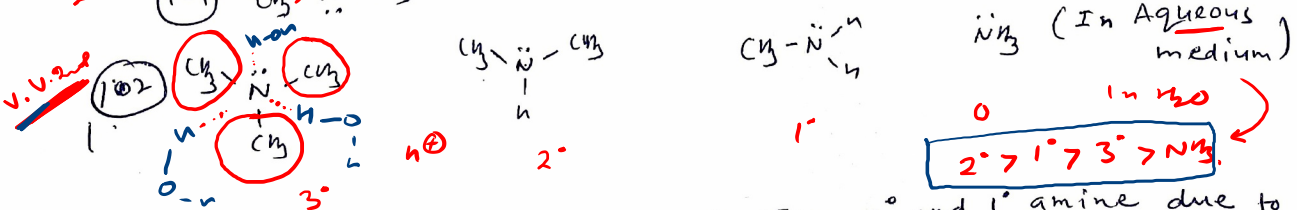
(97)

(98) 

(99) 

(100)

(101) 

(102) 

Note: 3° amine $(CH_3)_3N$ is less basic than 2° and 1° amine due to steric hindrance and H-bonding with H-O-H

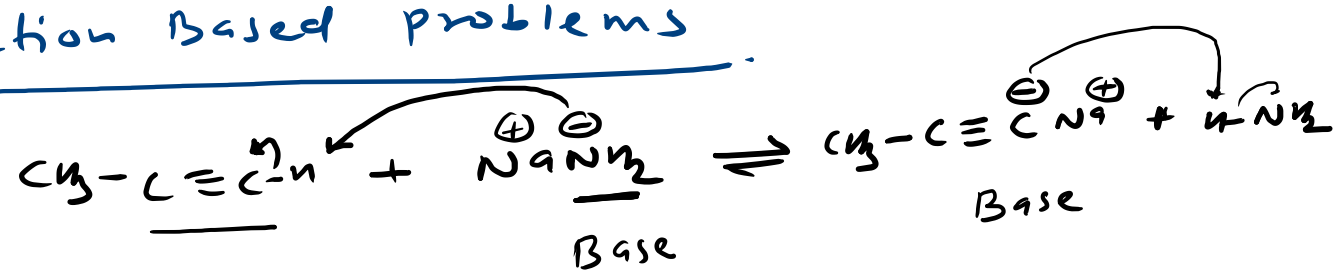
(102) $2° > 1° > 3° > NH_3$

2 1 3 0
Methyl

of Ethyl $2° > 3° > 1° > NH_3$ in aq medium.

is taken in place of Methyl

Reaction Based problems

(103) 

$\ominus NH_2$ < $CH_3-C\equiv C^{\ominus}$ stability

I

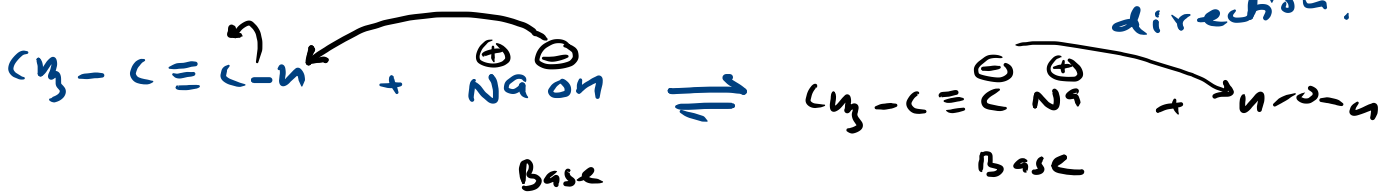
3

II

3.25

EN

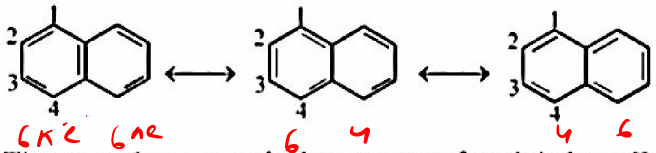
This reaction is favoured in fwd direction.

(104) 

$\ominus OH$ > $CH_3-C\equiv C^{\ominus}$

3.5 3.25

This reaction favoured in backward direction

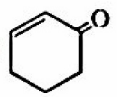
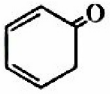
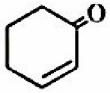
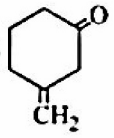
Q.37 

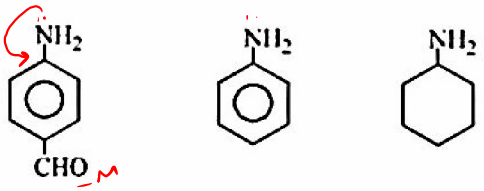
These are three canonical structures of naphthalene. Examine them and find correct statement among the following :

(A) All C – C bonds are of same length (B) C1 – C2 bond is shorter than C2 – C3 bond.

(C) C1 – C2 bond is longer than C2 – C3 bond (D) None

Q.38 Which of the following has longest C – O bond :

(A)  (B)  (C)  (D) 

Q.39 

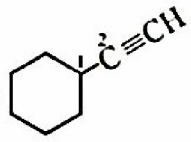
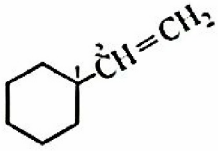
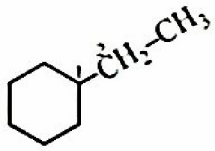
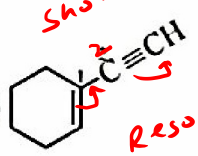
I II III IV

$CH_2 = NH$

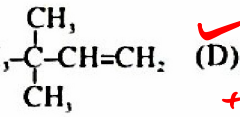
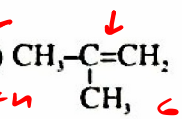
Among these compounds, the correct order of C – N bond lengths is :

(A) IV > I > II > III (B) III > I > II > IV (C) III > II > I > IV (D) III > I > IV > II

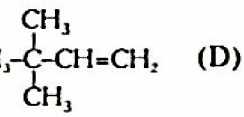
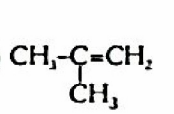
Q.40 C1 – C2 bond is shortest in

(A)  (B)  (C)  (D) 

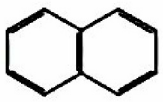
Q.41 Which of the following molecule has longest C=C bond length ?

(A) $CH_2=C=CH_2$ (B) $CH_3-CH=CH_2$ (C)  (D) 

Q.42 Which of the following molecule has shortest C=C bond length ?

(A) $CH_2=C=CH_2$ (B) $CH_3-CH=CH_2$ (C)  (D) 

Q.43 C—C and C=C bond lengths are unequal in :

(A) Benzene (B) 1,3-buta-di-ene (C) 1,3-cyclohexa-di-ene (D) 

Q.44 Among the following molecules, the correct order of C – C bond length is (C_6H_6 is benzene)

(A) $C_2H_6 > C_2H_4 > C_6H_6 > C_2H_2$ (B) $C_2H_6 > C_6H_6 > C_2H_4 > C_2H_2$

(C) $C_2H_4 > C_2H_6 > C_2H_2 > C_6H_6$ (D) $C_2H_6 > C_2H_4 > C_2H_2 > C_6H_6$

module\1000 BA\Other comp\VEE\Advanced\Nurture\Physics\Module01\Eng\ELECTRONIC DISPLACEMENT EFFECT

Q.45 $CH_3O - CH = CH - NO_2$ I

$CH_2 = CH - NO_2$ II

$CH_2 = CH - Cl$ III

$CH_2 = CH_2$ IV

Which of the following is the correct order of C – C bond lengths among these compounds :

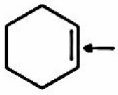
(A) I > II > III > IV (B) IV > III > II > I (C) I > III > II > IV (D) II > III > I > IV

Q.46 Which of the following is (are) the correct order of bond lengths :

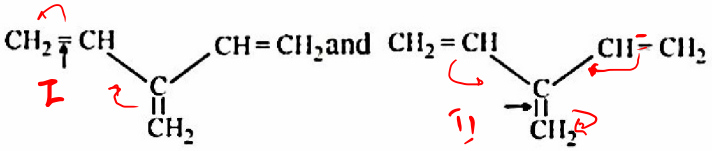
(A) C – C > C = C > C ≡ C > C ≡ N (B) C = N > C = O > C = C

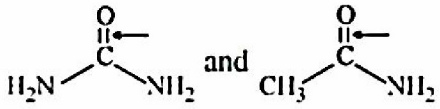
(C) C = C > C = N > C = O (D) C – C > C = C > C ≡ C > C – H

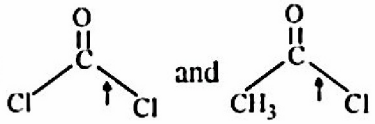
Q.47 In which of the following pairs, indicated bond having less bond dissociation energy :

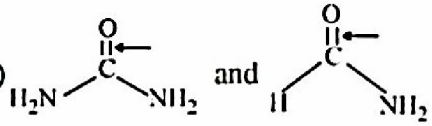
(a)  and $CH_2 \equiv CH_2$

(b) $CH_3 - C \equiv CH$ and $HC \equiv CH$

(c) 

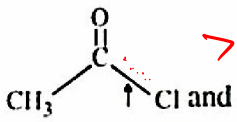
(d) 

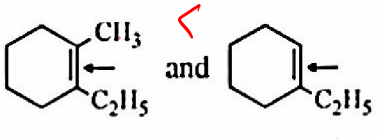
(e) 

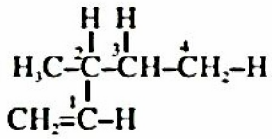
(f) 

Q.48 In which of the following pairs, indicated bond is of greater strength :

(a) $CH_3 - CH_2 - Br$ and $CH_3 - CH_2 - Cl$ (b) $CH_3 - CH = CH - Br$ and $CH_3 - \underset{\underset{Br}{|}}{CH} - CH_3$

(c)  and $CH_3 - CH_2 - Cl$ (d) $CH_2 = CH - CH = CH_2$ and $CH_2 = CH_2 - CH_2 - CH_3$

(e) $CH_2 = CH - CH = CH_2$ and $CH_2 = CH - NO_2$ (f) 

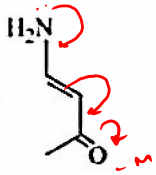
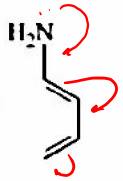
Q.49. 

the correct order of bond dissociation energy (provided bond undergoes homolytic cleavage):

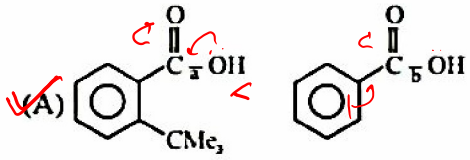
(A) C^2–H > C^3– H > C^4 –H > C^1 – H (B) C^2–H > C^3– H > C^1 –H > C^4 – H

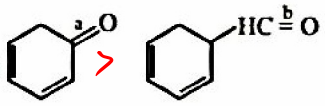
(C) C^1–H > C^4– H > C^2 –H > C^3 – H (D) C^1–H > C^4– H > C^3 –H > C^2 – H

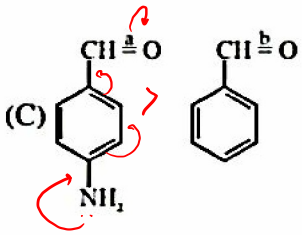
Q.50 Compare the C–N bond-length in the following species:

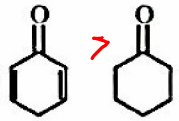
(A)  (B)  (C)

Q.51 In which case, C – O bond length is shorter for Ist compound :

(A) 

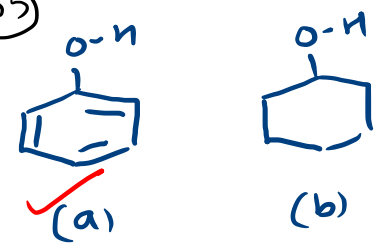
(B) 

(C) 

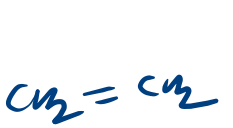
(D) 

prob Identify the compound(s) which react with Dil NaOH, (or dissolve with Dil NaOH.)

(105)



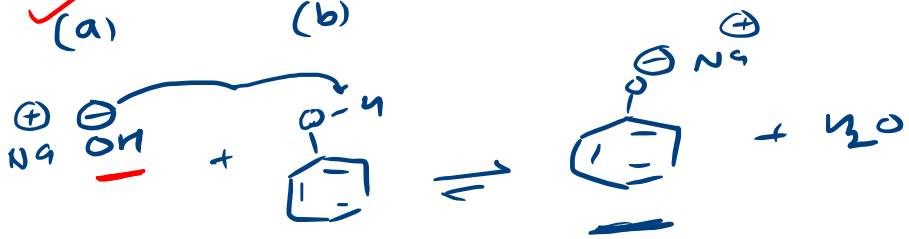
(a) ✓ (b)

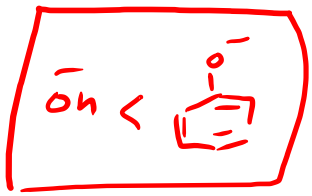


(c)

(d)

(e)



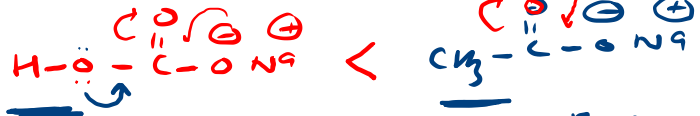


Stability

Reactions with $NaHCO_3$

(106)

S. A S. Base W. Base W.A



+M Base SB Stability +H Base WB

Reaction favoured in Fwd direction

S.A + S.B → W.B + W.A

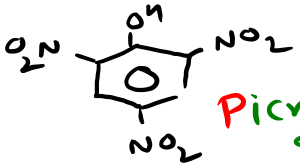
VIP The compound which is more acidic than H_2CO_3 (or para nitrophol) will release CO_2 with $NaHCO_3$.

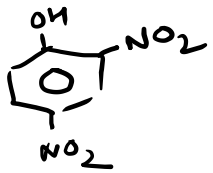
prob. (107) Identify the compound(s) which release CO_2 with $NaHCO_3$.

(a) All $R-\overset{O}{\overset{\|}{C}}-OH$

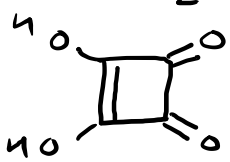
(b) All $R-SO_3H$

(c) All

(d)  Picric acid

(e) 

(f)

(g)  Squaric acid

$NaHCO_3$ → $\uparrow CO_2 + H_2O$

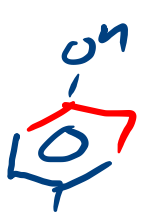
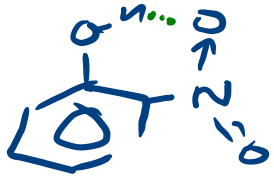
(h)

(i) Carbolic acid

(j) $R-O-H$

$NaHCO_3$ → ~~CO_2~~

They do not release CO_2 with $NaHCO_3$

 > 

NO_2 Acidic nature

Reaction with Na

108

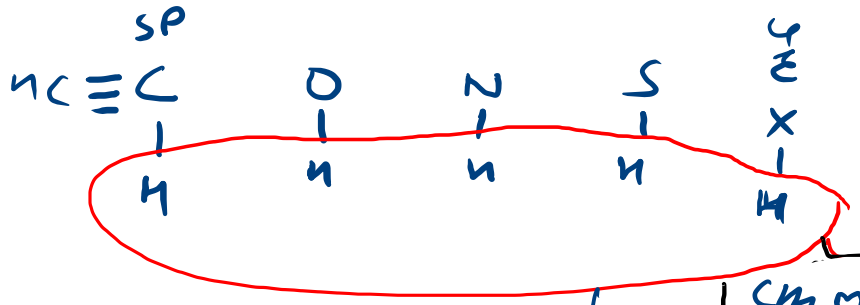
(a) $R-\overset{O}{\overset{\|}{C}}-O-H \xrightarrow[\text{or } K\ \Delta]{Na\ \Delta} R-\overset{O}{\overset{\|}{C}}-O^{\ominus}Na^{\oplus} + H\cdot$

$H\cdot \downarrow H\cdot \rightarrow \frac{1}{2} H_2$

(b) $R-\overset{O}{\overset{\|}{C}}-O-H \xrightarrow{CH_3MgBr} R-\overset{O}{\overset{\|}{C}}-O^{\ominus}MgBr^{\oplus} + CH_4\uparrow$

(c) $R-\overset{O}{\overset{\|}{C}}-O-H \xrightarrow{NaNH_2} R-\overset{O}{\overset{\|}{C}}-O^{\ominus}Na^{\oplus} + NH_3$

109



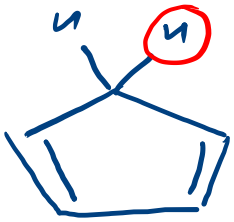
Na → (a) Aromatic

Na → $\frac{1}{2} H_2$

CH_3MgBr → CH_4

$NaNH_2$ → NH_3

110



Na/K or → $\frac{1}{2} H_2\uparrow$ + $Na^{\oplus}$ Aromatic

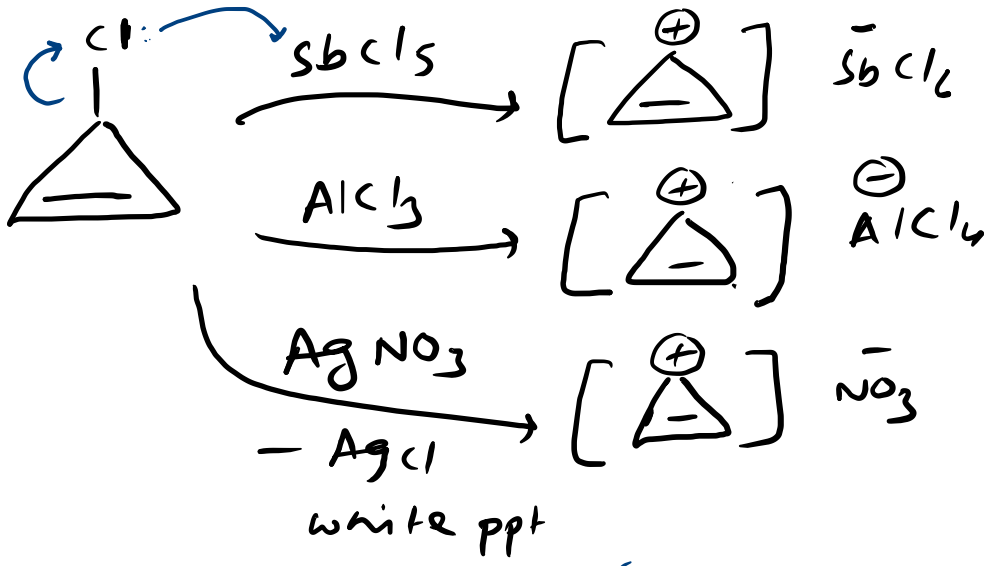
$NaH/KH \rightarrow H_2\uparrow$ 〃

$CH_3MgBr \rightarrow CH_4$ + $MgBr^{\oplus}$

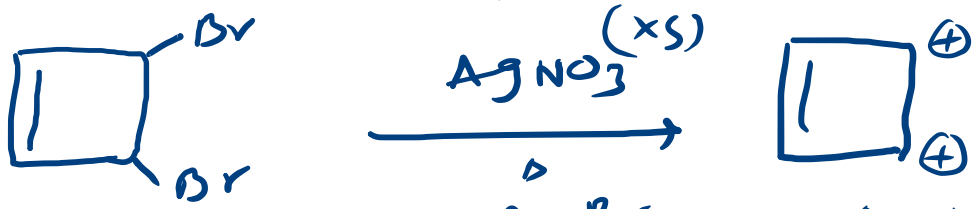
$NaNH_2 \rightarrow NH_3$ $Na^{\oplus}$

$R^{\ominus}Li^{\oplus}$ → $R-H$ $Li^{\oplus}$

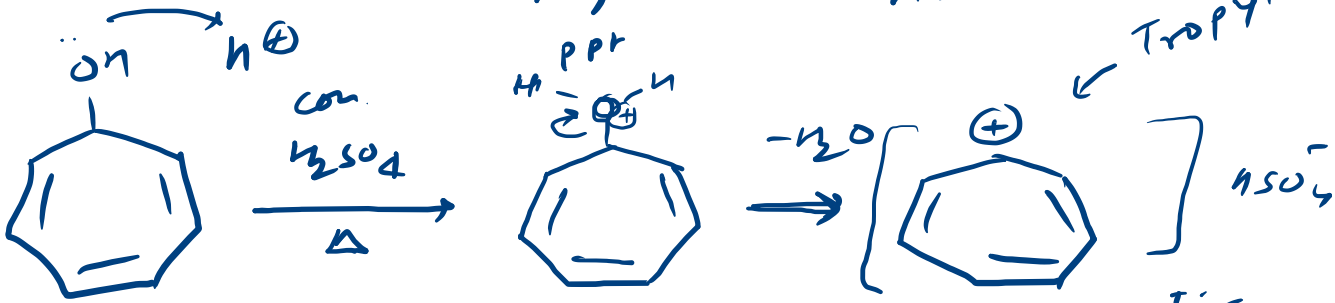
(111)



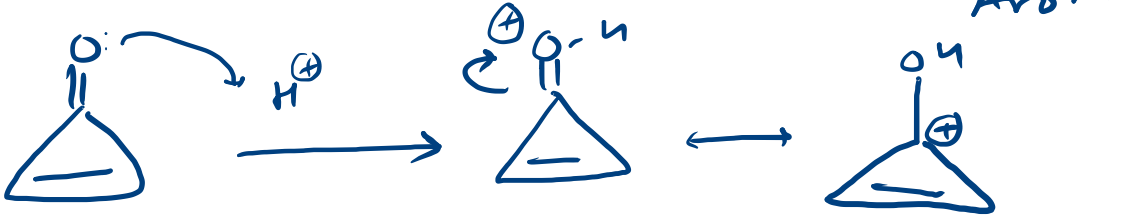
(112)



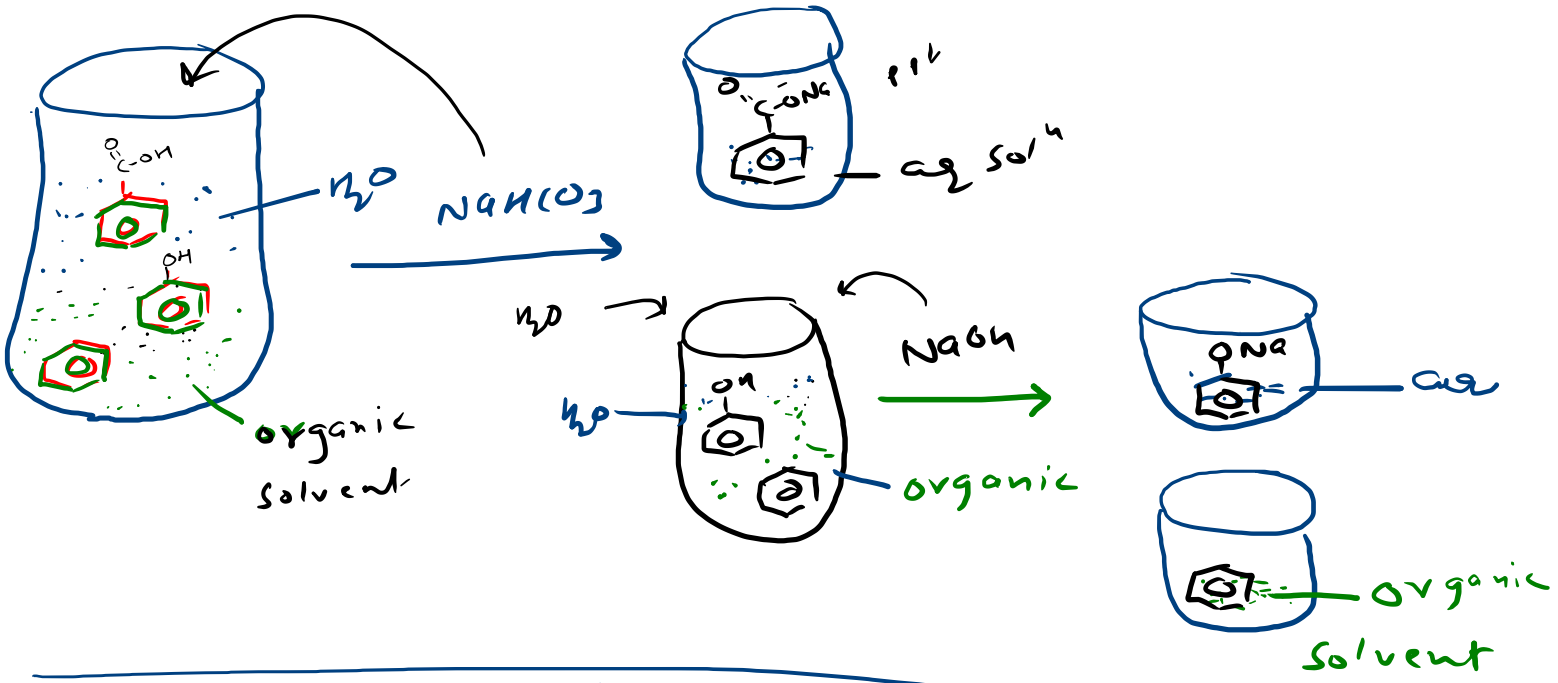
(113)



(114)



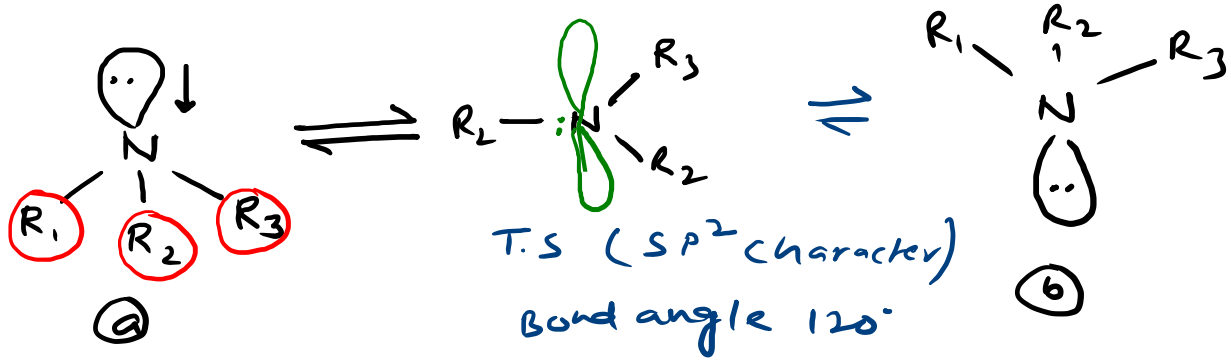
Separation of organic compounds from their mixture

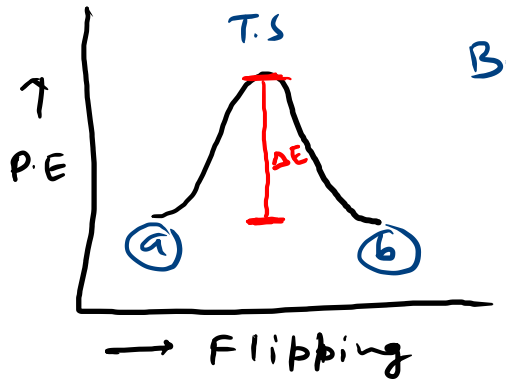


AMINE INVERSION (Flipping or umbrella flipping)

Inversion of Lone pair of sp^3 Nitrogen is known as amine inversion (flipping)

116
LLP.

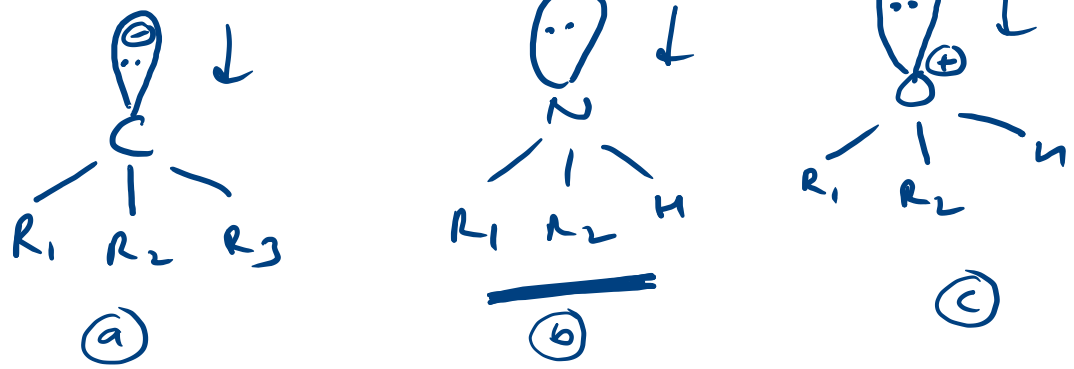




Bulky group easily undergoes flipping

Note only 2nd period elements show flipping

eg (117)

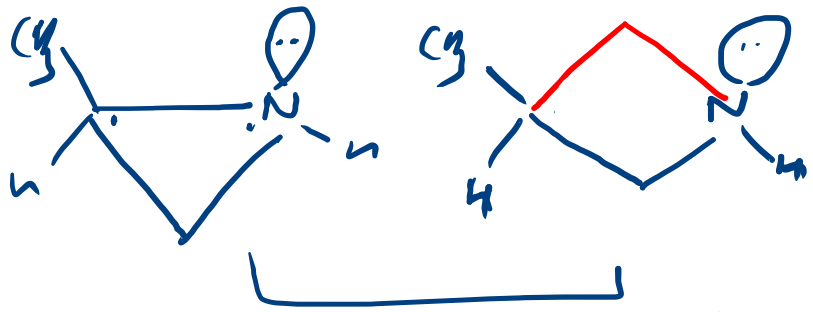


(a) (b) (c)

sp^3 atom of 3rd, 4th, 5th ... period elements do not show flipping (Reason Screening effect)

Note sp^3 atom in 3 membered, 4 membered ring or bridge head atom do not show flipping

(118)



Not show flipping

Reason angle strain.

(T.S of 120° not possible)

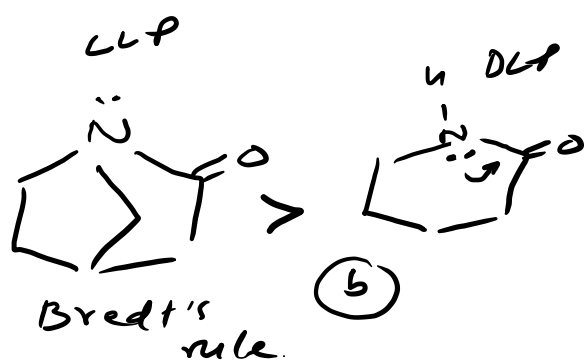
(119) Basic strength

(a) (b)

(a) No flipping

(b) Flipping

(120) compare Basicity



(a) (b)

Bredt's rule

The End

ISOMERISM

Different compounds having same molecular formula (MF), but different in atleast one physical or chemical property are known as isomers and their relationship is known as isomerism.

✓ Isomers have same Molecular formula (MF)

Isomers have same molecular weight (MW)

Isomers have same vapour density (V.D)

Isomers have same DBE / DOU / IHD etc.

✓ Isomers are different in complete IUPAC name (i.e including stereo)

Isomers are different in atleast ~~one~~ one physical or chemical property.

Note: Real isomers have same MF but different in complete IUPAC name and can be separated from each other at room temperature.

Note: diff Conformations (or conformers) are not real isomers, because they interconvert into each other very fast at Room temp. so it is difficult to separate them at RT.

Index of Hydrogen deficiency (IHD) or Degree of unsaturation (DOU) or Double bond equivalent (DBE)

No of H_2 molecules required to convert the given compound into open chain saturated neutral compound is known as IHD or DOU or DBE

Case (1) If compound is given by M.F

(a) FOR Hydrocarbons

(1) $C_4H_8 \xrightarrow[\text{IHD/DOU}]{1\ H_2} C_4H_{10}$ (C_nH_{2n+2})

(2) C_6H_{14} IHD = 0

(3) $C_6H_{12} \xrightarrow[\text{DBE}]{1\ H_2} C_6H_{14}$

(4) $C_6H_6 \xrightarrow[\text{DBE}]{4\ H_2} C_6H_{14}$

(5) $C_{100}H_{192} \xrightarrow{5\ H_2} C_{100}H_{202}$

(b) If halogen is present in MF

$-X$
$+H$

(6) $C_6H_5Cl \xrightarrow[+H]{-Cl} C_6H_6 \xrightarrow{4H_2} C_6H_{14}$

(7) $C_4H_4Br,F \xrightarrow[+2H]{-Br,-F} C_4H_6 \xrightarrow{2H_2} C_4H_{10}$

(c) If oxygen is present in MF

Ignore Oxygen

(8) $C_4H_8O \xrightarrow{-O} C_4H_8 \xrightarrow[DBE=1]{1H_2} C_4H_{10}$

DBE 1

(9) $C_6H_6O \xrightarrow{-O} C_6H_6 \xrightarrow[DBE]{4H_2} C_6H_{14}$

DBE = 4

(c) If Nitrogen is present

$-N$
$-H$

(10) $C_5H_{11}N \xrightarrow[-H]{-N} C_5H_{10} \xrightarrow[DBE=1]{1H_2} C_5H_{12}$

Case ② If halide, oxygen, Nitrogen are present

$-X \quad +H$

$-O$

$-N \quad -H$

(11) $C_6H_6NOF \xrightarrow[-O]{-F \; +H} C_6H_6 \xrightarrow{4H_2} C_6H_{14}$

DBE = 4

$+H \quad +H$ Halide

$O \quad O$ Oxygen

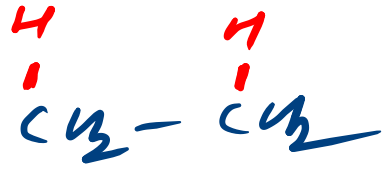
$-H \quad N$ Nitrogen

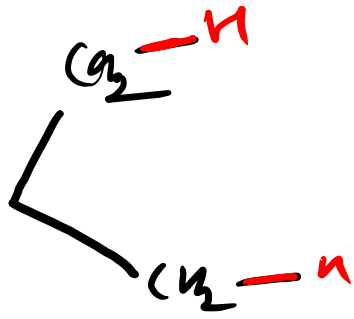
negative

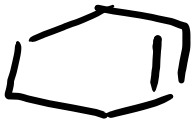
Case (2) If compound is given by Structure

1 π bond = 1 DBE or 1 IHD

1 Ring = 1 DBE

eg (12) $CH_2=CH_2 \xrightarrow{1H_2}$ 

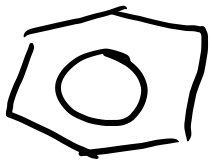
(13) $\xrightarrow{1H_2}$ 

(14) 

$\pi = 1$

$R = 1$

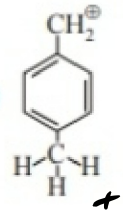
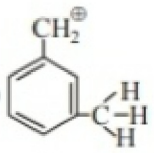
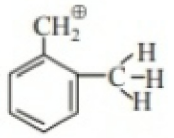
DOU 2

(15)  ⟶ DBE

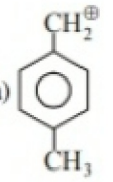
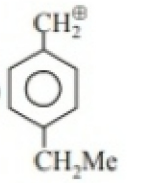
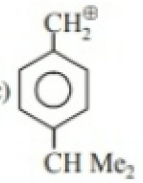
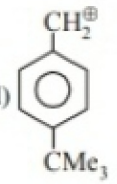
$\pi = 3$

$R = 1$

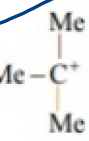
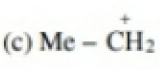
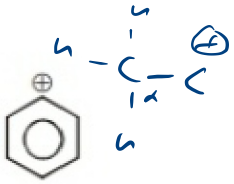
4.

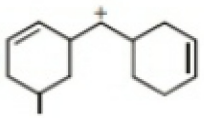
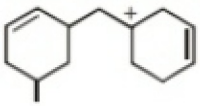
(xi) (a)  (b)  (c) 

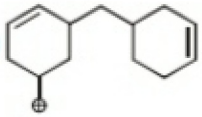
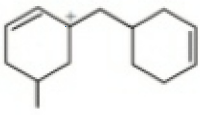
c > a > b.

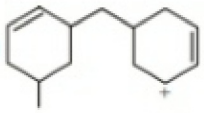
(xii) (a)  (b)  (c)  (d) 

b > a > c > d

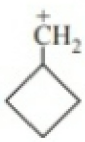
(xiii) (a)  (b) (c)  (d) 

(xiv)(a)  (b) 

(c)  (d) 

(e) 

d > e > b > a > c

(xv) (a) (b)  (c)

a > c > b

(xvi) (a) (b) (c)

b > c > a

(16)

RH

π =

R =

(17)

Anthracene

$\pi = 7$

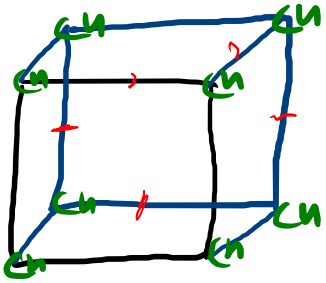
$R = 3$

10

Case (3)

DOU in 3D structure

(18)



DBE = 5

Cubane

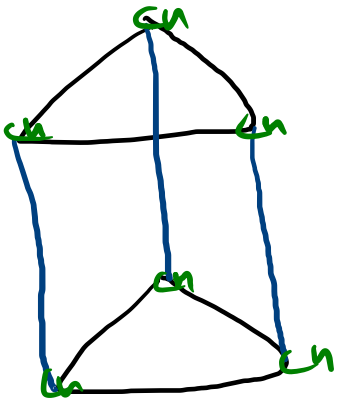
DOU = R - 1

6 - 1 = 5

$C_8H_8 \xrightarrow{5H_2} C_8H_{18}$

IHD

(19)



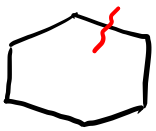
Prismane

DOU = 5 - 1 = 4

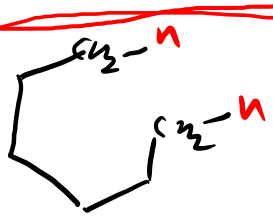
$C_6H_6 \xrightarrow{4H_2} C_6H_{14}$

IHD

(20)



$\xrightarrow{+H_2}$



calculation of DBE by formula

$$DOU = \frac{(2n_1 + 2) - n_2}{2}$$

n_1 = No. of C in given MF

n_2 = No. of H in given MF

eg. (21) C_6H_6

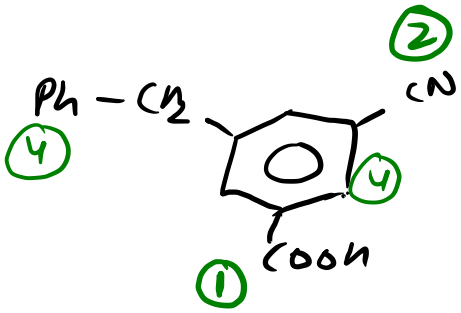
$$DOU = \frac{(2 \times 6 + 2) - 6}{2}$$

$$= \frac{14 - 6}{2} = \frac{8}{2} = 4$$

$C_6H_6 \xrightarrow{4H_2} C_6H_{14}$

calculation of DBE

(22)



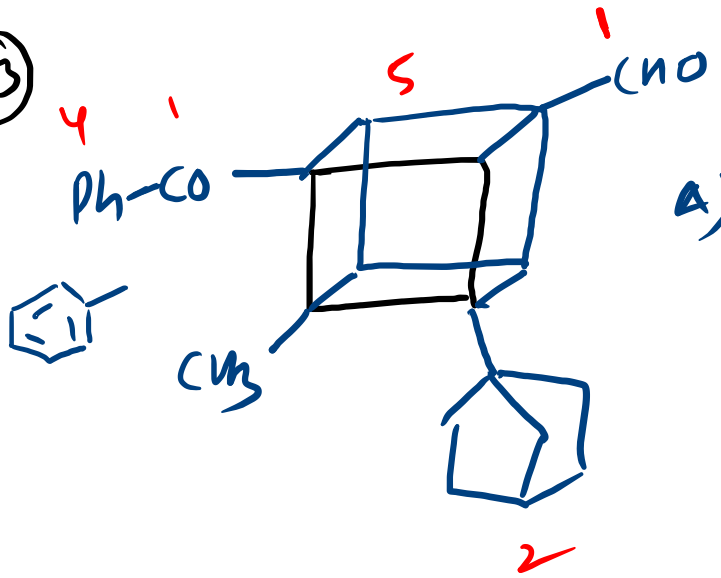
$-C \equiv N$

$\Rightarrow$

$4 + 4 + 2 + 1 = 11$

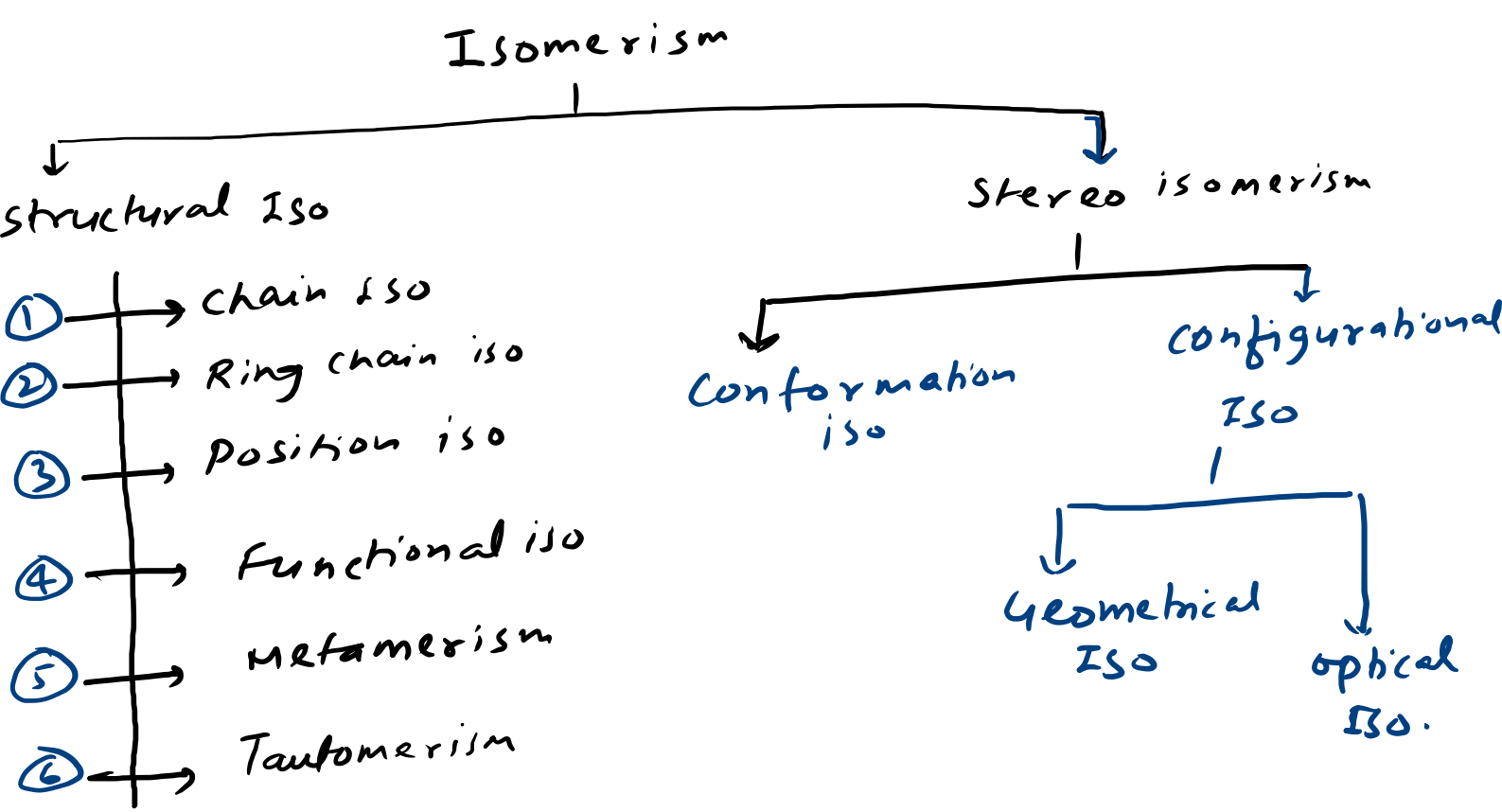
$O=\overset{1}{C}-OH$

(23)



A) 12 B) 13 C) 14 D) 15

4 + 5 + 2 + 1 + 1 = 13



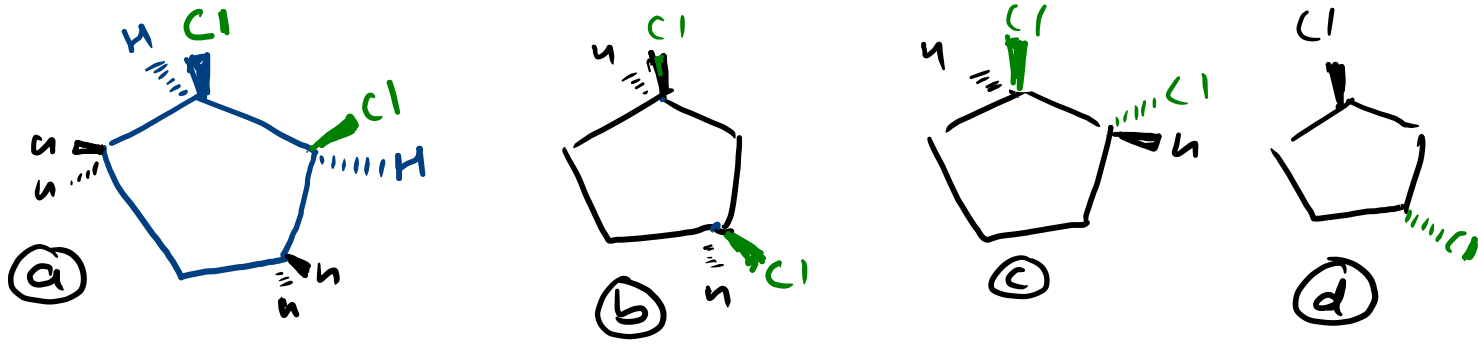
Structural Isomers.

* Same MF
* different in structure (different in pattern of connectivity of atoms to each other)

Stereo isomers

* Same MF
* Same structure
* But different in orientation in 3D (in space)

(24)



wedge (i.e above the plane)

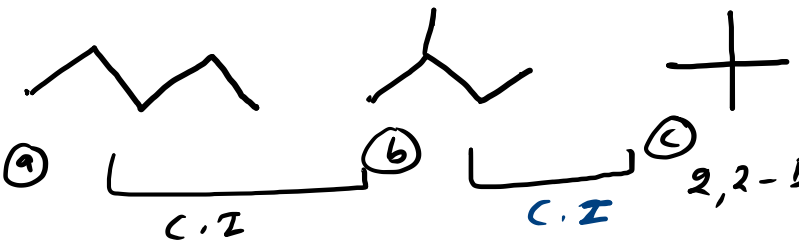
dash (i.e below the plane)

Structural iso ⇒ a & b , a & d, (Position iso) b & c , d & c,

Stereo isomers ⇒ a & c , b & d (G.Iso)

① Chain isomers :- Different compound having same MF, but different in carbon chain.

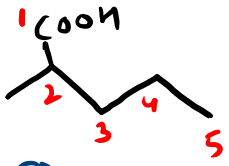
eg. (25) C_5H_{12}



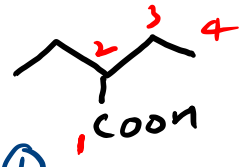
2,2-Dimethyl propane

a & c => Chain Iso

(26)



(a)



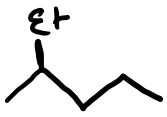
(b)

2-Ethyl butanoic acid

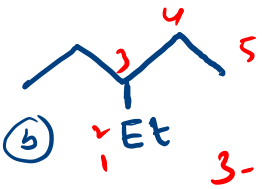
2-Methyl pentanoic acid

a & b => Chain iso.

(27)



(a)



(b)

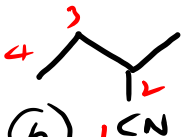
3-Ethyl pentane

3-Methyl hexane

C.I.

(28)

(a)



(b)

pentane nitrile

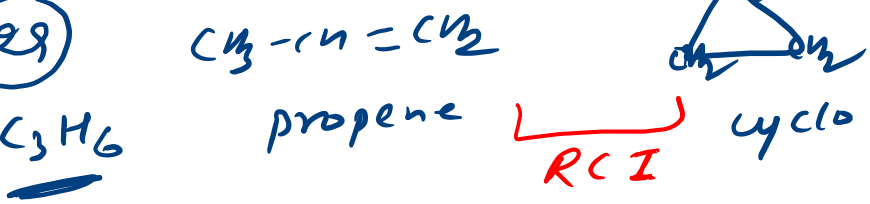
C.I

2-methyl butane nitrile

② Ring chain isomers :- (RCI)

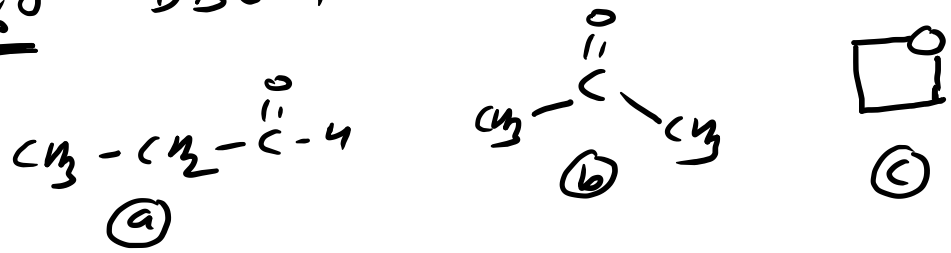
different compounds having same MF, but different in open & close chain structures are know as RCI.

ex. (29) C_3H_6



(Note : Atleast 1 DBE must be present in compounds to show RCI)

(30) C_3H_6O DBE = 1

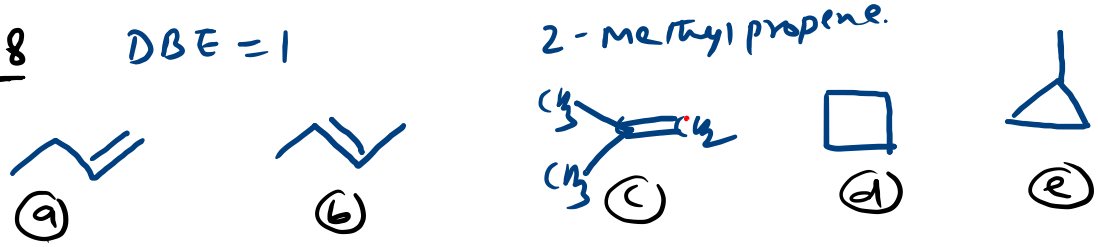


a & c ⇒ RCI > F.I

b & c ⇒ RCI > F.I

(31) C_4H_8 DBE = 1

2-methyl propene.



Chain isomers ⇒ (ac), (bc), (de)

Ring chain isomers ⇒ (ad), (ae), (bd), (be), (cd), (ce)

position isomers ⇒ a & b

(3) Position isomers (P.I)

Different compounds having same MF but different in position of Functional group/ Multiple bond or Substitution group with same carbon chain are known as P.I.

eg (32)

butan-1-ol

(a)

butan-2-ol

(b)

2-methyl propan-2-ol

(c)

a & b ⇒ P.I

a & c ⇒ C.I

b & c ⇒ C.I

(33) C_5H_{10}

(a)

(b)

(c)

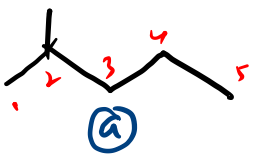
(d)

(a) & b = P.I

a & d ⇒ RCI

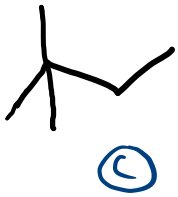
a & c ⇒ C.I

(34)



(a)

(b)



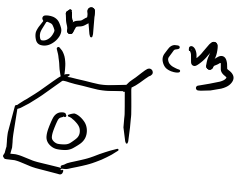
(c)

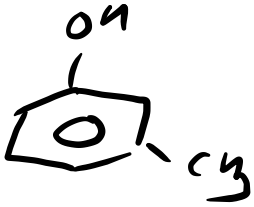
a & b ⇒ P.I

a & c ⇒ C.I

b & c ⇒ C.I

(35)





P.I

P.I

a & c

⇓

P.I

Functional isomers (F.I)

Different compounds having same MF, but different in Functional groups are known as functional isomers.

Functional groups which show Functional ISO.

(1) Alcohol ↔ Ether

$R-OH$ $R-O-R$

(2) Thio Alcohol ↔ thio Ether

$R-SH$ $R-S-R$

(3) Aldehyde ↔ ketone

$R-\overset{O}{\overset{\|}{C}}-H$ $R-\overset{O}{\overset{\|}{C}}-R$

(4) C.Acid ↔ Ester

$R-\overset{O}{\overset{\|}{C}}-O-H$ $R-\overset{O}{\overset{\|}{C}}-O-R$

(5) cyanide ↔ isocyanide

$R-CN$ $R-N\equiv C:$ / $-N\equiv\overset{\oplus}{C}:^{\ominus}$

$-NC$

(6) Nitro ↔ Nitrite

$R-N(=O)=O$ $R-O-N=O$

(7) 1°-Amine ↔ 2°-Amine ↔ 3°Amine

$R-NH_2$ $R-\underset{H}{N}-R$ $R-\underset{R}{\overset{|}{N}}-R$

(8) Diene ↔ Alkyne

C_4H_6

(9) Amide ↔ Oxime

$R-\overset{O}{\overset{\|}{C}}-NH_2$ $R-CH=N-OH$

⇅ Tautomers

(10) Alcoholic -OH ↔ Phenolic -OH

CH_2-OH C_7H_8O OH CH_3

(11) Amide ↔ Nitroso

$R-\overset{O}{\overset{\|}{C}}-NH_2$ $R-CH_2-N=O$

(12) Sulphonic acid ↔ Sulphonic ester

$R-\overset{O}{\overset{\|}{\underset{O}{\underset{\|}{S}}}}-OH$ $R-\overset{O}{\overset{\|}{\underset{O}{\underset{\|}{S}}}}-O-R$

(13) Sulphinic acid ↔ Sulphinic ester

$R-\overset{O}{\overset{\|}{S}}-OH$ $R-\overset{O}{\overset{\|}{S}}-O-R$

(36) CH_3-CH_2-OH (a) CH_3-O-CH_3 (b)

(37) $CH_3-CH_2-CH_2-NH_2$ (a) 1° amine $CH_3-CH_2-NH-CH_3$ (b) 2° $CH_3-N(CH_3)-CH_3$ (c) 3° amine

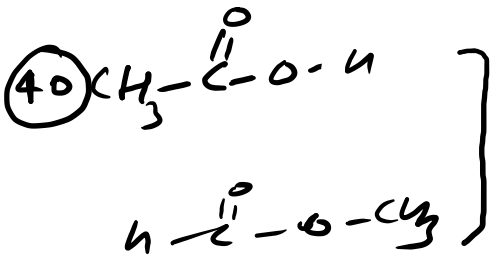
F.I F.I

(38) CH_3-NO_2 $CH_3-O-N=O$

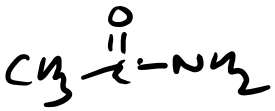
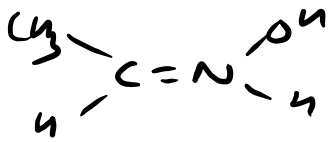
F.I

(39)

F.I

(40) 

F.I

(41)  

C_2H_5NO Amide Oxime

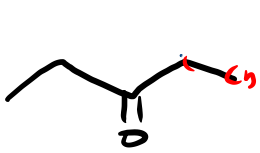
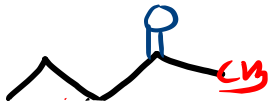
F.I

Radio city > Top > F > M > C > P

near

Ring chain Tauto Functional ↓ Meta ↓ chain ↓ position

---

eg (42)  

pentan-3-one pentan-2-one.

Metamers > P.I

Metamers

R — FG — R

Different compounds having same Molecular formula but different in Alkyl gr. (or substituted Alkyl group) attached to different side of polyvalent bridging functional group are known as Metamers.

Functional group which can show Metamerism

v. Imp

① Ether R—O—R ② thio Ether R—S—R

v. Imp ③ Ketone $R-\overset{O}{\overset{\|}{C}}-R$ v. Imp ④ Ester $R-\overset{O}{\overset{\|}{C}}-O-R$

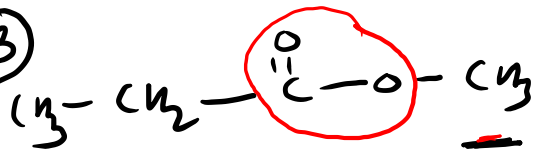
Imp ⑤ Acid anhydride $R-\overset{O}{\overset{\|}{C}}-O-\overset{O}{\overset{\|}{C}}-R$ ⑥ Sulphonic ester $R-\overset{O}{\overset{\|}{\underset{O}{\underset{\|}{S}}}}-O-R$

Imp ⑦ 2° Amine $R-\overset{H}{\underset{|}{N}}-R$ Imp ⑧ 3° Amine $R-\underset{R}{\overset{|}{N}}-R$

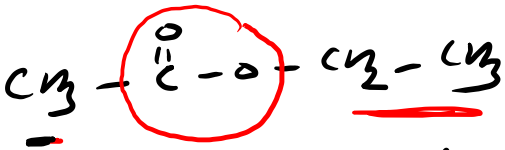
Imp ⑨ 2° Amide $R-\overset{O}{\overset{\|}{C}}-\underset{H}{N}-R$ ⑩ 3° Amide $R-\overset{O}{\overset{\|}{C}}-N\begin{matrix}R\\R\end{matrix}$

⑪ thio ketone $R-\overset{S}{\overset{\|}{C}}-R$ ⑫ Sulphinic ester $R-\overset{O}{\overset{\|}{S}}-O-R$

(43)



Methyl propanoate



Ethyl Ethanoate

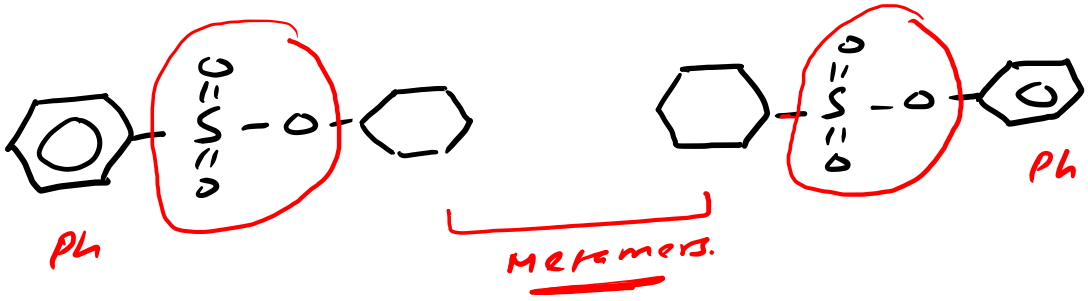
Note F. Group remain same in metamers

" " C. Z

" " P. I

" "

(44)



(45)

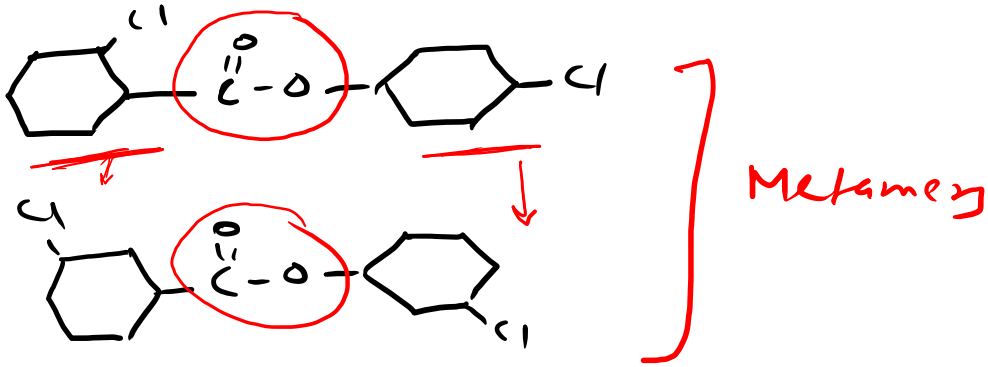
Ester

C. Acid

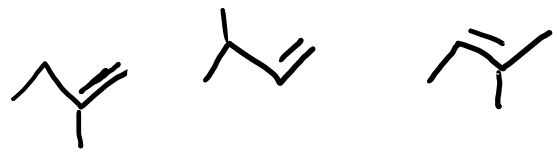
F. I

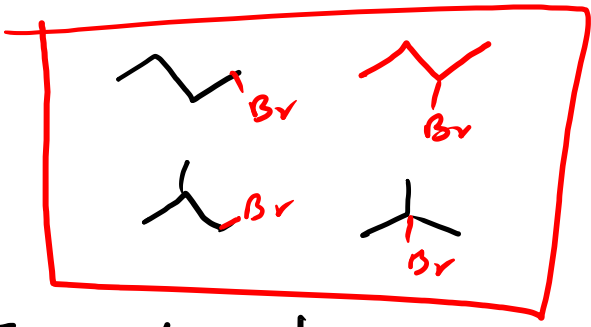
$C_2H_4O_2$

(46)



S.No	MF	DBE	C.I	RCI	PI	FI	Meta	Tauto	Total Structural Iso.	GI
47	C_4H_{10}	0	✓	×	×	×	×	×	2	×
48	C_5H_{12}	0	✓	×	×	×	×	×	3	×
49	C_4H_9Br	0	✓	×	✓	×	×	×	4	×
50	$C_4H_{10}O$	0	✓	×	✓	✓	✓	×	7	×
51	C_5H_{10}	1	✓	✓	✓	✓	×	×	10	✓
52	C_3H_6									
53.	C_3H_4									
54	$C_4H_{11}N$									
55	C_2H_4O									
56	C_3H_6O									





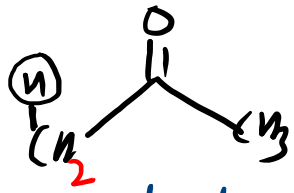
RCI > FI FRSR.

EAR

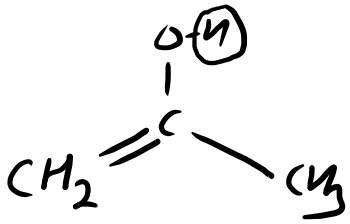
Tautomerism / Dynamic isomerism / Desmotropism

compounds having same MF, but different in structure due to oscillation of an atom (usually $H^{\oplus}$) b/w two polyvalent atoms are know as Tautomers, and this phenomenon is known as Tautomerism.

eg (57)

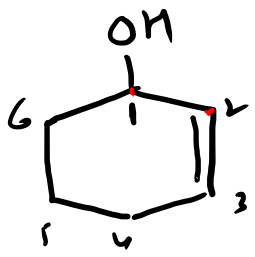


keto form



enol form

(58)



cyclo hex-2-enol ✓

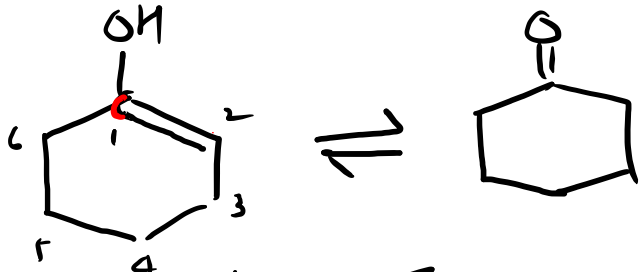
or

cyclo hex-2-en-1-ol ✓

] enol compound ✓

it is not enol form

(59)



✓ enol form ✓

✓ also enol compound cyclo hex-1-enol

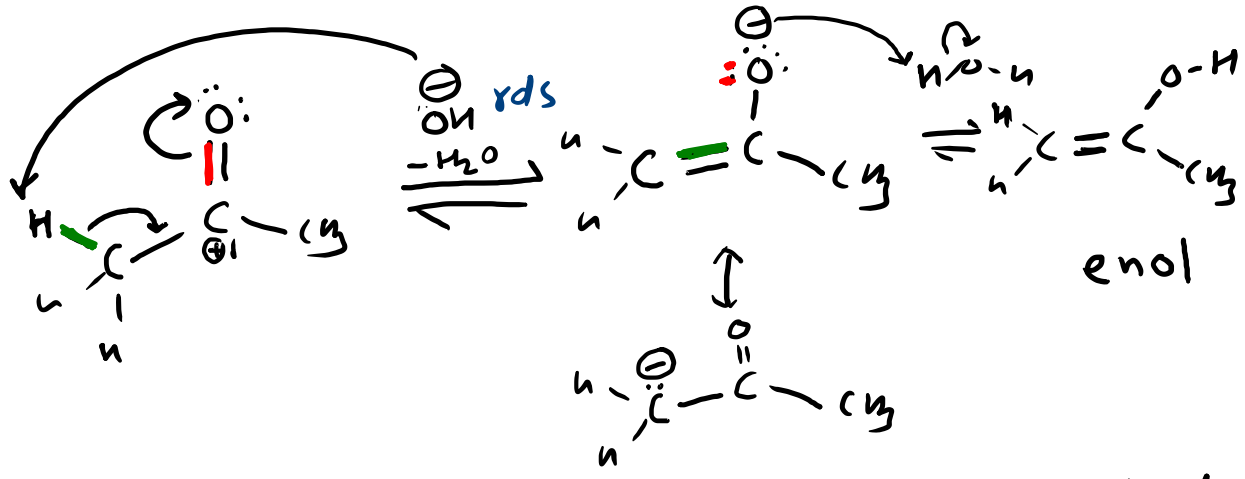
Note: Tautomers exist in dynamic equilibrium, so it is also called dynamic isomerism.

Note: Interconversion of Tautomers takes several weeks, so their interconversion is catalysed by Base or By acid.

Base catalysed Tautomerism

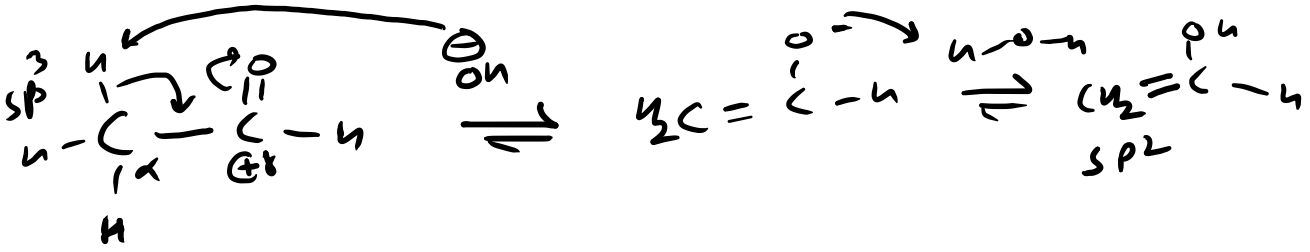
(60)

Dil NaOH or $\bar{O}H$ / HOH.



VVIP carbonyl compound having α-H or extended α-H attached to sp^3 carbon and converted into stable enol form, will show tautomersm

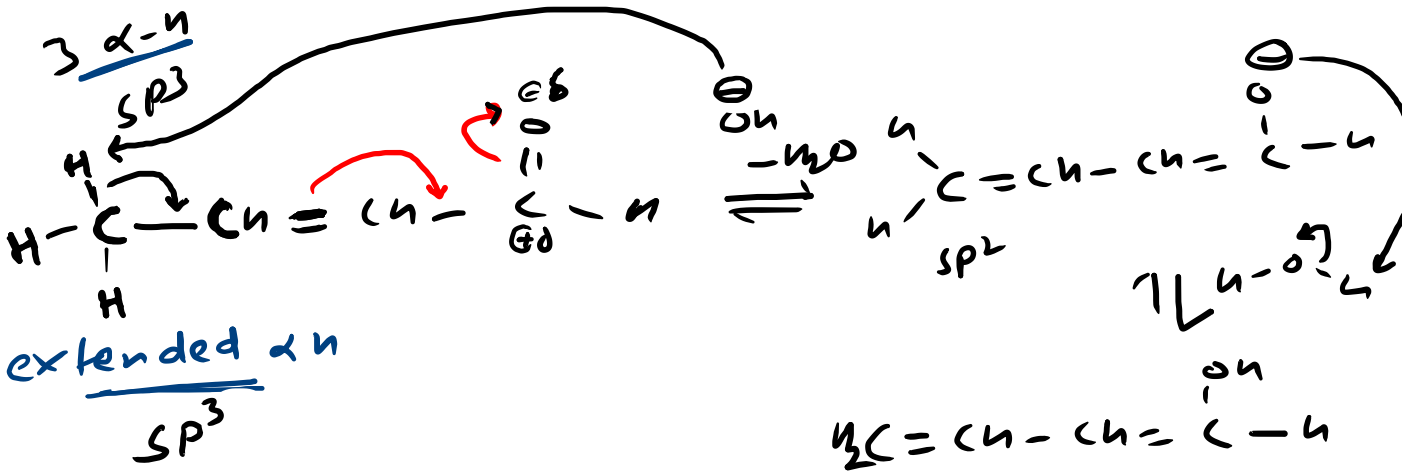
(61)



(62)

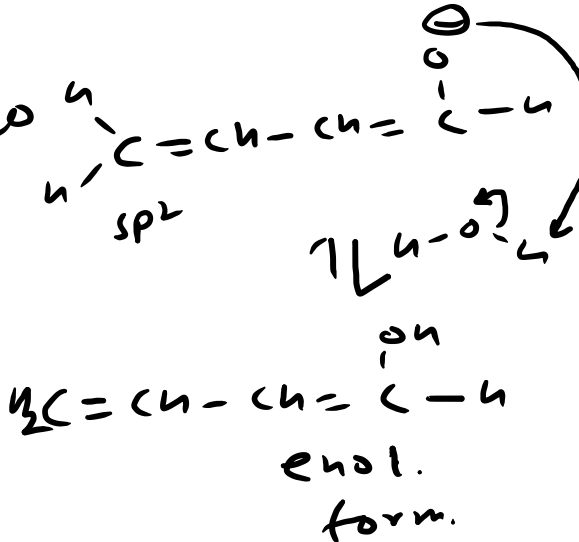
3 α-H

sp3



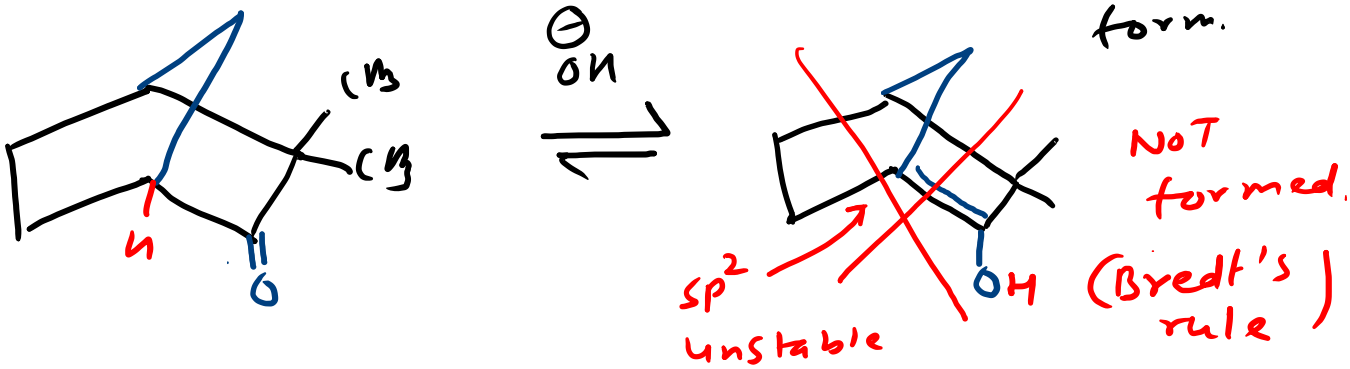
3 extended αH

sp3



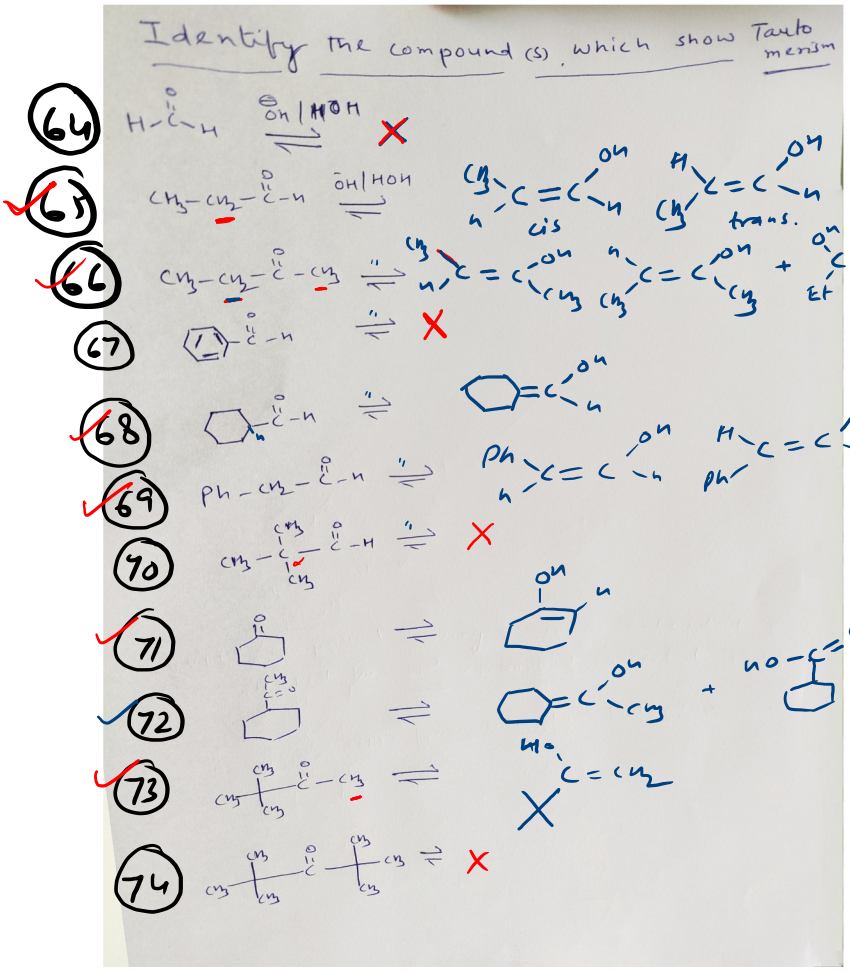
enol. form.

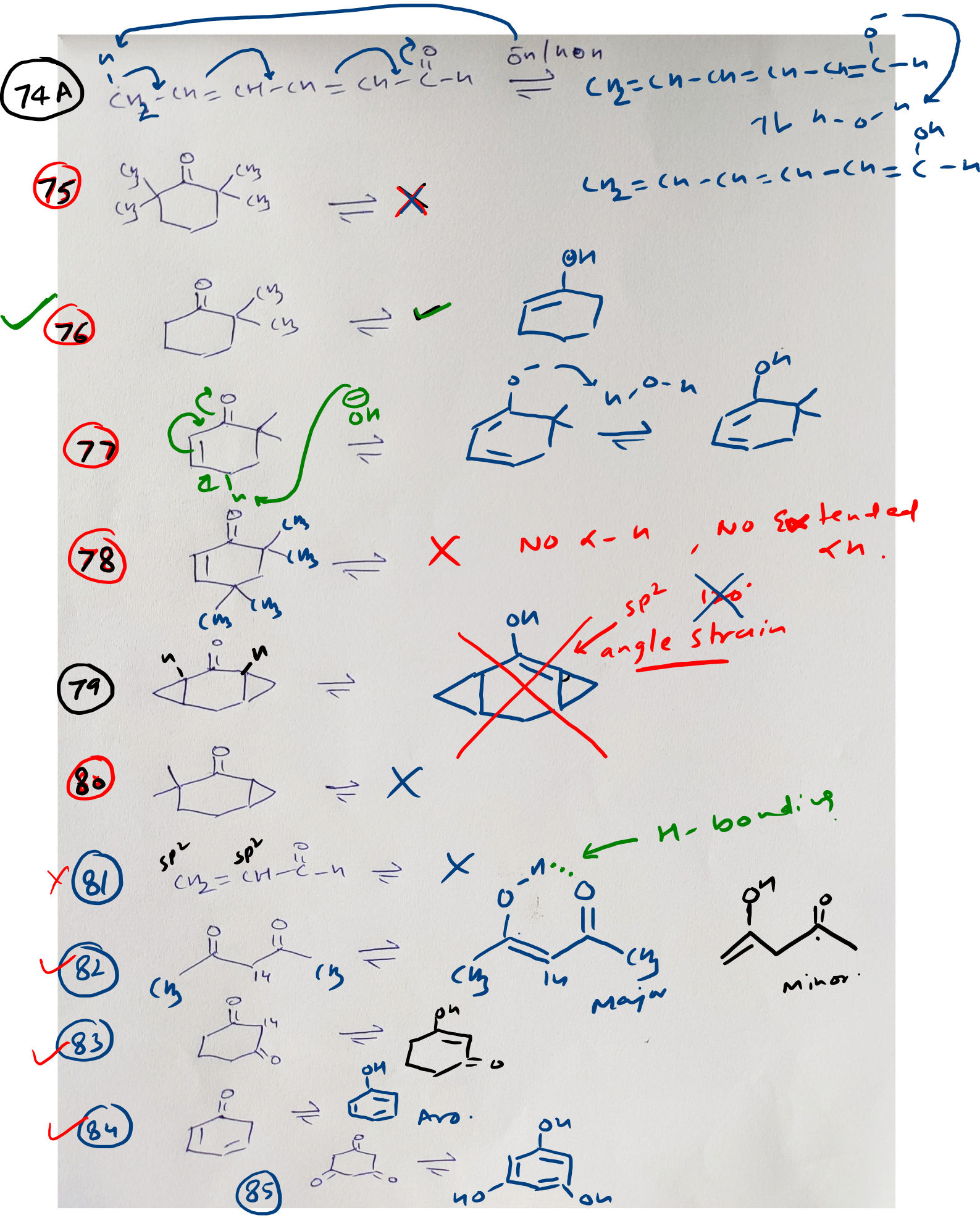
(63)



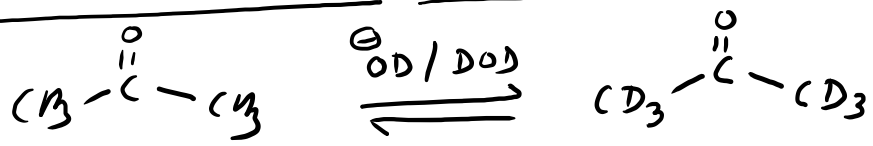
NOT formed. (Bredt's rule)

sp2 unstable


Identify the compound (s) which show Tautomerism
64
65
66
67
68
69
70
71
72
73
74
cis
trans.


74A
Enol/keto
75
76
77
78
NO α-H, NO extended αH.
sp2
angle strain
79
80
81
sp2 sp2
H-bonding
82
Major
Minor
83
84
Aro.
85

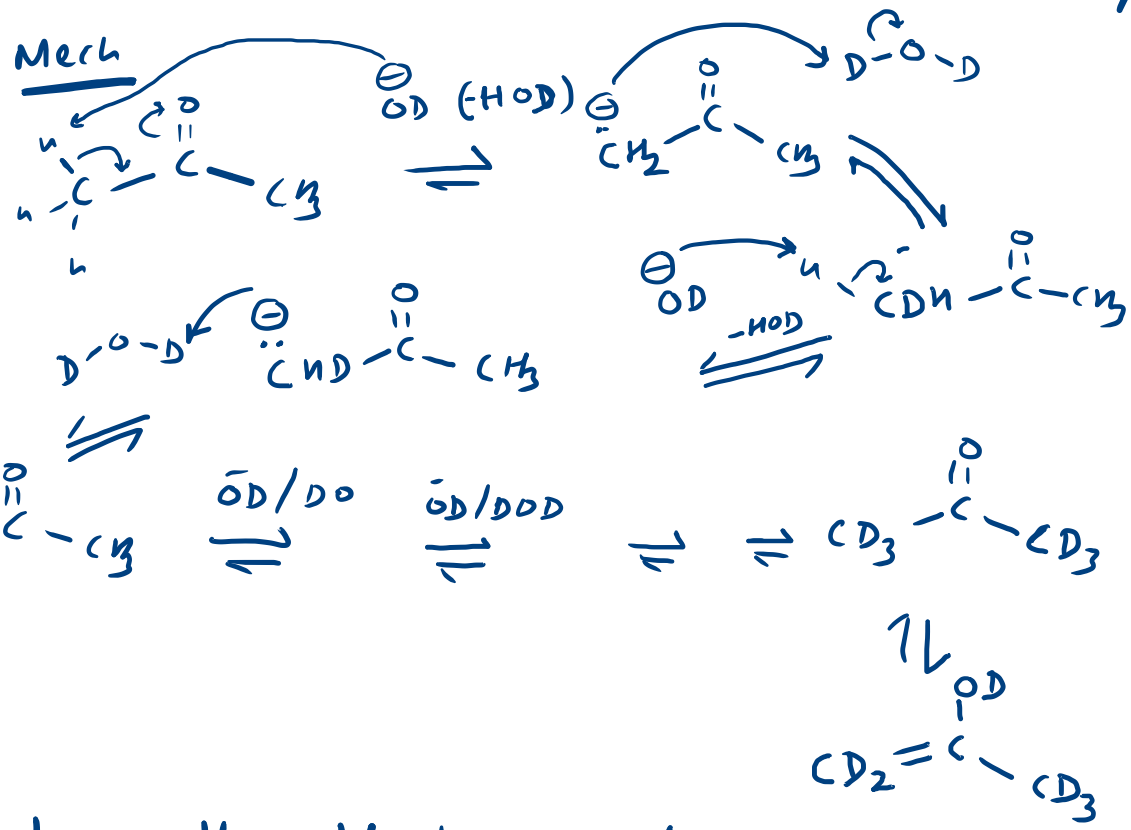
Deuterium exchange Mechanism

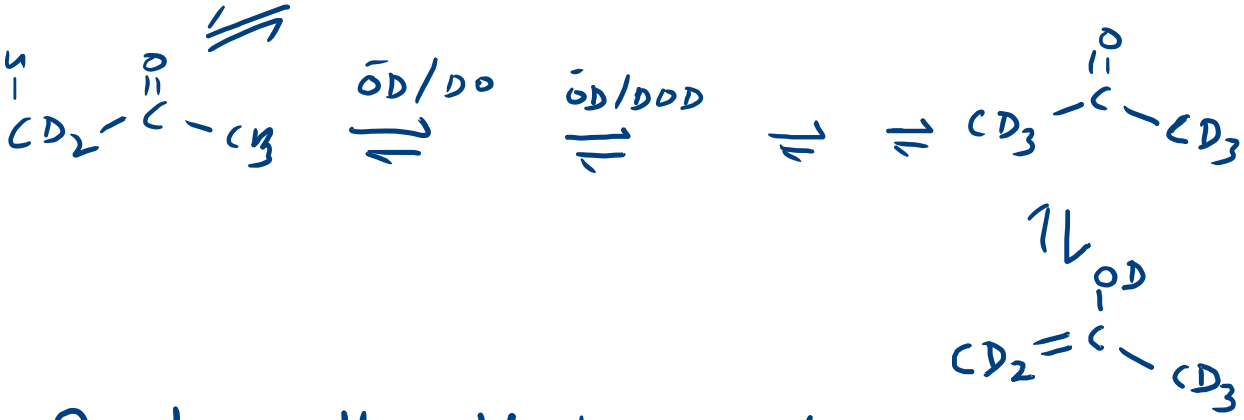
(86) 

6 H are replaced by D.

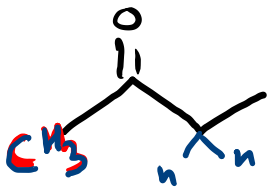
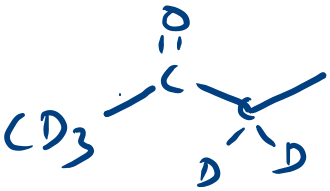
Dil NaOD or $^{-}OD / DOD$

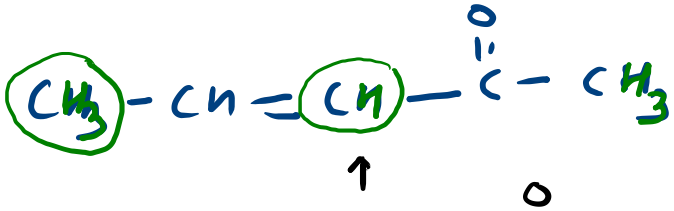
Mech

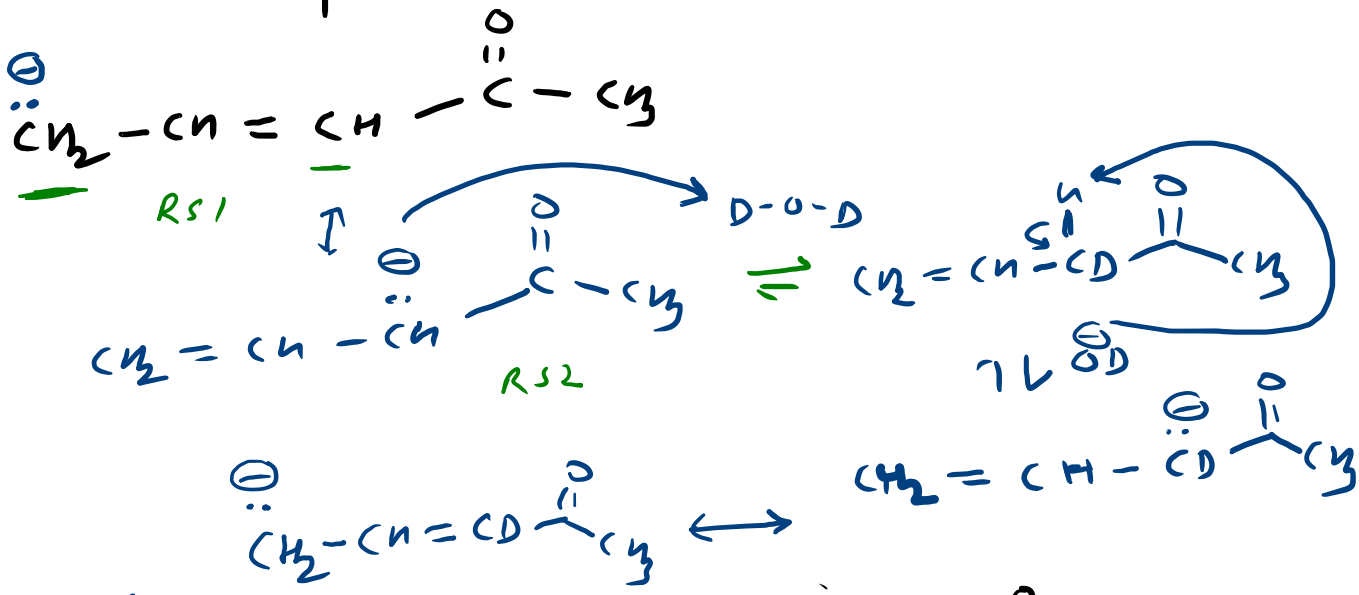


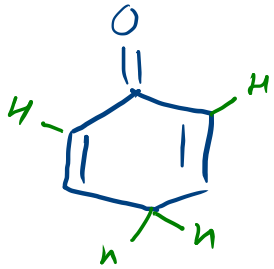


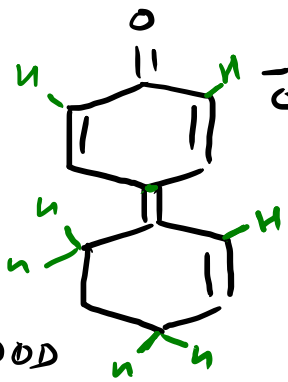
How many H are replaced by D when compound is introduced to $\bar{O}D/DOD$ (or Dil NaOD)

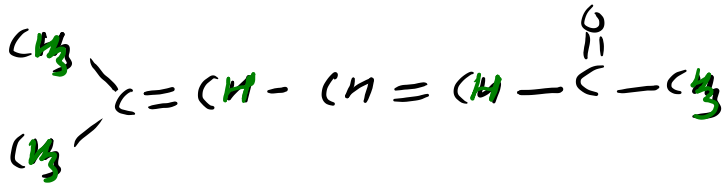
(87)  $\bar{O}D/DOD$  (5)

(88)  $\bar{O}D/DOD$ (7)

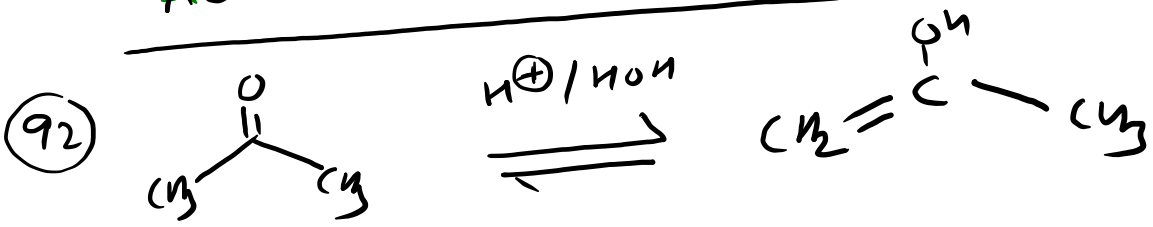


(89)  $\bar{O}D/DOD$ (4)

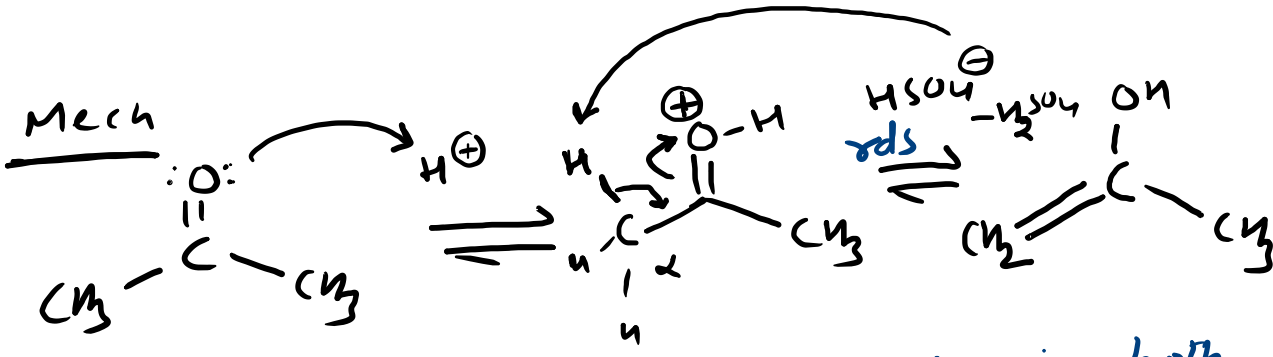
(90)  $\bar{O}D/DOD$ (7)

(91)  $\bar{O}D/DOD$ (11)

Acid catalysed Tautomerism

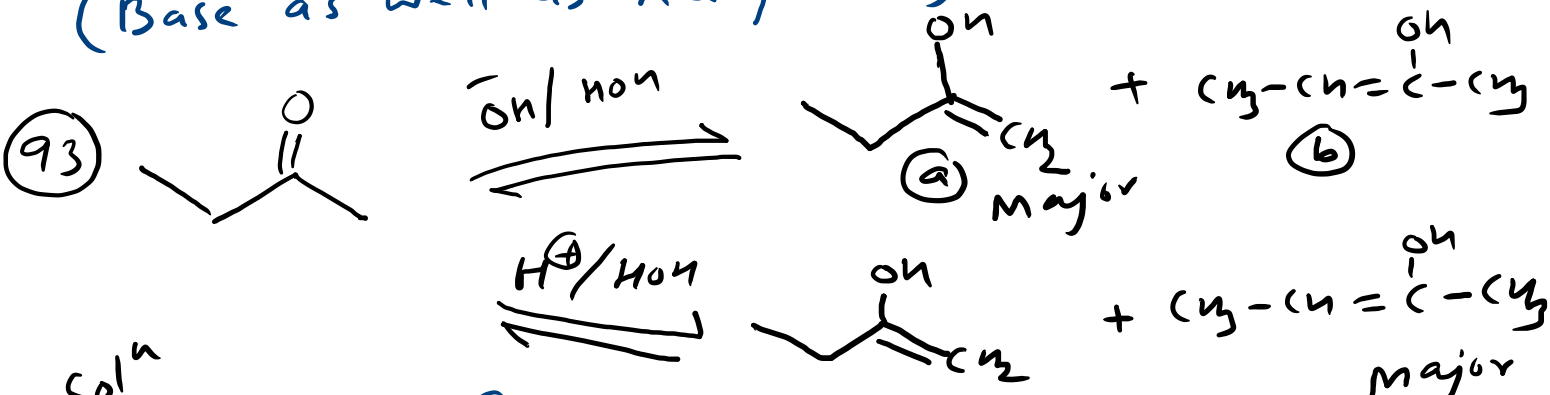
(92) 

Dil H_2SO_4 or $H^{\oplus}$/HOH or $H_3O^{\oplus}$

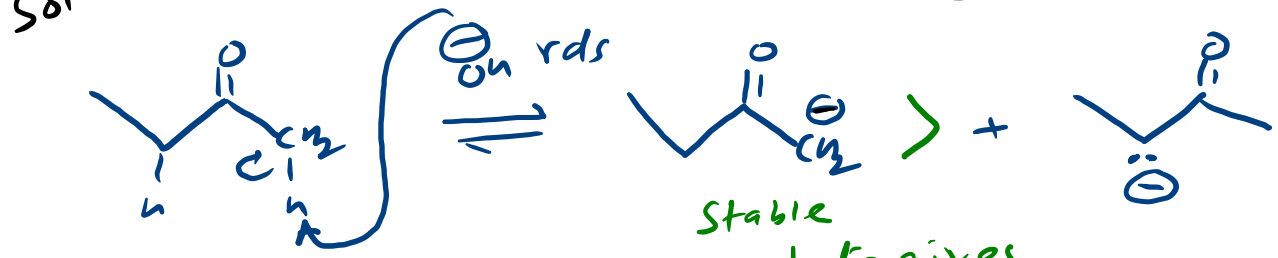
Mech 

rds

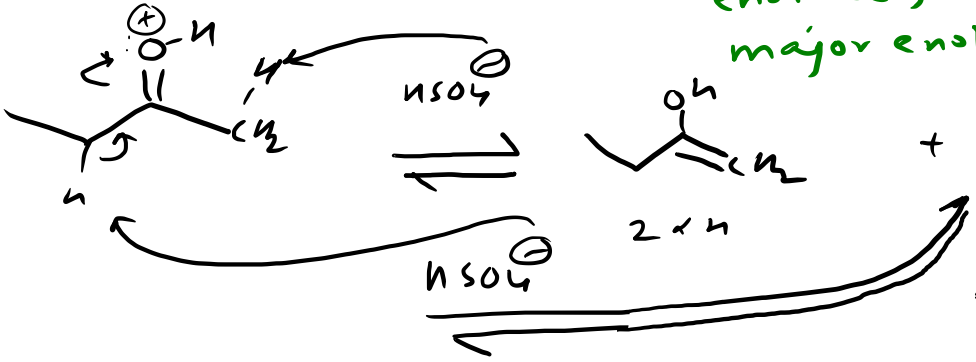
Note: Removal of Acidic H is rds in both (Base as well as Acid) catalysed Tautomerism.

(93) 

major

Sol^n 

Stable enolate gives major enol in Base catalysed mech.



2 α H

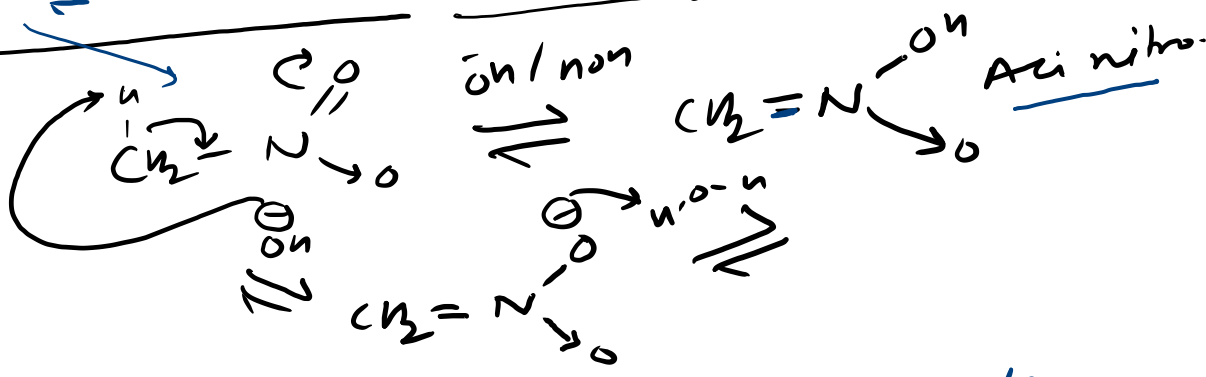
6 α H

Stable enol form will be major in acid catalysed Mech.

Others Tautomeric Systems

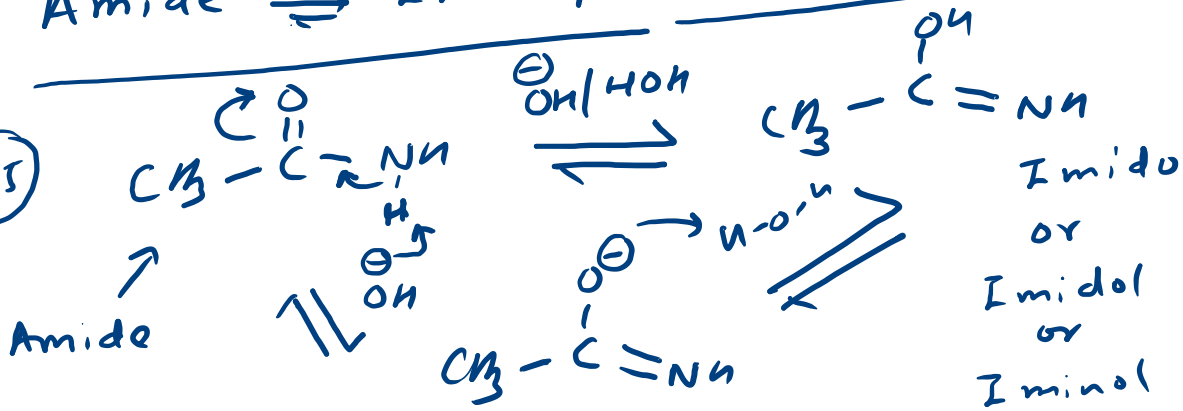
Triad ① Nitro ⇌ Aci nitro System of Tautomerism

(94)



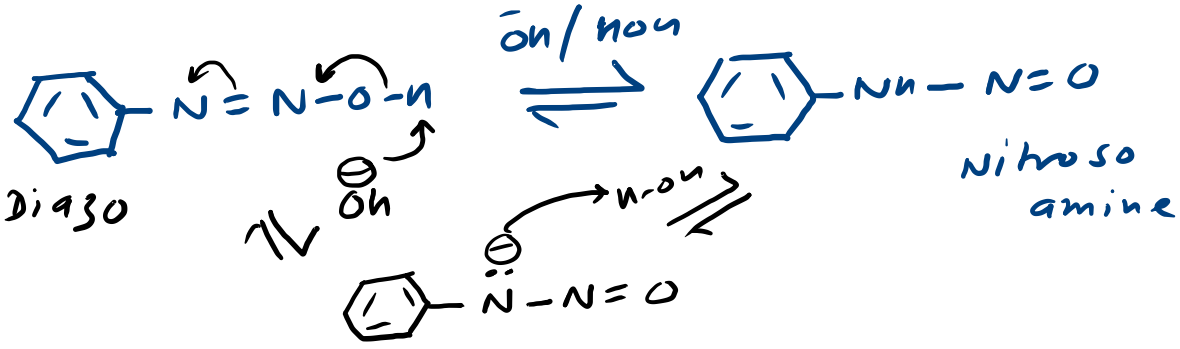
Triad ② Amide ⇌ Imido / Imidol / Iminol system

(95)



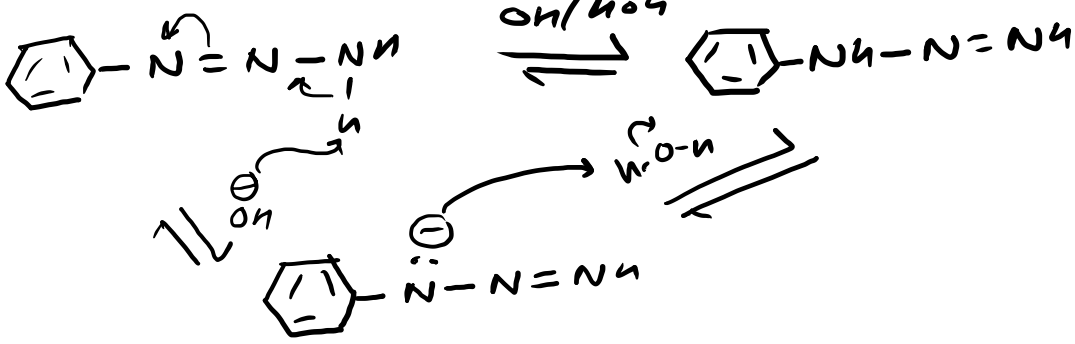
Triad ③ Diazo ⇌ Nitroso amine System of Tauto

(96)



Triad ④ Triazene System of Tautomerism

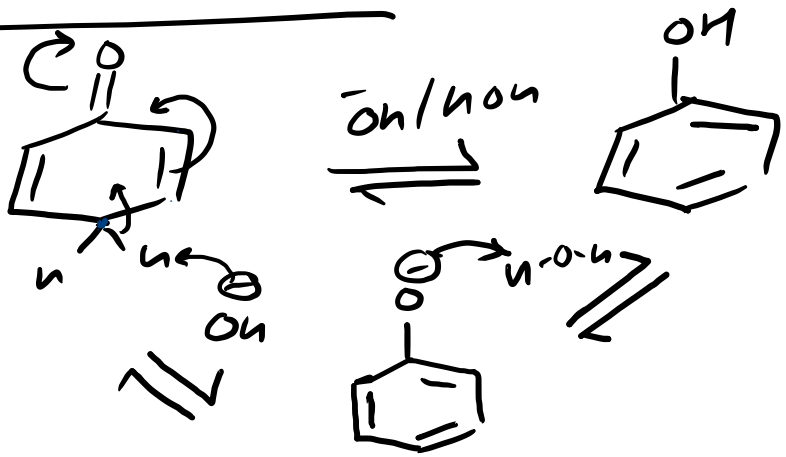
(97)



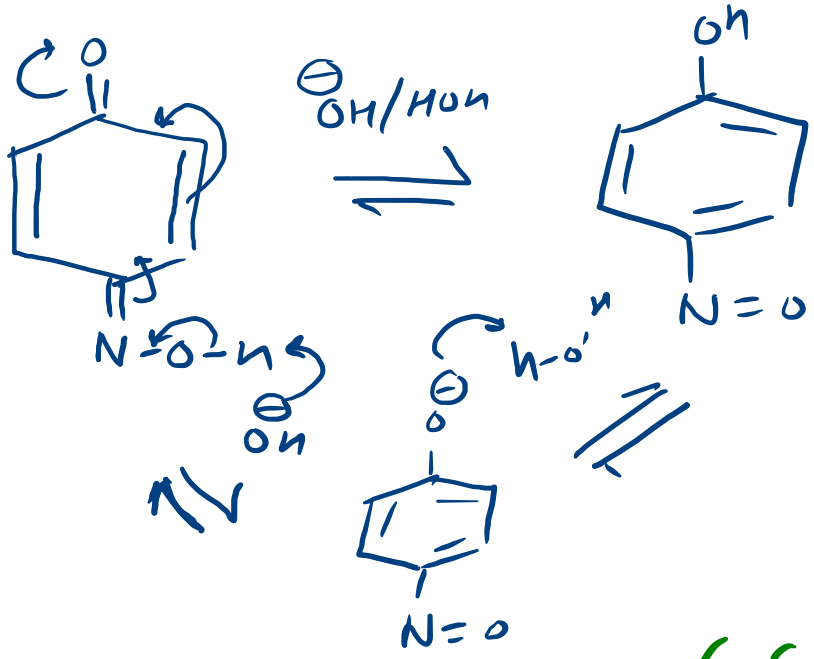
(5) Para Tautomerism (space Tautomerism)

Distance > 3

eg (98)

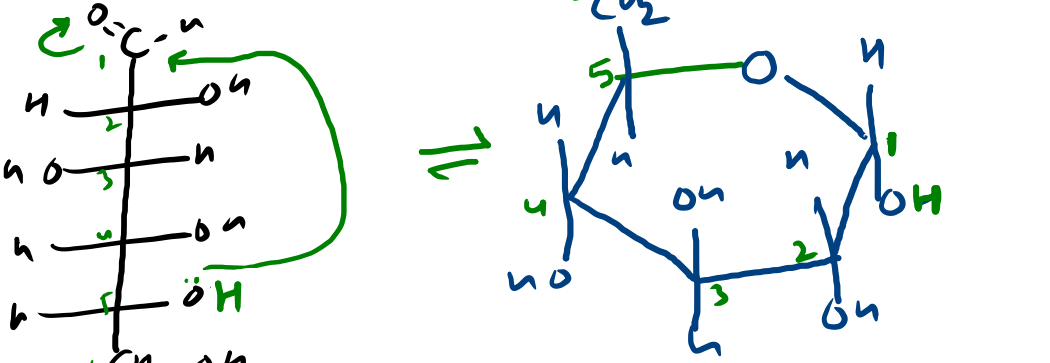


(99)



(6) Ring Chain Tautomerism (Space Tauto)

(100)



D- Glucose

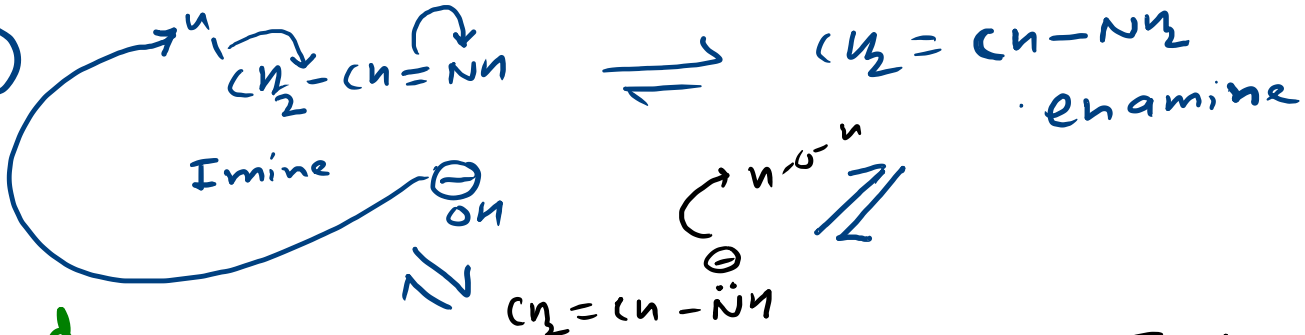
α- D-Glucose

Ring chain Tautomers.

Tricd.

(7) Imine ⇌ Enamine System of Tautomerism

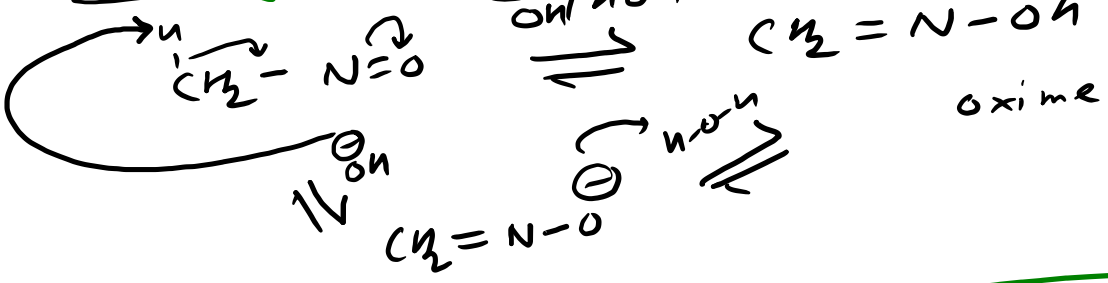
(101)



Triad

(8) Nitroso ⇌ oxime System of Tauto

(102)



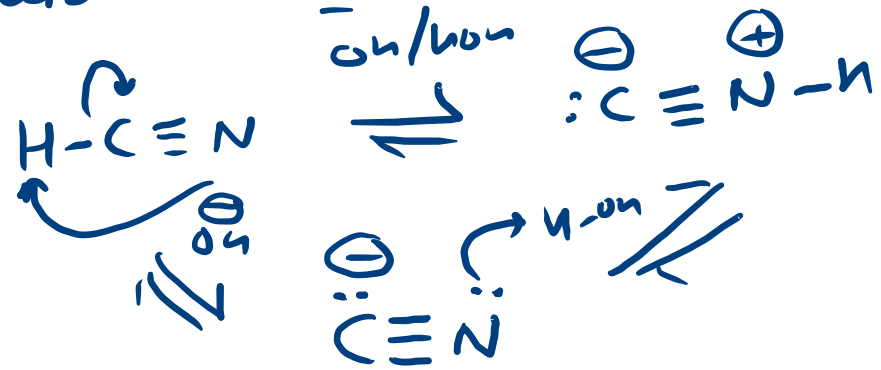
Diad (oscillation of H⊕ 1 ↔ 2)

Triad (" " " 1 ↔ 3)

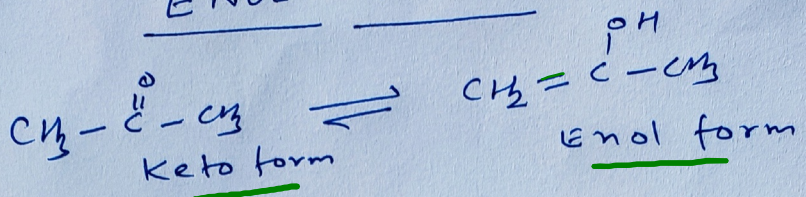
Space Tauto (" " " more than 3)

Diad

(103)



ENOL CONTENT



Note: Generally keto content is more than enol content.

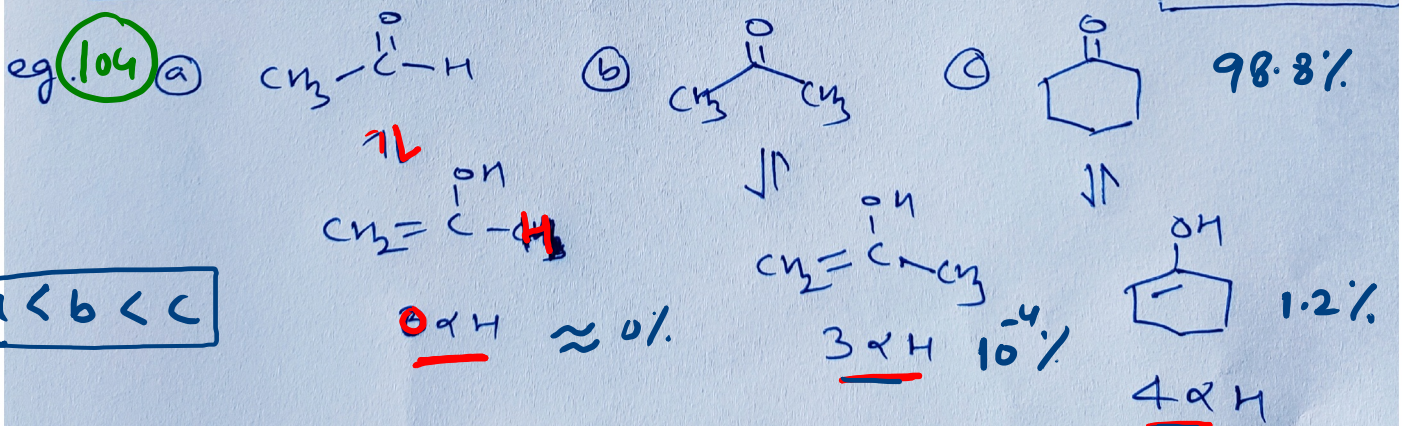
Reason ① Higher B.E of >C=O in keto form. (-52 K.cal)

② H attached to C (in keto form) is less acidic than H - attached to O (in enol form), so backward reaction is more than forward reaction

③ Unsaturated bond b/w carbons in enol form is less stable and more reactive, so backward reaction is more than forward reaction.

① Enol contentent In Mono carbonyl ~~compound~~ Compounds

eg. 104 ⓐ ⓑ ⓒ

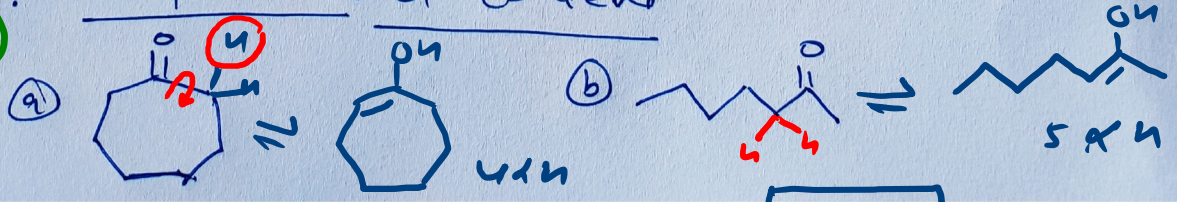


a < b < c

Note: more stable the enol form, more forward reaction, more will be the enol content.

eg. compare enol content

105



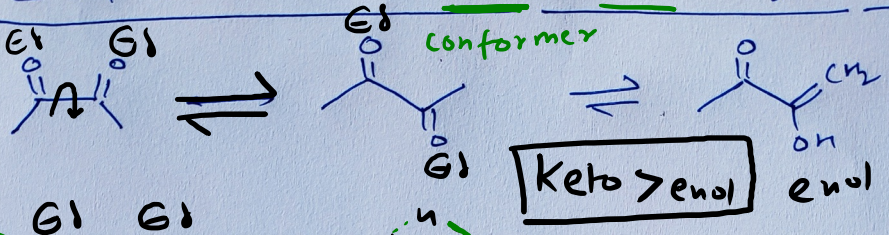
H of cyclic is more acidic

enol content a > b

cyclic

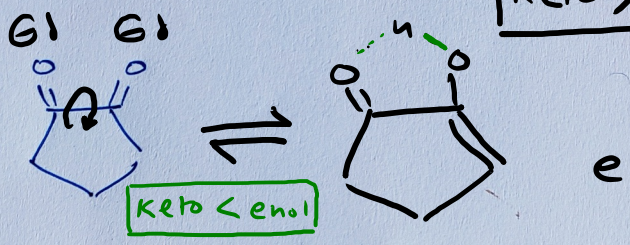
② Enol content in 1,2-Di carbonyl compounds

eq. 106



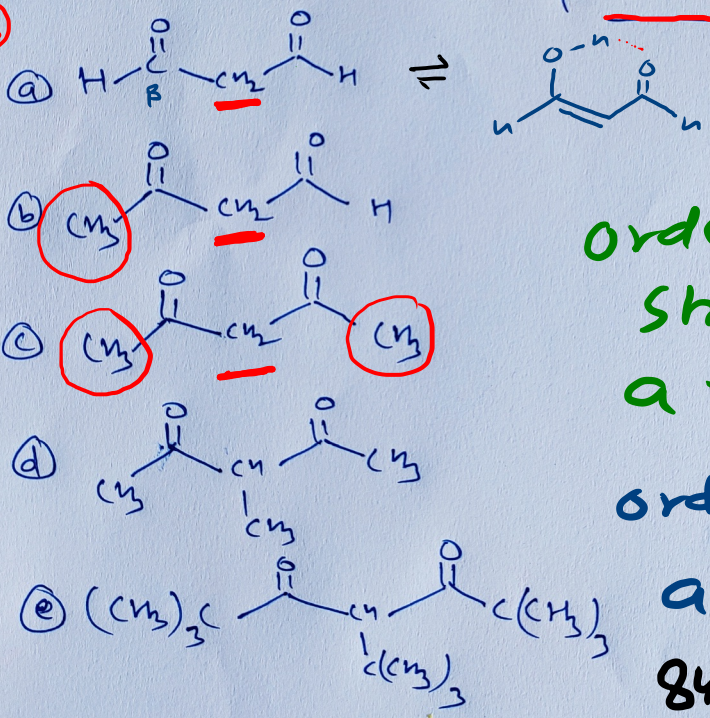
eq. 107

Restricted rotation



③ Enol content in 1,3-Di carbonyl compound or β-carbonyl compounds

eq. 108



order of acidic strength

a > b > c > d > e

order of enol content

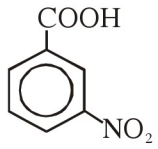
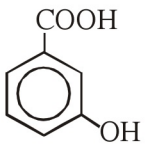
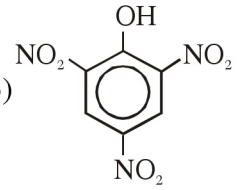
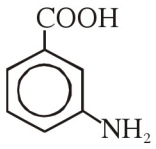
a > b > c > d > e

84% 80% 7% 44% ≈0%

Note: More acidic the H, more will be the forward reaction, which results more enol content;

EXERCISE # (JEE-MAIN)

1. Picric acid is - **[AIEEE-2002]**

(1)  (2)  (3)  (4) 

2. Which of the following speices acts both as bronsted acid & base – **[AIEEE-2002]**

(1) NH_3 (2) OH^- (3) HSO_4^- (4) 1 and 3 both

3. The correct order of increasing basic nature for the bases NH_3, CH_2NH_2 and $(CH_3)_2NH$ is- **[AIEEE-2003]**

(1) $CH_3NH_2 < NH_3 < (CH_3)_2NH$ (2) $(CH_3)_2NH < NH_3 < CH_3NH_2$

(3) $NH_3 < CH_3NH_2 < (CH_3)_2NH$ (4) $CH_3NH_2 < (CH_3)_2NH < NH$

4. Consider the acidity of the carboxylic acids- **[AIEEE-2004]**

(i) PhCOOH (ii) o–$NO_2C_6H_4COOH$

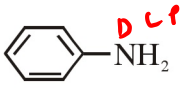
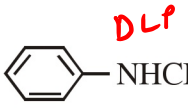
(iii) p-$NO_2C_6H_4COOH$ (iv) m-$NO_2C_6H_4COOH$

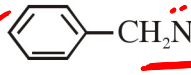
which of the following is the correct order of acidity-

(1) i > ii > iii > iv (2) ii > iv > iii > i

(3) ii > iv > i > iii (4) ii > iii > iv > i

5. Which of the following is the strongest base - **[AIEEE-2004]**

(1)  (2) 

(3) (4) 

6. Among the following acids which has the lowest pk_a value- **[AIEEE-2005]**

(1) CH_3CH_2COOH (2) $(CH_3)_2CHCOOH$

(3) HCOOH (4) CH_3COOH

7. Amongest the following the most basic compound is- **[AIEEE-2005]**

(1) p-nitro aniline (2) Acetanilide

(3) Aniline (4) Benzylamine

8. What is the conjugate base of OH^-? **[AIEEE-2005]**

(1) H_2O (2) O_2 (3) O^{2-} (4) O^-

9. Among the following acids which has the lowest pK_a value? **[AIEEE-2005]**

(1) HCOOH (2) CH_3COOH

(3) CH_3CH_2COOH (4) $(CH_3)_2CH–COOH$

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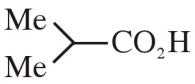
10. The correct order of increasing acid strength of the compounds is **[AIEEE-2006]**

(a) CH_3CO_2H (b) $MeOCH_2CO_2H$

(c) CF_3CO_2H (d) 

(1) d < a < c < b (2) d < a < b < c

(3) a < d < c < b (4) b < d < a < c

11. Which one of the following is the strongest base in aqueous solution ? **[AIEEE-2007]**

(1) Trimethylamine (2) Aniline

(3) Dimethylamine (4) Methylamine

12. The correct order of increasing basicity of the given conjugate base (R=CH_3) is :- **[AIEEE-2010]**

(1) $RCO\bar{O} < HC \equiv \bar{C} < \bar{N}H_2 < \bar{R}$ (2) $RCO\bar{O} < HC \equiv \bar{C} < \bar{R} < \bar{N}H_2$

(3) $\bar{R} < HC \equiv \bar{C} < RCO\bar{O} < \bar{N}H_2$ (4) $RCO\bar{O} < \bar{N}H_2 < HC \equiv \bar{C} < \bar{R}$

13. The strongest acid amongst the following compounds is ? **[AIEEE-2011]**

(1) $CH_3CH_2CH(Cl)CO_2H$ (2) $ClCH_2CH_2CH_2COOH$

(3) CH_3COOH (4) HCOOH

14. The correct order of acid strength of the following compounds :-

A. Phenol B. p-Cresol

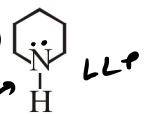
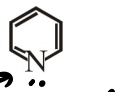
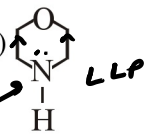
C. m-Nitrophenol D. p- Nitrophenol

is :- **[AIEEE-2011]**

(1) C > B > A > D (2) D > C > A > B

(3) B > D > A > C (4) A > B > D > C

15. In the following compounds : **[JEE(Main)-2012]**

(I)  (II)  (III)  (IV)

the order of basicity is as follows :

(1) IV > III > II > I (2) II > III > I > IV

(3) I > III > II > IV (4) III > I > II > IV

16. The most basic compound among the following is :- **[JEE(Main)-2012]**

(1) Acetanilide (2) Benzylamine

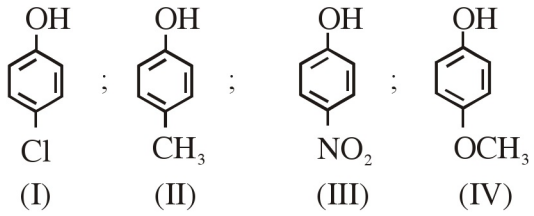
(3) p-Nitro aniline (4) Aniline

17. The order of basicity of amines in gaseous state is :- **[JEE(Main)-2013]**

(1) 3° > 2° > NH_3 > 1° (2) 1° > 2° > 3° > NH_3

(3) NH_3 > 1° > 2°> 3° (4) 3° > 2° > 1° > NH_3

18. Arrange the following compounds in order of decreasing acidity : **[JEE(Main)-2013]**



(1) II > IV > I > III (2) I > II > III > IV

(3) III > I > II > IV (4) IV > III > I > II

19. The conjugate base of hydrazoic acid is :- **[JEE(Main)-2014]**

(1) HN_3^- (2) N_3^- (3) N_2^- (4) N^{-3}

20. Which one of the following compounds will not be soluble in sodium bicarbonate ? **[JEE(Main)-2014]**

(1) Benzene sulphonic acid (2) Benzoic acid

(3) o-Nitrophenol (4) 2, 4, 6 - Trinitrophenol

21. Considering the basic strength of amines in aqueous solution, which one has the smallest pK_b value ? **[JEE(Main)-2014]**

(1) $(CH_3)_3N$ (2) $C_6H_5NH_2$ (3) $(CH_3)_2NH$ (4) CH_3NH_2

22. Among the following oxoacids, the correct decreasing order of acid strength is : **[JEE(Main)-2014]**

(1) $HClO_4 > HClO_3 > HClO_2 > HOCl$ (2) $HClO_2 > HClO_4 > HClO_3 > HOCl$

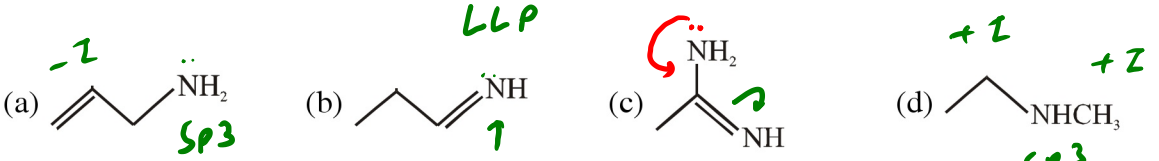
(3) $HOCl > HClO_2 > HClO_3 > HClO_4$ (4) $HClO_4 > HOCl > HClO_2 > HClO_3$

23. Among the following compounds, the increasing order of their basic strength is:-

(1) (II) < (I) < (III) < (IV) (2) (I) < (II) < (IV) < (III) **[JEE(Main)-On-Line 2017]**

(3) (II) < (I) < (IV) < (III) (4) (I) < (II) < (III) < (IV)

24. The increasing order of basicity of the following compounds is : **[JEE(Main)-2018]**



(1) (b) < (a) < (c) < (d) (2) (b) < (a) < (d) < (c)

(3) (d) < (b) < (a) < (c) (4) (a) < (b) < (c) < (d)

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25. The correct decreasing order for acid strength is :- **[JEE(Main)-Jan-2019]**

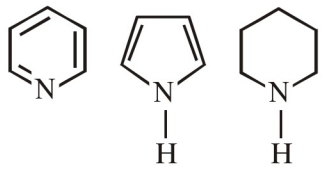
(1) $NO_2CH_2COOH > NCCH_2COOH > FCH_2COOH > ClCH_2COOH$

(2) $FCH_2COOH > NCCH_2COOH > NO_2CHCOOH > ClCH_2COOH$

(3) $NO_2CH_2COOH > FCH_2COOH > CNCH_2COOH > ClCH_2COOH$

(4) $CNCH_2COOH > O_2NCH_2COOH > FCH_2COOH > ClCH_2COOH$

26. Arrange the following amines in the decreasing order of basicity: **[JEE(Main)-Jan-2019]**



(I) (II) (III)

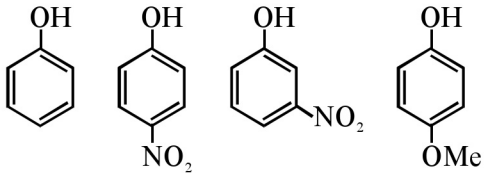
(1) I > II > III (2) III > II > I (3) I > III > II (4) III > I > II

27. The increasing basicity order of the following compounds is : **[JEE(Main)-Jan-2019]**

(A) $CH_3CH_2NH_2$ (B) $CH_3CH_2NH(CH_2CH_3)$ (C) $H_3C-N(CH_3)-CH_3$ (D) $Ph-NH(CH_3)$

(1) (D)<(C)<(A)<(B) (2) (A)<(B)<(D)<(C) (3) (A)<(B)<(C)< (D) (4) (D)<(C)<(B)<(A)

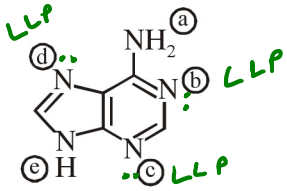
28. The increasing order of the pKa values of the following compounds is : **[JEE(Main)-Jan-2019]**



A B C D

(1) D < A < C < B (2) B < C < D < A (3) C < B < A < D (4) B < C < A < D

29. In the following compound, **[JEE(Main)-Jan-2019]**



the favourable site/s for protonation is/are :-

(1) (b), (c) and (d) (2) (a) (3) (a) and (e) (4) (a) and (d)

30. The correct order for acid strength of compounds **[JEE(Main)-Jan-2019]**

$CH{\equiv}CH$, $CH_3{-}C{\equiv}CH$ and $CH_2{=}CH_2$

is as follows :

(1) $CH \equiv CH > CH_2 = CH_2 > CH_3{-}C \equiv CH$

(2) $HC \equiv CH > CH_3 - C \equiv CH > CH_2 = CH_2$

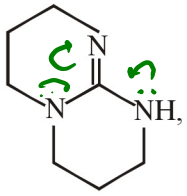
(3) $CH_3{-}C \equiv CH > CH_2 = CH_2 > HC \equiv CH$

(4) $CH_3{-}C \equiv CH > CH \equiv CH > CH_2 = CH_2$

31. The increasing order of pK_b for the following compounds will be : **[JEE(Main)-Jan-2020]**

$NH_2 - CH = NH$,

(A)



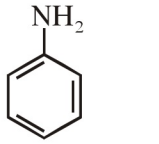
(B)

CH_3NHCH_3

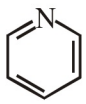
(C)

(1) (A) < (B) < (C) (2) (C) < (A) < (B) (3) (B) < (A) < (C) (4) (B) < (C) < (A)

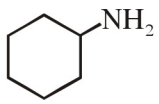
32. The decreasing order of basicity of the following amines is : **[JEE(Main)-Jan-2020]**



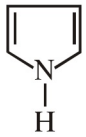
(I)



(II)



(III)



(IV)

(1) (I) > (III) > (IV) > (II) (2) (III) > (I) > (II) > (IV)

(3) (III) > (II) > (I) > (IV) (4) (II) > (III) > (IV) > (I)

33. The increasing order of basicity for the following intermediates is (from weak to strong)

(i) $H_3C - \overset{CH_3}{\underset{CH_3}{\overset{|}{\underset{|}{C}}}}^{\ominus}$

[JEE(Main)-Jan-2020]

(ii) $H_2C = CH - \overset{\ominus}{C}H_2$

(iii) $HC \equiv \overset{\ominus}{C}$ (iv) $\overset{\ominus}{C}H_3$ (v) $\overset{\ominus}{C}N$

(1) (v) < (i) < (iv) < (ii) < (iii)

(2) (iii) < (i) < (ii) < (iv) < (v)

(3) (v) < (iii) < (ii) < (iv) < (i)

(4) (iii) < (iv) < (ii) < (i) < (v)

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